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Sheaf cohomology

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Introduction

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PART I

Category theory

Introduction

Why categories? Historical introduction?

Chapter 1

Categories, morphisms and duality

1.1 Categories

Definition 1.1.1 (Category). A **category** C consists of:

- a class $\mathrm{Ob}(\mathsf{C})$ of **objects**.
- for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ a collection of sets $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ of **morphisms from** X **to** Y.
- for each object $X \in \mathrm{Ob}(\mathsf{C})$ a morphisms $\mathbb{1}_X \in \mathrm{Hom}_{\mathsf{C}}(X, X)$ called the **identity** morphism.
- for each pair of morphisms $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$ a morphism $g \circ f \in \mathrm{Hom}_{\mathsf{C}}(X, Z)$ called their **composite**.

satisfying the following axioms:

- Composition is associative: for each quadruple $X, Y, Z, W \in \mathrm{Ob}(\mathsf{C})$ of objects, if $f \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$, $g \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathsf{C}}(W, X)$ then

$$(f \circ g) \circ h = f \circ (g \circ h)$$

- Composition satiesfies the unit laws: for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ if $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ then

$$\mathbb{1}_Y \circ f = f = f \circ \mathbb{1}_X$$

Definition 1.1.2 (Small category). A category C is **small** if its collection of objects $\mathrm{Ob}(\mathsf{C})$ is a set.

Remark 1.1.1 (Size issues). We will be sketchy about foundational issues (set-theoretic issues, universes, ...).
Therefore all categories will be $\mathcal{U}$-categories for a large enough *universe* $\mathcal{U}$ (i.e: for any pair of objects $X, Y \in \mathsf{C}$ the Hom-set $\mathrm{Hom}(X, Y)$ is $\mathcal{U}$-small) and small categories will be $\mathcal{U}$-small (the set of objects $\mathrm{Ob}(\mathsf{C})$ is $\mathcal{U}$-small).
We will refer as "big" categories those that are $\mathcal{V}$-categories for a larger universe $\mathcal{V}$. By Grothendieck's axiom (for any set X there exists a universe $\mathcal{U}$ such that $X \in \mathcal{U}$) every category C is a $\mathcal{V}$-category for a big enough universe $\mathcal{V}$, and this universe can be chosen so that C is $\mathcal{V}$-small.
For a basic introduction of what an *universe* is and a more carefull treatment of this type of size issues see [KS06, Section 1.1]. More details can also be found at [SGA4]

Notation. To indicate that X is an object of a category C we write

$$X \in \mathsf{C} \text{ instead of } X \in \mathrm{Ob}(\mathsf{C})$$

To indicate that $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a morphism from X to Y we write

$$f : X \to Y \text{ or } X \xrightarrow{f} Y$$

We will sometimes abbreviate the composition $f \circ g$ as fg.

Example 1.1.1.

- The category Set has the collection of all sets as objects and functions as morphisms.
- The category Grp has groups as objects and group homomorphisms as morphisms.
- The category Ab has abelian groups as objects and group homomorphisms as morphisms. It is a full subcategory of Grp.
- The category Ring has rings as objects and ring homomorphisms as morphisms.[a]
- The category Rng has (not necessarilly unital) rings as objects and ring homomorphisms as morphism.
- The category Top has topological spaces as objects and continuous functions as morphisms.
- For a fixed ring R, the category $\mathsf{Mod}(R)$ has (left) R-modules as objects and R-linear maps as morphisms.
- For a fixed field k, the category Vect_k has k-vector spaces as objects and k-linear maps as morphisms.

[a] We adopt the convetion that all rings have an unit and all ring homomorphisms preserve it.

Examples of non-concrete categories?

Example 1.1.2 (Discrete category)**.** Given a category C we can define the **discrete category** C_d considering the same collection of objects as C but only the identity morphisms between them.

Definition 1.1.3 (Subcategory)**.** Given a category C, a **subcategory** D consists of a subcollection of objects and a subcollection of morphisms in C such that

- If $f : X \to Y$ is a morphism in D then X, Y are objects in D.
- If $f : X \to Y$ and $g : Y \to Z$ are morphisms in D, the composition $gf : X \to Z$ is in D.
- If X is an object in D then the identity map $\mathbb{1}_X$ is in D.

We say that D is a **full subcategory** if for all objects X, Y in D every morphism $f : X \to Y$ in C is also in D.

Example 1.1.3.

- Ab is a full subcategory of Grp.
- The category of finite sets FinSet is a full subcategory of Set.
- Ring is a subcategory of Rng that is not full.
- If k is a field, Vect_k is a full subcategory of $\mathsf{Mod}(k)$.

Definition 1.1.4 (Product category)**.** Let C, D be two categories. The **product category** $\mathsf{C} \times \mathsf{D}$ is the category consisting of

- As objects: ordered pairs (C, D), where C is an object in C and D is an object in D, i.e:
$$\mathrm{Ob}_{\mathsf{C}\times\mathsf{D}} = \mathrm{Ob}_{\mathsf{C}} \times \mathrm{Ob}_{\mathsf{D}}$$
- As morphisms: ordered pairs $(C \xrightarrow{f} C', D \xrightarrow{g} D')$, where $C \xrightarrow{f} C'$ is a morphism in C and $D \xrightarrow{g} D'$ is a morphism in D, i.e: for all objects $(C, D), (C', D')$ in $\mathsf{C} \times \mathsf{D}$
$$\mathrm{Hom}_{\mathsf{C}\times\mathsf{D}}((C, D), (C', D')) = \mathrm{Hom}_{\mathsf{C}}(C, C') \times \mathrm{Hom}_{\mathsf{D}}(D, D')$$
- Composition is defined *componentwise*: if $(C \xrightarrow{f} C', D \xrightarrow{g} D')$ and $(C' \xrightarrow{f'} C'', D' \xrightarrow{g'} D'')$ are two morphisms in $\mathsf{C} \times \mathsf{D}$ then
$$(f', g') \circ (f, g) \coloneqq (f' \circ f, g' \circ g) : C \times D \to C'' \times D''$$

Example 1.1.4 (Category of open sets)**.** Let X be a topological space.
The category $\mathsf{Op}(X)$ consisting of

- As objects, the open subsets of X.
- As morphisms, inclussion of open sets: $U \to V$ if and only if $U \subseteq V$

is small.

1.1.1 Duality

Definition 1.1.5 (Opposite category)**.** Let C be a category. The **opposite category** C^{op} consists of:

- The same objects as C: $\mathrm{Ob}(\mathsf{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathsf{C})$.
- For each pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.
- If $f^{\mathrm{op}} : X \to Y$, $g^{\mathrm{op}} : Y \to Z$ are morphisms in C^{op} their composition is defined by
$$f^{\mathrm{op}} \circ g^{\mathrm{op}} = (g \circ f)^{\mathrm{op}}$$
where $f : Y \to X$ and $g : Z \to Y$ are the associated morphisms in C.

Example 1.1.5. Consider a preordered set $(P, \leq)$ viewed as a small category where a unique morphism $x \to y$ exists if and only if $x \leq y$.
The opposite category P^{op} corresponds to the same set P equipped with the reversed order $\geq$. That is, a morphism $x \to y$ exists in P^{op} if and only if $y \leq x$ in P.

Remark 1.1.2. Let $f^{\mathrm{op}} : Y \to X$ and $g^{\mathrm{op}} : Y \to X$ be two morphisms in the opposite category C^{op}. Then $f^{\mathrm{op}} = g^{\mathrm{op}}$ in C^{op} if and only if $f = g$ in C.

Definition 1.1.6 (Dual statement)**.** A categorical statement P^* is defined as the **dual** of a statement P in a category C if P^* holds in the opposite category C^{op} if and only if P holds in C.

Remark 1.1.3. Note that by definition for any category C we have that $(\mathsf{C}^{\mathrm{op}})^{\mathrm{op}} = \mathsf{C}$. Therefore if a statement is true for all categories C, its dual is true for all categories C^{op}. Since C^{op} is an arbitrary category the dual statement is also true for all categories.

In practice, this means that proving an statement directly proves its dual for free.

1.2 Types of morphisms

Definition 1.2.1 (Monomorphism and epimorphism). Let $f : X \to Y$ be a morphism in a category C. We say that f is a

- **Monomorphism** (or is **monic**) if for every pair of parallel morphisms $g, h : Z \to X$ in C, the equality $f \circ g = f \circ h$ implies $g = h$.
- **Epimorphism** (or is **epic**) if for every pair of parallel morphisms $g, h : Y \to Z$ in C, the equality $g \circ f = h \circ f$ implies $g = h$.

Example 1.2.1.

- In Set, an monomorphism (resp. epimorphism) is an injective (resp. surjective) function.
- In Grp, an monomorphism (resp. epimorphism) is an injective (resp. surjective) group homomorphism.
- In Top, an monomorphism (resp. epimorphism) is an injective (resp. surjective) continuous map.

Lemma 1.2.1 (Monomorphisms and epimorphisms are dual). Let $f : X \to Y$ be a morphism in a category C. Consider the associated morphism $f^{\mathrm{op}} : Y \to X$ in the opposite category C^{op}.
Then f is a monomorphism (resp. epimorphism) in C if and only if f^{op} is an epimorphism (resp. epimorphism) in C^{op}.

Proof. Consider of pair of morphisms $g, h : Z \rightrightarrows X$ in C and their opposites $g^{\mathrm{op}}, h^{\mathrm{op}} : X \rightrightarrows Z$ in C^{op}.

Note that, by definition of composition in C^{op} we have

$$g^{\mathrm{op}} \circ f^{\mathrm{op}} = (f \circ g)^{\mathrm{op}}$$

Therefore we have that $f \circ g = f \circ h$ implies $g = h$ if and only if $(f \circ g)^{\mathrm{op}} = (f \circ h)^{\mathrm{op}}$ implies $g = h$ if and only if $g^{\mathrm{op}} \circ f^{\mathrm{op}} = h^{\mathrm{op}} \circ f^{\mathrm{op}}$ implies $g^{\mathrm{op}} = h^{\mathrm{op}}$. □

Definition 1.2.2 (Isomorphism). A morphism $f : X \to Y$ in a category C is an **isomorphism** if there exists another morphism $g : Y \to X$ in C such that

$$g \circ f = \mathbb{1}_X \quad \text{and} \quad f \circ g = \mathbb{1}_Y$$

The morphism g is called the **inverse** of f and is denoted by f^{-1}.

Example 1.2.2.

- In Set, an isomorphism is a bijection.
- In Grp, an isomorphism is a bijective group homomorphism.
- In Top, an isomorphism is a homeomorphism.

Proposition 1.2.1. Let $f : X \to Y$ be an isomorphism. Then f is both a monomorphism and epimorphism.

Proof. Suppose $f : X \to Y$ is an isomorphism in a category C. Then there exists a morphism $f^{-1} : Y \to X$ in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Consider a pair of morphisms $g, h : Z \rightrightarrows X$ such that $f \circ g = f \circ h$. Precomposing with f^{-1} we get

$$f^{-1} \circ (f \circ g) = f^{-1} \circ (f \circ h) \implies (f^{-1} \circ f) \circ g = (f^{-1} \circ f) \circ h \implies g = h$$

Therefore f is a monomorphism.

Analogously, if we consider pair of morphism $g', h' : Y \rightrightarrows Z$ such that $g' \circ f = h' \circ f$ this time by postcomposing with f^{-1} we get $g' = h'$, i.e: f is a epimorphism. □

Example 1.2.3. In general, a morphism $f : X \to Y$ that is both a monomorphism and epimorphism is not an isomorphism.
Indeed, in the category Top of topological spaces consider a set with more than one point X endowed with the discrete X_d and trivial X_t topology.
Consider the identity map $f : X_d \to X_t$, which is continuous. Since f is injective we have that it is a monomorphism and, since it is surjective, it is an epimorphism. However since there does not exists a continuous function $g : X_t \to X_d$, f cannot be an isomorphism.

Proposition 1.2.2. Let $f : X \to Y$ and $g : Y \to Z$ be isomorphisms in a category C. Then:

1. The inverse f^{-1} of f is unique.
2. The identity $\mathbb{1}_X$ is an isomorphism for every $X \in \mathsf{C}$.
3. $g \circ f$ is an isomorphism with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
4. If $g \circ f$ is an isomorphism and f is an isomorphism, then g is an isomorphism. Analogously if $g \circ f$ and g are isomorphisms, then f is an isomorphism.

Proof.

1. Suppose that $h_1, h_2 : Y \to X$ are inverses of f. We have
$$h_1 = h_1 \circ \mathbb{1}_Y = h_1 \circ (f \circ h_2) = (h_1 \circ f) \circ h_2 = \mathbb{1}_X \circ h_2 = h_2$$
2. For all $X \in \mathsf{C}$ by definition
$$\mathbb{1}_X \circ \mathbb{1}_X = \mathbb{1}_X$$
i.e: $\mathbb{1}_X$ is its own left and right inverse.
3. We only need to check that $f^{-1} \circ g^{-1}$ is both a left and right inverse of $g \circ f$. On the one hand
$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ f = \mathbb{1}_X$$
On the other
$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ g^{-1} = \mathbb{1}_Z$$

4. Let $h \coloneqq (g \circ f)^{-1} : Z \to X$. We are going to check that $f \circ h : Z \to Y$ is the inverse of g. On the one hand

$$g \circ (f \circ h) = (g \circ f) \circ h = \mathbb{1}_Z$$

On the other, since $h \circ (g \circ f) = \mathbb{1}_X$ precomposing with f and postcomposing with f^{-1} we have

$$\begin{aligned} f \circ (h \circ g \circ f) \circ f^{-1} = f \circ \mathbb{1}_X \circ f^{-1} = \mathbb{1}_Y &\implies (f \circ h) \circ g \circ (f \circ f^{-1}) = \mathbb{1}_Y \\ &\implies (f \circ h) \circ g = \mathbb{1}_Y \end{aligned}$$

□

Proposition 1.2.3. Let D be a subcategory of a category C. If $f : X \to Y$ is an isomorphism in D then f is also an isomorphism in C.

Proof. Since $f : X \to Y$ is an isomorphism in D there exists an inverse $f^{-1} : Y \to X \in \mathrm{Hom}_D(Y, X)$. Therefore since $\mathrm{Hom}_{\mathsf{D}}(Y, X) \subseteq \mathrm{Hom}_{\mathsf{C}}(Y, X)$, f^{-1} is an inverse of f in C. □

Proposition 1.2.4. Let D be a full subcategory of a category C. If $f : X \to Y$ is an isomorphism in C and $X, Y \in \mathsf{D}$ then f is also an isomorphism in D.

Proof. Since $f : X \to Y$ is an isomorphism in C there exists an inverse $f^{-1} : Y \to X \in \mathrm{Hom}_C(Y, X)$.

Since D is a full subcategory of D and $X, Y \in \mathsf{D}$ we have that $f^1 \in \mathrm{Hom}_{\mathsf{D}}(Y, X)$. Therefore f admits an inverse in D. □

1.3 Special objects

Definition 1.3.1 (Initial and terminal objects)**.** An object $I \in \mathsf{C}$ is **initial** if for every object $X \in \mathsf{C}$ there is a unique morphism $I \to X$.
Dually, an object $T \in \mathsf{C}$ is **terminal** if for every object $X \in \mathsf{C}$ there is a unique morphism $X \to T$.

Proposition 1.3.1 (Uniqueness of initial and terminal objects)**.** Let X, Y be two initial (resp. terminal) objects in a category C. Then there exists a unique isomorphism $X \xrightarrow{\sim} Y$ in C.

Proof. Since X is initial in C there exists a unique morphism $f : X \to Y$. Analogously there exist a unique morphism $g : Y \to X$.

Note that $g \circ f : X \to X$. Therefore since X is initial $g \circ f = \mathbb{1}_X$. Analogously since $f \circ g : Y \to Y$ we have $f \circ g = \mathbb{1}_Y$.

Therefore f and g are inverses of each other, i.e: $X \cong Y$. Moreover they are unique by definition. □

Definition 1.3.2 (Zero object)**.** An object $0 \in \mathsf{C}$ is a **zero** object if it is both initial and terminal.
Note that for any two objects $X, Y \in \mathsf{C}$ there exist maps $X \to 0 \to Y$. We call this composite map the **zero map** between X and Y and denote it $X \xrightarrow{0} Y$.

Corollary 1.3.1 (Uniqueness of zero objects)**.** Let $0, \widetilde{0}$ be two zero objects in a category C. Then there exists a unique isomorphism $0 \xrightarrow{\sim} \widetilde{0}$ in C.

Proof. Since 0 and $\widetilde{0}$ are in particular initial we can apply Proposition 1.3.1. □

Chapter 2

Functors and natural transformations

2.1 Functors

Definition 2.1.1 (Functor). Let C, D be two categories. A (covariant) **functor** $F : \mathsf{C} \to \mathsf{D}$ consists of:

- for each object $X \in \mathsf{C}$, an object $F(X) \in \mathsf{D}$.
- for each morphism $f : X \to Y$ in C, a morphism $Ff : F(X) \to F(Y)$ in D,

such that:

- $F(\mathbb{1}_X) = \mathbb{1}_{F(X)}$ for every $X \in \mathsf{C}$.
- $F(g \circ f) = Fg \circ Ff$ for every composable pair of morphisms f, g in C.

A functor is **contravariant** if it reverses the order of composition: $F(g \circ f) = Ff \circ Fg$. This is equivalent to a covariant functor $F : \mathsf{C}^{\mathrm{op}} \to \mathsf{D}$.
Diagramatically, a covariant functor is represent by

$$\begin{array}{rccc} F: & \mathsf{C} & \longrightarrow & \mathsf{D} \\ & X & & F(X) \\ & {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle Ff} \\ & Y & & F(Y) \end{array}$$

And a contravariant functor by

$$\begin{array}{rccc} F: & \mathsf{C} & \longrightarrow & \mathsf{D} \\ & X & & F(X) \\ & {\scriptstyle f}\downarrow & \longmapsto & \uparrow{\scriptstyle Ff} \\ & Y & & F(Y) \end{array}$$

Remark 2.1.1. We will disfavour the covariant/contravariant distinction. Therefore all functors will be covariant (unless otherwise specified) and contravariant functors will be covariant functors from the opposite category.

Example 2.1.1 (Identity functor). Let C be a category. There exists an identity functor

$$\begin{array}{rccc} \mathbb{1}: & \mathsf{C} & \longrightarrow & \mathsf{C} \\ & X & & X \\ & {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle f} \\ & Y & & Y \end{array}$$

Example 2.1.2 (Constant functor). Let C be a category. For every object $C \in \mathsf{C}$ there exists a constant functor

$$\begin{array}{rccc} \mathrm{ct}_C : & \mathsf{C} & \longrightarrow & \mathsf{C} \\ & X & & C \\ & f \downarrow & \longmapsto & \downarrow \mathbb{1}_C \\ & Y & & C \end{array}$$

Example 2.1.3 (Bifunctor). Let C, C' and D be three categories. Recall that we can construct the product category $\mathsf{C} \times \mathsf{C}'$.
A **bifunctor** is a functor of the form $F : \mathsf{C} \times \mathsf{C}' \to \mathsf{D}$. This means that for any $X \in \mathsf{C}$ and $X' \in \mathsf{C}$, $F(X, -) : \mathsf{C}' \to \mathsf{D}$ and $F(-, X') : \mathsf{C} \to \mathsf{D}$ are functors.
Furthermore for any morphism $f : X \to Y$ in C and $g : X' \to Y'$ in C' we have a commutative diagram

$$\begin{array}{ccc} F(X,X') & \xrightarrow{F(X,g)} & F(X,Y') \\ {\scriptstyle F(f,X')}\downarrow & \searrow^{F(f,g)} & \downarrow{\scriptstyle F(f,Y')} \\ F(Y,X') & \xrightarrow[F(Y,g)]{} & F(Y,Y') \end{array}$$

Example 2.1.4. Let C be a For every object $C \in \mathsf{C}$ there exists a *covariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(C,-) : & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f \downarrow & \longmapsto & \downarrow f\circ \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

and a *contravariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(-,C) : & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f \downarrow & \longmapsto & \uparrow \circ f \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

We can combine this two into a bifunctor

$$\begin{array}{rccc} \mathrm{Hom}(-,-) : & \mathsf{C}^{\mathrm{op}} \times \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & (X,Y) & & \mathrm{Hom}(X,Y) \\ & (f,g) \downarrow & \longmapsto & \downarrow g\circ - \circ f \\ & (Z,W) & & \mathrm{Hom}(Z,W) \end{array}$$

Example 2.1.5. Let D be a subcategory of a category C. There exist an inclussion functor

$$\begin{array}{rccc} \iota : & \mathsf{D} & \hookrightarrow & \mathsf{C} \\ & X & & X \\ & f \downarrow & \longmapsto & \downarrow f \\ & Y & & Y \end{array}$$

Example 2.1.6 (Preimage functor). Let X, Y be two topological spaces. Consider their categories of open sets $\mathsf{Op}(X), \mathsf{Op}(Y)$.

For every continuous map $f : X \to Y$ we have a preimage functor

$$\begin{array}{ccc} f^{-1} : \mathsf{Op}(Y) & \longrightarrow & \mathsf{Op}(X) \\ U & & f^{-1}(U) \\ \downarrow & \longmapsto & \downarrow \\ V & & f^{-1}(V) \end{array}$$

Proposition 2.1.1 (Functors preseve isomorphisms)**.** Let C, D be two categories and $F : \mathsf{C} \to \mathsf{D}$ a functor between them. If f is an isomorphism in C then Ff is an isomorphism in D.

Proof. Let $f : X \to Y$ be a isomorphism in C. Then there exists an inverse $f^{-1} : Y \to X$ in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Applying F to this equalities we have

$$\begin{aligned} F(f^{-1} \circ f) = F\mathbb{1}_X &\implies Ff^{-1} \circ Ff = \mathbb{1}_{F(X)} \\ F(f \circ f^{-1}) = F\mathbb{1}_Y &\implies Ff \circ Ff^{-1} = \mathbb{1}_{F(Y)} \end{aligned}$$

i.e: Ff has an inverse Ff^{-1} in D. □

Definition 2.1.2 (Functor types)**.** A functor $F : \mathsf{C} \to \mathsf{D}$ is:

- **faithful** if for every pair of objects $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$ is injective.
- **full** if for every pair $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$ is surjective.
- **fully faithful** if it is both full and faithful.
- **essentially surjective** if for every object $Y \in \mathsf{D}$ there exists an object $X \in \mathsf{C}$ such that $F(X) \cong Y$.

Example 2.1.7. A subcategory D of a category C is full if and only if the inclussion functor $\iota : \mathsf{D} \to \mathsf{C}$ is full.

2.2 Natural transformations

Definition 2.2.1. Let $F, G : \mathsf{C} \to \mathsf{D}$ be two functors. A **natural transformation** $\psi : F \implies G$ consists of a family of morphisms $\psi_X : F(X) \to G(X)$ in D for each object $X \in \mathsf{C}$, called the **components** of ψ, such that for every morphism $f : X \to Y$ in C, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\psi_X} & G(X) \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ F(Y) & \xrightarrow[\psi_Y]{} & G(Y) \end{array}$$

Diagramatically, a natural transformations is represented by

$$\begin{array}{ccc} & \overset{F}{\curvearrowright} & \\ \mathsf{C} & \Downarrow \psi & \mathsf{D} \\ & \underset{G}{\curvearrowright} & \end{array}$$

If each component ψ_X is an isomorphism in D, we say that ψ is a **natural**

isomorphism, and we write $F \cong G$.

Definition 2.2.2 (Equivalent categories)**.** Two categories C, D are **equivalent** if there exist functors $F : \mathsf{C} \to \mathsf{D}$ and $G : \mathsf{D} \to \mathsf{C}$ such that $G \circ F \cong \mathbb{1}_{\mathsf{C}}$ and $F \circ G \cong \mathbb{1}_{\mathsf{D}}$. In this case, F (and G) is called an **equivalence of categories**.

Theorem 2.2.1 ([Rie, Theorem 1.5.9])**.** A functor $F : \mathsf{C} \to \mathsf{D}$ defining an equivalence of categories is fully faithful and essentially surjective. Assuming the axiom of choice, any functor with these properties defines an equivalence of categories.

Chapter 3

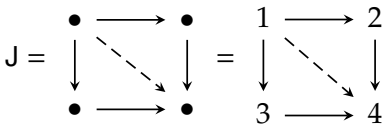
Limits and colimits

3.1 Diagrams, cones and cocones

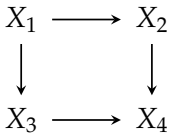
Definition 3.1.1 (Diagram). Let J be a small category.
A **diagram** X in a category C with shape J is a functor $X : \mathsf{J} \to \mathsf{C}$. For every $j \in \mathsf{J}$ we write $X_j := X(j)$.

Example 3.1.1 (Constant diagram). A diagram $X : \mathsf{J} \to \mathsf{C}$ is **constant** if it is a constant functor const_c.
In this case the data of the diagram corresponds to the data given by a single object $c \in \mathsf{C}$.

Example 3.1.2. Consider the category

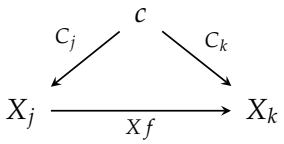


A diagram with shape J is a commutative square



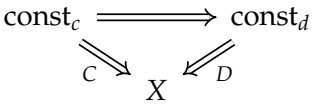
3.1.1 Cones and cocones

Definition 3.1.2 (Cone). A **cone** C over a diagram $X : \mathsf{J} \to \mathsf{C}$ with *apex* c is a natural transformation $C : \mathrm{const}_c \Rightarrow X$ from the constant diagram to X. The components $\{C_j : c \to X_j\}_{j \in \mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cone C over $X : \mathsf{J} \to \mathsf{C}$ with *apex* c is a collection of morphisms $C_j : c \to X_j$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes



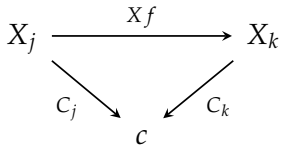
If $C : \mathrm{const}_c \Rightarrow X$ and $D : \mathrm{const}_d \Rightarrow X$ are two cones over X a morphism of cones is a natural transformation $\mathrm{const}_c \Rightarrow \mathrm{const}_d$ making the following diagram

commute

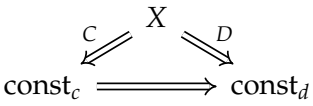


Therefore for any diagram $X : \mathsf{J} \to \mathsf{C}$ we have a category of cones $\Delta \downarrow X$.

Definition 3.1.3 (Cocone). A **cocone** C over a diagram $X : \mathsf{J} \to \mathsf{C}$ with *nadir* c is a natural transformation $C : X \Rightarrow \text{const}_c$ from the constant diagram to X. The components $\{C_j : X_j \to c\}_{j \in \mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cocone C over $X : \mathsf{J} \to \mathsf{C}$ with *nadir* c is a collection of morphisms $C_j : X_j \to c$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes

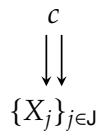


If $C : X \Rightarrow \text{const}_c$ and $D : X \Rightarrow \text{const}_d$ are two cocones over X a morphism of cocones is a natural transformation $\text{const}_c \Rightarrow \text{const}_d$ making the following diagram commute

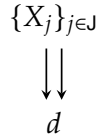


Therefore for any diagram $X : \mathsf{J} \to \mathsf{C}$ we have a category of cocones $X \downarrow \Delta$.

Notation. For a diagram $X : \mathsf{J} \to \mathsf{C}$ we will denote a cone $C : \text{const}_c \Rightarrow X$ by

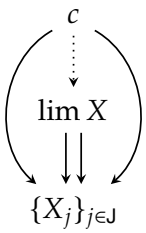


and a cocone $D : X \Rightarrow \text{const}_d$ by

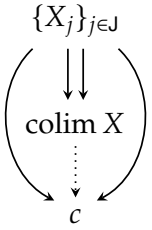


3.2 Limits and colimits

Definition 3.2.1 (Limit). A **limit** of a diagram $X : \mathsf{J} \to \mathsf{C}$ is a terminal object in the category of cones $\Delta \downarrow X$.
Less abstractly, a limit is a cone $\lim X : \text{const}_{\lim X} \Rightarrow X$ such that for any other cone $C : \text{const}_c \Rightarrow X$ there exists a unique morphism $c \to \lim X$ making the following diagram commute



Definition 3.2.2 (Colimit). A **colimit** of a diagram $X : \mathsf{J} \to \mathsf{C}$, denoted $\operatorname{colim} X$, is a terminal object in the category of cocones $X \downarrow \Delta$.
Less abstractly, a colimit is a cocone $\operatorname{colim} X : X \Rightarrow \operatorname{const}_{\operatorname{colim} X}$ such that for any other cocone $C : X \Rightarrow \operatorname{const}_c$ there exists a unique morphism $\operatorname{colim} X \to c$ making the following diagram commute



Proposition 3.2.1 (Uniqueness of limits and colimits). If a diagram $X : \mathsf{J} \to \mathsf{C}$ has a limit/colimit, it is unique up to unique isomorphism.

Proof. Direct from Proposition 1.3.1. □

Definition 3.2.3 (Functors and limits). Let $F : \mathsf{C} \to \mathsf{D}$ be a functor and $X : \mathsf{J} \to \mathsf{C}$ a diagram. We say that

- F **preserves** limits if whenever (L, φ) is the limit of X in C, $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D.
- F **reflects** limits if whenever (L, φ) is a cone over X such that $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D then (L, φ) is the limit of X in C.

Replacing cones with cocones and limits with colimits we get dual statements.

3.2.1 Examples of limits and colimits

We now define some special cases of limits and colimits that are generally useful. However, it is important to take into account that this constructions do not exist in general for all categories.

Products and coproducts

Let J be a small and discrete category.

Since the only morphisms in J are the identity morphisms, a diagram $X : \mathsf{J} \to \mathsf{C}$ consists in a collection of objects $\{X_j\}_{j \in \mathsf{J}}$.

Therefore a cone with apex c over X is an object $c \in \mathsf{C}$ together with morphisms $c \to X_j$ for each $j \in \mathsf{J}$ and a cocone with nadir $c \in \mathsf{C}$ over X is an object c together with morphisms $X_j \to c$ for each $j \in \mathsf{J}$.

Note that any collection of objects $\{X_j\}_{j \in \mathsf{J}}$ in C may be considered as a diagram $X : \mathsf{J} \to \mathsf{C}$.

Definition 3.2.4 (Product). Let $\{X_j\}_{j\in \mathsf{J}}$ be a collection of objects in a category C.
We define the **product** of $\{X_j\}_{j\in \mathsf{J}}$, denoted $\prod_{j\in \mathsf{J}} X_j$, as the limit of the diagram $X : \mathsf{J} \to \mathsf{C}$.
Less abstractly, a **product** is an object $\prod_{j\in \mathsf{J}} X_j \in \mathsf{C}$ together with maps $\pi_j : \prod_{j\in \mathsf{J}} X_j \to X_j$ for every $j \in \mathsf{J}$ such that, for any map $Y \to X_j$ there exists a unique morphism $Y \to \prod_{j\in \mathsf{J}} X_j$ making the following diagram commute

$$\begin{array}{ccccc} Y & & \prod_{j\in \mathsf{J}} X_j & \longrightarrow & X_j \end{array}$$

Definition 3.2.5 (Coproduct). Let $\{X_j\}_{j\in \mathsf{J}}$ be a collection of objects in a category C.
We define the **coproduct** of $\{X_j\}_{j\in \mathsf{J}}$, denoted $\coprod_{j\in \mathsf{J}} X_j$, as the colimit of the diagram $X : \mathsf{J} \to \mathsf{C}$.
Less abstractly, a **coproduct** is an object $\coprod_{j\in \mathsf{J}} X_j \in \mathsf{C}$ together with maps $\iota_j : X_j \to \coprod_{j\in \mathsf{J}} X_j$ for every $j \in \mathsf{J}$ such that, for any map $X_j \to Y$ there exists a unique morphism $\coprod_{j\in \mathsf{J}} X_j \to Y$ making the following diagram commute

$$\begin{array}{ccccc} Y & & \coprod_{j\in \mathsf{J}} X_j & \longleftarrow & X_j \end{array}$$

Example 3.2.1 (Products in Set). In Set the categorical product coincides with the cartesian product.

Example 3.2.2 (Coproducts in Set). In Set the categorical coproduct coincides with the disjoint union.

Equalisers and coequalisers

Consider the *parallel pair* category

$$\mathsf{J} = \quad \bullet \rightrightarrows \bullet$$

A diagram $X : \mathsf{J} \to \mathsf{C}$ corresponds to a pair of morphisms $f, g : A \rightrightarrows B$ in C.

Therefore a cone $C : \mathrm{const}_c \Rightarrow X$ is a morphism $c : C \to A$ such that $fc = gc$:

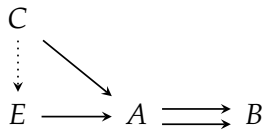
$$C \xrightarrow{c} A \underset{g}{\overset{f}{\rightrightarrows}} B$$

Analogously a cocone $D : X \Rightarrow \mathrm{const}_d$ is a morphism $d : B \to D$ such that $df = dg$:

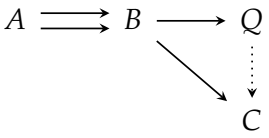
$$A \underset{g}{\overset{f}{\rightrightarrows}} B \xrightarrow{d} D$$

Note that any pair of morphisms $f, g : A \rightrightarrows B$ can be considered as a diagram $X : \mathsf{J} \to \mathsf{C}$.

Definition 3.2.6 (Equaliser). Let $f, g : A \rightrightarrows B$ be two morphisms in C, considered as a diagram $X : \mathsf{J} \to \mathsf{C}$. The **equaliser** of f and g is the limit of this diagram.
Less abstractly, it is an object $E \in \mathsf{C}$ together with a morphism $\mathrm{eq} : E \to A$ such that for any morphism $c : C \to A$ with $fc = gc$ there exists a unique map $C \to E$ the following diagram commute



Definition 3.2.7 (Coequaliser). Let $f, g : A \rightrightarrows B$ be two morphisms in C, considered as a diagram $X : \mathsf{J} \to \mathsf{C}$. The **coequaliser** of f and g is the limit of this diagram. Less abstractly, it is an object $Q \in \mathsf{C}$ together with a morphism $\mathrm{coeq} : B \to Q$ such that for any morphism $d : B \to C$ with $df = dg$ there exists a unique map $Q \to C$ making the following diagram commute



Proposition 3.2.2. Equaliser arrows are monomorphisms. Dually, coequaliser arrows are epimorphisms.

Proof. Let $\mathrm{eq} : E \to X$ be a equaliser. Consider two morphisms $f, g : C \rightrightarrows E$ in C such that $\mathrm{eq} \circ f = \mathrm{eq} \circ g$. By the uniqueness of the map $\mathrm{eq} : E \to X$ we have $f = g$. □

Example 3.2.3 (Equalisers in Set). In Set the equaliser of two maps $f, g : A \rightrightarrows B$ is $E = \{a \in A \mid f(a) = g(a)\}$ together with the inclussion $E \hookrightarrow A$.

Example 3.2.4 (Coequalisers in Set). In Set the coequaliser of two maps $f, g : A \rightrightarrows B$ is $Q = {}^{B}/_{\sim}$ where $\sim$ is the smallest equivalence relation such that for all $a \in A \quad f(a) \sim g(a)$ together with the natural projection

$$\begin{aligned} \mathrm{coeq} : B &\to Q \\ b &\mapsto [b] \end{aligned}$$

A special case of equalisers and coequalisers are kernels and cokernels.

Definition 3.2.8 (Kernel). Let C be a category with a zero object $0 \in \mathsf{C}$. The **kernel** $\ker f$ of a morphism $f : A \to B$ is the equaliser

$$\ker f \longrightarrow A \underset{0}{\overset{f}{\rightrightarrows}} B$$

Less abstractly, it is an object $\ker f$ together with a morphism $i : \ker f \to X$ such that for every $C \in \mathsf{C}$ and every morphism $g : C \to X$ such that $f \circ g = 0$ there is a unique morphism $\phi : C \to \ker f$ such that $g = p \circ \phi$

Definition 3.2.9 (Cokernel). Let C be a category with a zero object $0 \in \mathsf{C}$. The **cokernel** $\mathrm{coker}\, f$ of a morphism $f : A \to B$ is the coequaliser

$$A \underset{0}{\overset{f}{\rightrightarrows}} B \longrightarrow \mathrm{coker}\, f$$

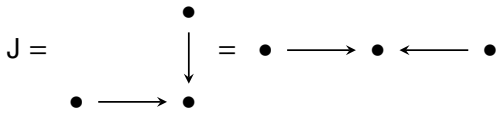
Less abstractly, it is an object $\mathrm{coker}\, f$ together with a morphism $p : Y \to \mathrm{coker}\, f$ such that for every $C \in \mathsf{C}$ and every morphism $g : Y \to C$ such that $g \circ f = 0$ there is a unique morphism $\phi : \mathrm{coker}\, f \to C$ such that $g = \phi \circ p$.
We denote

$${}^{A}/_{B} := \mathrm{coker}(A \to B)$$

Remark 3.2.1. By Proposition 3.2.2, kernels are monomorphisms and cokernels are epimorphisms.

Pullbacks and pushouts

Consider the category



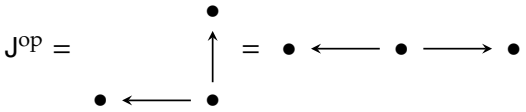
A diagram $X : \mathsf{J} \to \mathsf{C}$ consists of a pair of morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$.

Therefore a cone $\text{const}_C \Rightarrow X$ consists of an object $C \in \mathsf{C}$ together with morphisms $C \to X$ and $C \to Y$ making the following diagram commute.

$$\begin{array}{ccc} C & \longrightarrow & A \\ \downarrow & & \downarrow f \\ B & \xrightarrow[g]{} & Z \end{array}$$

Note that any pair of morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$ can be considered as a diagram $X : \mathsf{J} \to \mathsf{C}$.

Consider now the opposite category



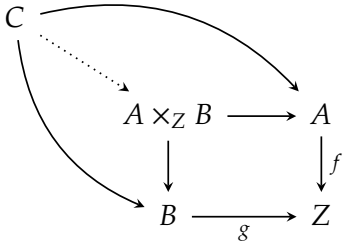
A diagram $Y : \mathsf{J}^{\text{op}} \to \mathsf{C}$ consists of a pair of morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$.

Therefore a cocone $X \Rightarrow \text{const}_D$ consists of an object $D \in \mathsf{D}$ together with morphisms $A \to D$ and $B \to D$ making the following diagram commute

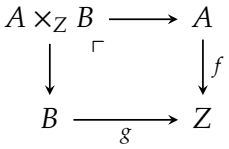
$$\begin{array}{ccc} Z & \xrightarrow{f} & A \\ {\scriptstyle g}\downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

Note that any pair of morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$ can be considered as a diagram $X : \mathsf{J} \to \mathsf{C}$.

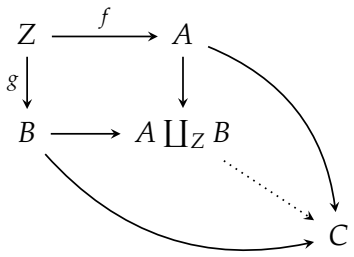
Definition 3.2.10 (Pullback)**.** Let $A \xrightarrow{f} Z \xleftarrow{g} B$ be a pair of morphisms in C, considered as a diagram $X : \mathsf{J} \to \mathsf{C}$.
The **pullback** (or **fibre product**) of f and g is the limit of this diagram, denoted by $A \times_Z B$.
Less abstractly, it is an object $A \times_Z B \in \mathsf{C}$ together with a pair of morphisms $\pi_A : A \times_Z B \to A, \pi_B : A \times_Z B \to B$ such that for any pair of morphisms $C \to A, C \to B$ making the square commute, there exists a unique morphism $C \to A \times_Z B$ such that



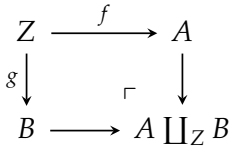
Pullback diagrams are often denoted by



Definition 3.2.11 (Pushout). Let $A \xleftarrow{f} Z \xrightarrow{g} B$ be a pair of morphisms in C, considered as a diagram $X : \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$.
The **pushout** of f and g is the colimit of this diagram, denoted by $X \coprod_Z Y$.
Less abstractly, it is an object $A \coprod_Z B \in \mathsf{C}$ together with a pair of morphisms $\iota_A : A \to A \coprod_Z B, \iota_B : B \to A \coprod_Z B$ such that for any pair of morphisms $A \to C, B \to C$ making the square commute, there exists a unique morphism $A \coprod_Z B \to C$ such that



Pushout diagrams are often denoted by



Example 3.2.5 (Pullbacks in Set). In Set the pullback of two morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$ is the set

$$A \times_Z B = \{(a, b) \in A \times B \mid f(a) = g(b)\}$$

together with the projection maps

$$\begin{aligned} \pi_A : A \times_Z B &\to A \\ (a, b) &\mapsto a \end{aligned} \qquad \begin{aligned} \pi_B : A \times_Z B &\to B \\ (a, b) &\mapsto b \end{aligned}$$

Example 3.2.6 (Pushouts in Set). In Set the pushout of two morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$ is the set

$$A \coprod_Z B = {}^{A \coprod B}\!/_{\sim}$$

where $\sim$ is the equivalence relation generated by $f(a) \sim g(a)$ for all $a \in A$ together

the inclussion maps

$$\iota_A : A \to A \coprod_Z B \qquad \iota_B : B \to A \coprod_Z B$$
$$a \mapsto [a, 0] \qquad b \mapsto [b, 1]$$

Remark 3.2.2. Notice how pullbacks in Set can be seen as a equaliser in Set from the product maps $f \circ \pi_A, g \circ \pi_B : A \times B \to Z$. From the pushouts in Set we get a *dual observation*.
Indeed, it is not difficult to check that if C has products then the pullback

$$\begin{array}{ccc} A \times_Z B & \longrightarrow & A \\ \downarrow & \ulcorner & \downarrow f \\ B & \xrightarrow[g]{} & Z \end{array}$$

is equivalent to the equaliser

$$A \times_Z B \longrightarrow A \times B \underset{g\circ\pi_B}{\overset{f\circ\pi_A}{\rightrightarrows}} Z$$

Dually if C has coproducts the pushout

$$\begin{array}{ccc} Z & \xrightarrow{f} & A \\ {\scriptstyle g}\downarrow & \ulcorner & \downarrow \\ B & \longrightarrow & A \coprod_Z B \end{array}$$

is equivalent to the coequaliser

$$Z \underset{\iota_B\circ g}{\overset{\iota_A\circ f}{\rightrightarrows}} A \coprod B \longrightarrow A \coprod_Z B$$

For a proof see [Rie, Lemmas 3.4.16, 3.4.17].

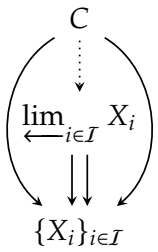
Inverse and direct limits

Definition 3.2.12 (Directed set)**.** A **directed set** is a preordered set $(\mathcal{I}, \leq)$ such that for every $i, j \in \mathcal{I}$, there exists $k \in \mathcal{I}$ with $i \leq k$ and $j \leq k$. We can view this as a category following Example 1.1.5.

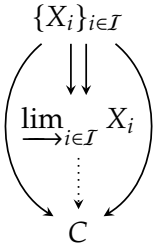
Example 3.2.7. Let $(X, \mathcal{T})$ be a topological space. Then $(\mathcal{T}, \subseteq)$ forms a directed set. This means that the category of open sets $\mathsf{Op}(X)$ can be seen as a directed set.

Definition 3.2.13 (Direct and inverse systems)**.** A **direct system** in C over $\mathcal{I}$ is a diagram $X : \mathcal{I} \to \mathsf{C}$. Less abstractly, it consists of a family of objects $\{X_i\}_{i\in\mathcal{I}}$ and morphisms $f_{ij} : X_i \to X_j$ for all $i \leq j$, such that $f_{ii} = \mathrm{id}_{X_i}$ and $f_{jk} \circ f_{ij} = f_{ik}$ for all $i \leq j \leq k$.
Dually, an **inverse system** in C over $\mathcal{I}$ is a diagram $X : \mathcal{I}^{\mathrm{op}} \to \mathsf{C}$. Less abstractly, it consists of a family of objects $\{X_i\}_{i\in\mathcal{I}}$ and morphisms $p_{ij} : X_j \to X_i$ for all $i \leq j$, such that $p_{ii} = \mathrm{id}_{X_i}$ and $p_{ij} \circ p_{jk} = p_{ik}$ for all $i \leq j \leq k$.

Definition 3.2.14 (Inverse limit)**.** The **inverse limit** (or **projective limit**) of an inverse system $X : \mathcal{I} \to \mathsf{C}$ is the limit of this diagram, denoted as $\varprojlim_{i\in\mathcal{I}} X_i$.
Less abstractly, it is an object $\varprojlim_{i\in\mathcal{I}} X_i \in \mathsf{C}$ together with morphisms $\pi_i : \varprojlim_{i\in\mathcal{I}} X_i \to X_i$ such that $p_{ij} \circ \pi_j = \pi_i$ for all $i \leq j$. For any other object C and morphisms $c_i : C \to X_i$ compatible with the system $(p_{ij} \circ c_j = c_i)$, there exists a unique morphism $C \to \varprojlim_{i\in\mathcal{I}} X_i$ making the following diagram commute:



Definition 3.2.15 (Direct limit)**.** The **direct limit** (or **inductive limit**) of a direct system $X : \mathcal{I} \to \mathsf{C}$ is the colimit of this diagram, denoted as $\varinjlim_{i\in\mathcal{I}} X_i$.
Less abstractly, it is an object $\varinjlim_{i\in\mathcal{I}} X_i \in \mathsf{C}$ together with morphisms $\iota_i : X_i \to \varinjlim_{i\in\mathcal{I}} X_i$ such that $\iota_j \circ f_{ij} = \iota_i$ for all $i \leq j$. For any other object C and morphisms $c_i : X_i \to C$ compatible with the system ($c_j \circ f_{ij} = c_i$), there exists a unique morphism $\varinjlim_{i\in\mathcal{I}} X_i \to C$ making the following diagram commute:



Example 3.2.8 (Inverse limits in Set)**.** In Set, if $(\{X_i\}_{i\in\mathcal{I}}, \{p_{ij}\}_{i\leq j})$ is an inverse system indexed by a directed set $(\mathcal{I}, \leq)$, its inverse limit is

$$\varprojlim_{i\in\mathcal{I}} X_i = \left\{ (x_i)_{i\in\mathcal{I}} \in \prod_{i\in\mathcal{I}} X_i \mid \forall i \leq j \quad x_i = p_{ij}(x_j) \right\}$$

together with the natural projection maps $\pi_k((x_i)_{i\in\mathcal{I}}) = x_k$.

Example 3.2.9 (Direct limits in Set)**.** In Set, if $(\{X_i\}_{i\in\mathcal{I}}, \{f_{ij}\}_{i\leq j})$ is a direct system indexed by a directed set $(\mathcal{I}, \leq)$, its direct limit is

$$\varinjlim_{i\in\mathcal{I}} X_i = \coprod_{i\in\mathcal{I}} X_i \Big/ \sim$$

where $\sim$ is the equivalence relation defined by $x_i \sim x_j$ if and only if there exists $k \in \mathcal{I}$ with $k \geq i, j$ such that

$$f_{ik}(x_i) = f_{jk}(x_j)$$

together with the natural insertion maps $\iota_k(x_k) = [x_k]$.

We finish this chapter by giving a condition for the existence of finite limits and colimits

Theorem 3.2.1. A category has all finite limits if and only if it has a terminal object, products and equalisers.
Dually it has all finite colimits if and only if it has a initial object, coproducts and coequalisers.

Proof.

$\Longrightarrow$ Direct since all of these are particular cases of limits.

$\Longleftarrow$ Consider a diagram $X : \mathsf{J} \Rightarrow \mathsf{C}$.
Since products exist in C there exists

$$P := \prod_{j \in \mathsf{J}} X_j$$

For every morphism $u : j \to k$ in J we can consider two maps from P to X_k: the projection $\pi_k : P \to X_k$ and the composition $Xu \circ \pi_j : P \to X_j \to X_k$. Therefore we can consider a pair of parallel morphisms

$$\prod_{j \in \mathsf{J}} X_j \underset{t}{\overset{s}{\rightrightarrows}} \prod_{u : j \to k \in \mathsf{J}} X_k$$

where s is defined in each component as π_k and t is $Xu \circ \pi_j$ in each component. The equaliser of s and t (which exists by hypothesis) is then the limit of X. Indeed, let $\mathrm{eq} : L \to \prod_{j \in \mathsf{J}} X_j$ be this equaliser. This comes with a set of morphisms $\{\phi_j : L \to X_j\}_{j \in \mathsf{J}}$ such that

$$\phi_j = \pi_j \circ \mathrm{eq}$$

By definition $s \circ \mathrm{eq} = t \circ \mathrm{eq}$, so for every $u : j \to k$ in J we have

$$\pi_k \circ \mathrm{eq} = (Xu \circ \pi_j) \circ \mathrm{eq}$$

Therefore

$$\phi_k = Xu \circ \phi_j$$

and so L is a cone over X.
To see that it is a limit cone consider another cone $(W, \{\psi_j : W \to X_j\})$ such that for any $u : j \to k$ we have $\psi_k = Xu \circ \psi_j$. By definition of the product there exists a unique morphism $h : W \to \prod_{j \in \mathsf{J}} W_j$ such that $\pi_j \circ h = \psi_j$.
Note that $s \circ h = t \circ h$. Indeed the components of $s \circ h$ are $\pi_k \circ h = \psi_k$ and the components of $t \circ h$ are $Xu \circ \pi_j \circ h = Xu \circ \psi_j = \psi_k$.
Therefore by the definition of equaliser there exists a unique morphism $\widetilde{h} : W \to L$ such that $\mathrm{eq} \circ \widetilde{h} = h$. Finally since for any $j \in \mathsf{J}$

$$\phi_j \circ \widetilde{h} = \pi_j \circ \mathrm{eq} \circ \widetilde{h} = \pi_j \circ h = \psi_j$$

we have that W factors uniquely through L and so L is the limit of X.

□

Chapter 4

More on functors

4.1 Representable functors

Definition 4.1.1. A functor $F : \mathsf{C} \to \mathsf{Set}$ is **representable** if there exists an object $X \in \mathsf{C}$ such that

$$F \cong \mathrm{Hom}_{\mathsf{C}}(X, -)$$

where $\mathrm{Hom}_{\mathsf{C}}(X, -)$ is the covariant Hom functor.

Example 4.1.1. The covariant Hom functor is representable.

Definition 4.1.2 (Concrete category)**.** A category C is **concrete** if there exists a faithful functor $U : \mathsf{C} \to \mathsf{Set}$. Such a functor is often called a **forgetful functor**.

Theorem 4.1.1 (Yoneda - [Rie, Theorem 2.2.4])**.** Let C be a category, $X \in \mathsf{C}$ an object, and $F : \mathsf{C} \to \mathsf{Set}$ a functor. There is a bijection:

$$\Phi : \mathrm{Hom}(\mathrm{Hom}(X, -), F) \xrightarrow{\sim} F(X)$$

given by $\alpha \mapsto \alpha_X(\mathbb{1}_X)$. This bijection is natural in both X and F.

Corollary 4.1.1 (Yoneda embedding - [Rie, Corollary 2.2.8])**.** Let C be a category. The functor $Y : \mathsf{C}^{\mathrm{op}} \to [\mathsf{C}, \mathsf{Set}]$ given by $X \mapsto \mathrm{Hom}(X, -)$ is fully faithful.

4.2 Adjunctions

Definition 4.2.1 (Adjoint functors)**.** Let $L : \mathsf{C} \to \mathsf{D}$ and $R : \mathsf{D} \to \mathsf{C}$ be a pair of functors. We say that (L, R) is a pair of **adjoint functors** or that L is left-adjoint to R and R is right-adjoint to L if there exists a natural isomorphism of bifunctors

$$\mathrm{Hom}_{\mathsf{D}}(L(-), -) \cong \mathrm{Hom}_{\mathsf{C}}(-, R(-)) \tag{4.2.1}$$

We represent this diagramatically using

$$\mathsf{C} \underset{R}{\overset{L}{\rightleftarrows}} \mathsf{D} \quad (\perp)$$

Remark 4.2.1 (Unit and counit of an adjunction)**.** Let (L, R) be a pair of adjoint functors.

The natural isomorphism (4.2.1) gives the natural isomorphisms

$$\mathrm{Hom}_{\mathsf{D}}(L(-), L(-)) \cong \mathrm{Hom}_{\mathsf{C}}(-, (G \circ L)(-))$$
$$\mathrm{Hom}_{\mathsf{D}}((L \circ R)(-), -) \cong \mathrm{Hom}_{\mathsf{C}}(R(-), R(-))$$

In particular we get natural transformations

$$\eta : \mathbb{1}_{\mathsf{C}} \Rightarrow R \circ L \qquad \varepsilon : L \circ R \Rightarrow \mathbb{1}_{\mathsf{D}}$$

that satisfy the *triangular identities*

$$\begin{array}{ccccc} L & \xRightarrow{L\eta} & L \circ R \circ L & \quad R & \xRightarrow{\eta R} & R \circ L \circ R \\ & \searrow^{\mathbb{1}_F} & \Big\Downarrow \varepsilon F & & \searrow^{\mathbb{1}_R} & \Big\Downarrow F\varepsilon \\ & & L & & & R \end{array}$$

Conversely, if L, R are two pair of functors and ε, η are natural transformations as above then (L, R) form an adjoint pair. For a proof of this see [**riehlCategoryTheory**].

Remark 4.2.2. Left and right adjoints are dual concepts.

Proposition 4.2.1 (Adjoints are unique)**.** The right adjoint (resp. left adjoint) to a functor, provided it exists, is unique up to natural isomorphism

Proof. Let $L : \mathsf{C} \to \mathsf{D}$ be a functor.

Let $R_1, R_2 : \mathsf{D} \to \mathsf{C}$ be two functors which are right adjoint to L. This means that for all $X \in \mathsf{C}$ and $Y \in \mathsf{D}$ we have bijections

$$\phi_{X,Y} : \mathrm{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}}(X, R_1(Y))$$
$$\psi_{X,Y} : \mathrm{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Using this we can define a bijection

$$\Theta_{X,Y} = \psi_{X,Y} \circ \phi_{X,Y}^{-1} : \mathrm{Hom}_{\mathsf{C}}(X, R_1(Y)) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Since ψ and ϕ are natural in X so is Θ. Therefore for a fixed $Y \in \mathsf{D}$ we have a natural isomorphism

$$\begin{array}{ccccc} & & \mathrm{Hom}_{\mathsf{C}}(-, R_1(Y)) & & \\ \mathsf{C} & & \Big\Downarrow \Theta_{-,Y} & & \mathsf{Set} \\ & & \mathrm{Hom}_{\mathsf{C}}(-, R_2(Y)) & & \end{array}$$

By the Yoneda lemma $\Theta_{-,Y}$ induces a unique isomorphism in C

$$\alpha_Y : R_1(Y) \xrightarrow{\sim} R_2(Y)$$

Furthermore, since ϕ and ψ are natural this ensemble into a natural isomorphism

$$\alpha : R_1 \Rightarrow R_2$$

□

Theorem 4.2.1 ([Rie, Theorem 4.5.2]). Right adjoints preserve all limits that exist in their codomain.
Dually left adjoints preserve all colimits that exist in their domain.

4.3 Functor categories

Definition 4.3.1 (Functor category). Let C and D be categories. The **functor category** $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ has:

- **Objects**: functors $F : \mathsf{C} \to \mathsf{D}$.
- **Morphisms**: natural transformations $\eta : F \implies G$.

Composition and identities are defined componentwise.

Proposition 4.3.1. Let C and D be two categories and $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ be its functor category. If D has limits (resp. colimits) of shape J then $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ has limits (resp. colimits) of shape J. Furthermore, they are computed objectwise.

Proof. Consider a J-shaped diagram $X : \mathsf{J} \Rightarrow [\mathsf{C}, \mathsf{D}]$. We wish to construct a limit object $L \in \mathsf{Fct}(\mathsf{C}, \mathsf{D})$ together with a limit cone $\pi_j : L \to X_j$ for all $j \in \mathsf{J}$.

For any object $c \in \mathsf{C}$ the *evaluation* functor

$$\begin{array}{rccc} \mathrm{ev}_c : & [\mathsf{C}, \mathsf{D}] & \longrightarrow & \mathsf{D} \\ & F & & F(c) \\ & \eta \Downarrow & \longmapsto & \downarrow \eta_c \\ & G & & G(c) \end{array}$$

allows us to define a diagram $X^c := \mathrm{ev}_c \circ X : J \Rightarrow \mathsf{D}$. By hypothesis there exists

$$\lim_{j \in \mathsf{J}} X^c(j) =: L(c) \qquad \pi^c_j : L(c) \to X^c(j)$$

To make L into a functor we need to define its action on morphisms. Consider a morphism $f : c \to d$ in C. Since X_j is a functor for all $j \in \mathsf{J}$ we have an induced morphism $X_j f : X_j(c) \to X_j(d)$. Let's consider the family of morphism

$$\left\{ X_j f \circ \pi^c_j : L(c) \to X_j(d) \right\}_{j \in \mathsf{J}}$$

We want to show that this forms a cone over $L(d)$ with apex $L(c)$

Let $u : j \to k$ be a morphism in J. Since $Xu : X_j \Rightarrow X_k$ is a natural transformation we have a commutative diagram

$$\begin{array}{ccc} X_j(c) & \xrightarrow{(Xu)_c} & X_k(x) \\ {\scriptstyle X_j f}\downarrow & & \downarrow{\scriptstyle X_k f} \\ X_j(d) & \xrightarrow[(Xu)_d]{} & X_k(d) \end{array}$$

Since π^c is a limit over X^c we also now that $(Xu)_c \circ \pi^c_j = \pi^c_k$. Therefore we have

$$(Xu)_d \circ (X_j f \circ \pi^c_j) = ((Xu)_d \circ X_j f) \circ \pi^c_j = X_k f \circ (Xu)_c \circ \pi^c_j = X_k f \circ \pi^c_k$$

i.e: it is a valid cone over $L(d)$. By the universal property of $L(d) = \lim_{j \in \mathsf{J}} X^d(j)$ there exists a unique morphism $Lf : L(c) \to L(d)$ such that for all $j \in \mathsf{J}$ the following diagram commutes

$$\begin{array}{ccc} L(c) & \xrightarrow{Lf} & L(d) \\ {\scriptstyle \pi^c_j}\downarrow & & \downarrow{\scriptstyle \pi^d_j} \\ X^c(j) & \xrightarrow[X_j f]{} & X^d(j) \end{array}$$

The functiorality of L follows directly from the uniqueness of this induced map into $L(d)$. Moreover the above commutative square is the naturallity condition for π_j with respect to f. Therefore, for each $j \in \mathsf{J}$ the collection $\{\pi^c_j\}_{c \in \mathsf{C}}$ is a natural transformation $\pi_j : L \Rightarrow X_j$. It is an easy check to see that this collection forms a cone over X in the functor category $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$.

To finish the proof, we need to check the universal property. Consider another functor $M \in \mathsf{Fct}(\mathsf{C}, \mathsf{D})$ equipped with a cone $\tau_j : M \Rightarrow X_j$.

For each $c \in \mathsf{C}$ the evaluation $\tau^c_j := \mathrm{ev}_c \circ M : M(c) \to X^c(j)$ is a cone over X^c. This induces a unique morphism $\phi_c : M(c) \to L(c)$ such that $\pi^c_j \circ \phi_c = \tau^c_j$.

To show that $\phi \equiv \{\phi_c\}_{c \in \mathsf{C}}$ is a natural transformation let $f : c \to d$ be a morphism in C. We want to see that the following diagram commutes

$$\begin{array}{ccc} M(c) & \xrightarrow{\phi_c} & L(c) \\ {\scriptstyle Mf}\downarrow & & \downarrow{\scriptstyle Lf} \\ M(d) & \xrightarrow[\phi_d]{} & L(d) \end{array}$$

Postcompossing with π^d_j we get, on the one hand

$$\pi^d_j \circ (Lf \circ \phi_c) = (\pi^d_j \circ Lf) \circ \phi_c = X_j f \circ \pi^c_j \circ \phi_c = X_j f \circ \tau^c_j$$

And on the other

$$\pi^d_j \circ (\phi_d \circ Mf) = (\pi^d_j \circ \phi_d) \circ Mf = \tau^d_j \circ Mf$$

Since τ_j is a natural transformation we have

$$X_j f \circ \tau^c_j = \tau^d_j \circ Mf$$

Therefore since the projections agree for all $j \in \mathsf{J}$ by uniqueness of the maps π_j we get the commutativity of the diagram. Its uniqueness as a morphism of cones follows directly from the objectwise uniqueness of each ϕ_c. □

Chapter 5

Localization

5.1 Localization of categories

Definition 5.1.1. A **localization** of a category C by a class of morphisms $\mathcal{W}$ in C is a category $\mathsf{C}[\mathcal{W}^{-1}]$ together with a functor $Q : \mathsf{C} \to \mathsf{C}[\mathcal{W}^{-1}]$ called the *localization functor* satisfying:

1. For every morphism $w \in \mathcal{W}$ $\quad Qw$ is an isomorphism.
2. For any category D and functor $F : \mathsf{C} \to \mathsf{D}$ such that Fw is an isomorphism for all $w \in \mathcal{W}$ there exists a functor $F_{\mathcal{W}} : \mathsf{C}[\mathcal{W}^{-1}] \to \mathsf{D}$ and a natural isomorphism $F \cong F_{\mathcal{W}} \circ Q$:

$$\begin{array}{ccc} \mathsf{C} & \xrightarrow{F} & \mathsf{D} \\ {\scriptstyle Q}\downarrow & \nearrow_{F_{\mathcal{W}}} & \\ \mathsf{C}[\mathcal{W}^{-1}] & & \end{array}$$

3. The map between functor categories

$$- \circ Q : \mathsf{Fct}(\mathsf{C}[\mathcal{W}^{-1}], \mathsf{D}) \to \mathsf{Fct}(\mathsf{C}, \mathsf{D})$$

is fully faithful for every category D.
Less abstractly, for any two functors $G_1, G_2 : \mathsf{C}[\mathcal{W}^{-1}] \to \mathsf{D}$ the map

$$\mathrm{Hom}_{\mathsf{Fct}(\mathsf{C}[W^{-1}],\mathsf{D})}(G_1, G_2) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{Fct}(\mathsf{C},\mathsf{D})}(G_1 \circ Q, G_2 \circ Q)$$

is bijective.
This makes the functor $F_{\mathcal{W}}$ unique up to unique natural isomorphism.

Proposition 5.1.1.

1. If $\mathsf{C}[\mathcal{W}^{-1}]$ exists, it is unique up to unique equivalence of categories.
2. If $\mathsf{C}[\mathcal{W}^{-1}]$ exists then, denoting $\mathcal{W}^{\mathrm{op}}$ the image of $\mathcal{W}$ in C^{op} by the functor $\mathrm{op} : \mathsf{C} \to \mathsf{C}^{\mathrm{op}}$, $\mathsf{C}^{\mathrm{op}}[(\mathcal{W}^{\mathrm{op}})^{-1}]$ exists and $\mathsf{C}^{\mathrm{op}}[(\mathcal{W}^{\mathrm{op}})^{-1}] \cong \left(\mathsf{C}[\mathcal{W}^{-1}]\right)^{\mathrm{op}}$.

Proof.

1. Direct from the definition.
2. Assume $\mathsf{C}[\mathcal{W}^{-1}]$ exists. Set $\mathsf{C}^{\mathrm{op}}[(\mathcal{W}^{\mathrm{op}})^{-1}] := \left(\mathsf{C}[\mathcal{W}^{-1}]\right)^{\mathrm{op}}$ and define

$$Q^{\mathrm{op}} : \mathsf{C}^{\mathrm{op}} \to \mathsf{C}^{\mathrm{op}}[(\mathcal{W}^{\mathrm{op}})^{-1}], Q^{\mathrm{op}} := \mathrm{op} \circ Q \circ \mathrm{op}$$

The properties of Definition 5.1.1 are clearly satisfied.

□

Definition 5.1.2. A class of morphisms $\mathcal{S}$ of a category C is a **right multiplicative system** if it satisfies

- **S.1)** $\mathbb{1}_X \in \mathcal{S}$ for all $X \in \mathsf{C}$.
- **S.2)** For all $f, g \in \mathcal{S}$ if $g \circ f$ exists then $g \circ f \in \mathcal{S}$.
- **S.3)** Given $f : X \to Y$ in C and $s : X \to X'$ in $\mathcal{S}$ there exist $g : X' \to Y'$ in C and $t : Y \to Y'$ in $\mathcal{S}$ making the following diagram commute

$$\begin{array}{ccc} X' & \overset{g}{\dashrightarrow} & Y' \\ {\scriptstyle s}\uparrow & & \uparrow{\scriptstyle t} \\ X & \xrightarrow[f]{} & Y \end{array}$$

- **S.4)** For any pair of morphisms $f, g : X \rightrightarrows Y$ in C if their equaliser $\mathrm{eq} : E \to X$ exists and $\mathrm{eq} \in \mathcal{S}$ then the coequaliser $\mathrm{coeq} : Y \to C$ exists and $\mathrm{coeq} \in \mathcal{S}$. Visually

$$E \xrightarrow{\mathrm{eq}} X \underset{g}{\overset{f}{\rightrightarrows}} Y \overset{\mathrm{coeq}}{\dashrightarrow} C$$

We can define a dual system called **left multiplicative system** by reversing the arrows of S.3 and S.4.

Remark 5.1.1. Axioms S.1 and S.2 assert that $\mathcal{S}$ is the family of morphisms of a subcategory $\widetilde{\mathcal{S}}$ with $\mathrm{Ob}(\widetilde{\mathcal{S}}) = \mathrm{Ob}(\mathsf{C})$.

Definition 5.1.3. Suppose $\mathcal{S}$ satisfies S.1 and S.2 and let $X \in \mathsf{C}$. We define the categories $\mathcal{S}^X$ and $\mathcal{S}_X$ as

$$\begin{aligned} \mathrm{Ob}(\mathcal{S}^X) &:= \{s : X \to Y \mid s \in \mathcal{S}\} \\ \mathrm{Hom}_{\mathcal{S}^X}((s : X \to Y), (s' : X \to Y')) &:= \{h \in \mathrm{Hom}_{\mathsf{C}}(Y, Y') \mid h \circ s = s'\} \end{aligned}$$

$$\begin{aligned} \mathrm{Ob}(\mathcal{S}_X) &:= \{s : Y \to X \mid s \in \mathcal{S}\} \\ \mathrm{Hom}_{\mathcal{S}_X}((s : Y \to X), (s' : Y' \to X)) &:= \{h \in \mathrm{Hom}_{\mathsf{C}}(Y, Y') \mid s' \circ h = s\} \end{aligned}$$

Proposition 5.1.2. Let $\mathcal{S}$ be a right (resp. left) multiplicative system. Then the category $\mathcal{S}^X$ (resp. $(\mathcal{S}^x)^{\mathrm{op}}$) is directed.

Proof. Note that $\mathcal{S}^X$ is non-empty since $\mathbb{1}_X \in \mathcal{S}^X$.

Let $s : X \to Y, s' : X \to Y' \in \mathcal{S}^X$. On the one hand by S.3 there exists $t : Y \to Z$ and $t' : Y' \to Z$ such that $t' \circ s' = t \circ s$ and $t \in \mathcal{S}$. Therefore by S.2 $t \circ s \in \mathcal{S}$ and $(X \to Z) \in \mathcal{S}^X$.

On the other if $f, g : Y \to Y'$ are two morphisms in C with $f \circ s = g \circ s = s'$ by S.4 there exists $t : Y' \to W$ in $\mathcal{S}$ such that $t \circ f = t \circ g$. Therefore by S.2 $t \circ s' \in \mathcal{S}^X$. □

Definition 5.1.4. Let $\mathcal{S}$ be a right multiplicative system and $X, Y \in \mathsf{C}$. We set

$$\mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(X, Y) := \varinjlim_{(Y \to Y') \in \mathcal{S}^Y} \mathrm{Hom}_{\mathsf{C}}(X, Y')$$

Less abstractly a morphism $f \in \mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(X, Y)$ is represented by a *roof* diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y' \\ & & \uparrow{\scriptstyle s} \\ & & Y \end{array}$$

where $f : X \to Y'$ is in C and $s : Y \to Y'$ in $\mathcal{S}$. We denote this by (Y', f, t) or more simply by (f, t).

Lemma 5.1.1. Let $\mathcal{S}$ be a right multiplicative system. Let $Y \in \mathsf{C}$ and $s : X \to X'$ in $\mathcal{S}$. Then s induces an isomorphism

$$\mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(X', Y) \xrightarrow[\circ s]{\sim} \mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(X, Y)$$

Proof.

Injectivity Since $\mathcal{S}^Z$ is directed two morphisms $f, g \in \mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(X', Y)$ can be represented as

$$\begin{array}{ccc} X' & \underset{g}{\overset{f}{\rightrightarrows}} & Y' \\ & & \uparrow{\scriptstyle t} \\ & & Y \end{array}$$

with $t : Y \to Y'$ in $\mathcal{S}$.
If $f \circ s = g \circ s$ then by S.4 there exists $t' : Y' \to Y''$ in $\mathcal{S}$ making the following diagram commute

$$\begin{array}{ccccccc} X & \xrightarrow{s} & X' & \underset{g}{\overset{f}{\rightrightarrows}} & Y' & \xrightarrow{t'} & Y'' \\ & & & & \uparrow{\scriptstyle t} & \nearrow{\scriptstyle t' \circ t} & \\ & & & & Y & & \end{array}$$

Therefore (f, t) and (g, t) have the same image in $\mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(X', Y)$.

Surjectivity By S.3 we have the following commutative diagram

$$\begin{array}{ccccc} X' & \xrightarrow{g} & Y'' & & \\ \uparrow{\scriptstyle s} & & \uparrow{\scriptstyle t} & & \\ X & \xrightarrow[f]{} & Y' & \xleftarrow[t]{} & Y \end{array}$$

where $s, t, t' \in \mathcal{S}$. Therefore the map (f, t) is the image by $\circ s$ of $(g, t' \circ t)$.

□

An easy but very tedious comprovation shows the following lemma.

Lemma 5.1.2. Let $\mathcal{S}$ be a right multiplicative system. The *composition*

$$\mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(X, Y) \times \mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(Y, Z) \to \mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(X, Z)$$

defined as

$$\begin{aligned}
&\varinjlim_{Y\to Y'} \mathrm{Hom}_{\mathsf{C}}(X,Y') \times \varinjlim_{Z\to Z'} \mathrm{Hom}_{\mathsf{C}}(Y,Z') \\
&\cong \varinjlim_{Y\to Y'} \left(\mathrm{Hom}_{\mathsf{C}}(X,Y') \times \varinjlim_{Z\to Z'} \mathrm{Hom}_{\mathsf{C}}(Y,Z') \right) \\
&\tilde{\leftarrow} \varinjlim_{Y\to Y'} \left(\mathrm{Hom}_{\mathsf{C}}(X,Y') \times \varinjlim_{Z\to Z'} \mathrm{Hom}_{\mathsf{C}}(Y',Z') \right) \\
&\mapsto \varinjlim_{Y\to Y'} \varinjlim_{Z\to Z'} \mathrm{Hom}_{\mathsf{C}}(X,Z') \\
&\cong \varinjlim_{Z\to Z'} \mathrm{Hom}_{\mathsf{C}}(X,Z')
\end{aligned}$$

where $\tilde{\leftarrow}$ comes from Lemma 5.1.1, is associative.

Therefore by Lemma 5.1.2 we can define a category $\mathsf{C}[\mathcal{S}^{-1}]^r$ where morphisms are given by Definition 5.1.4. This category comes with a functor $Q_{\mathcal{S}} : \mathsf{C} \to \mathsf{C}[\mathcal{S}^{-1}]^r$, which when there is no risk of confussion we will simply denote by Q, associated with the map

$$\mathrm{Hom}_{\mathsf{C}}(X,Y) \to \varinjlim_{(Y\to Y')\in \mathcal{S}^Y} \mathrm{Hom}_{\mathsf{C}}(X,Y')$$

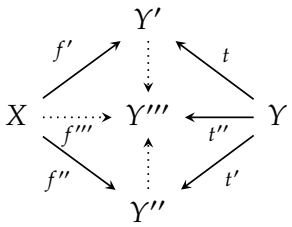
Lemma 5.1.3. If $s : X \to Y$ is in $\mathcal{S}$ then Qs is invertible.

Proof. By Lemma 5.1.1 for any $Z \in \mathsf{C}[\mathcal{S}^{-1}]^r$ the map $\mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(Y,Z) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}[\mathcal{S}^{-1}]^r}(X,Z)$ is bijective. □

Remark 5.1.2. A morphism $f : X \to Y$ in $\mathsf{C}[\mathcal{S}^{-1}]^r$ is given by an equivalence class of triplets (Y', t, f') with $t : Y \to Y'$ in $\mathcal{S}$ and $f : X \to Y'$:

$$X \xrightarrow{f} Y' \xleftarrow{t} Y$$

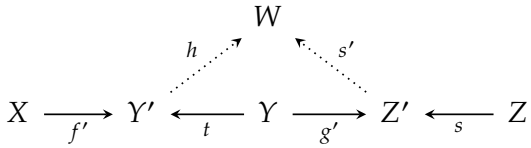
and $(Y', t, f') \sim (Y'', t', f'')$ if and only if there exists (Y''', t'', f''') and a commutative diagram



Note that the morphism (Y', t, f') in $\mathsf{C}[\mathcal{S}^{-1}]^r$ is $(Qt)^{-1} \circ Qf'$, i.e: $f = (Qt)^{-1} \circ f'$. For any pair of morphisms $f, g : X \rightrightarrows Y$ in C we have the equivalence

$$Qf = Qg \text{ in } \mathsf{C}[\mathcal{S}^{-1}]^r \iff \exists (s : Y \to Y') \in \mathcal{S} \mid s \circ f = s \circ g$$

The composition of two morphisms $(Y', t, f') : X \to Y$ and $(Z', s, g') : Y \to Z$ is defined by the diagram



where $t, s, s' \in \mathcal{S}$. In other words the composition is given by $(W, s' \circ s, h \circ f')$.

We will not prove the following

Theorem 5.1.1 ([KS06, Theorem 7.1.6]). Let $\mathcal{S}$ be a right multiplicative system. The category $\mathsf{C}[\mathcal{S}^{-1}]^r$ and the functor $Q_{\mathcal{S}}$ define a localization of C by $\mathcal{S}$.

Notation. We will write $\mathsf{C}[\mathcal{S}^{-1}]$ instead of $\mathsf{C}[\mathcal{S}^{-1}]^r$. This is justified by Theorem 5.1.1.

5.1.1 Saturated multiplicative systems

In this subsection C is a category, $\mathcal{S}$ is a right multiplicative system and $Q : \mathsf{C} \to \mathsf{C}[\mathcal{S}^{-1}]$ is the localization functor.

Proposition 5.1.3. For any morphism $f : X \to Y$ in C, Qf is an isomorphism in $\mathsf{C}[\mathcal{S}^{-1}]$ if and only if there exist $g : Y \to Z$ and $h : Z \to W$ such that $g \circ f \in \mathcal{S}$ and $h \circ g \in \mathcal{S}$.

Proof.

$\Longrightarrow$ Represent $(Qf)^{-1}$ by

$$X \xrightarrow{f} Y \xrightarrow{g} X' \xleftarrow{s} X$$

Then $(Qs)^{-1} \circ Qg$ is the inverse of Qf and $Qg \circ Qf = Q(g \circ f) = Qs$. By Remark 5.1.2 there exists $t : X' \to X''$ in $\mathcal{S}$ such that $t \circ g \circ f = t \circ s$. If by abuse of notation we replace g with $t \circ g$ we have found $g : Y \to Z$ such that $g \circ f \in \mathcal{S}$. Then since Qg is an isomorphism by hypothesis so is $Qg \circ Qf$.
An analogous argument applied to g instead of f shows that there exists $h : Z \to W$ such that $h \circ g \in \mathcal{S}$.

$\Longleftarrow$ In any category if we consider

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} W$$

where $g \circ f$ and $h \circ g$ are isomorphisms then f, g and h are isomorphisms. Therefore this implication is a particular case with Qf, Qg and Qh.

□

Definition 5.1.5. We say that $\mathcal{S}$ is **saturated** if it verifies

S.5) For any morphisms $f : X \to Y, g : Y \to Z$ and $h : Z \to W$ in C such that $g \circ f$ and $h \circ g$ are in $\mathcal{S}$, $f \in \mathcal{S}$.

Corollary 5.1.1. $\mathcal{S}$ is saturated if and only if a morphism f in C is in $\mathcal{S}$ if and only if Qf is an isomorphism.

Proof.

$\Longrightarrow$ If $f \in \mathcal{S}$ then by Lemma 5.1.1 Qf is an isomorphism.
Conversely if Qf is an isomorphism then by Proposition 5.1.3 and the definition of a saturated multiplicative system $f \in \mathcal{S}$.

$\Longleftarrow$ Consider $g : Y \to Z$ and $h : Z \to W$ in C such that $g \circ f$ and $h \circ g$ are in $\mathcal{S}$. Then by Proposition 5.1.3 Qf is an isomorphism, which by hypothesis implies that $f \in \mathcal{S}$.

□

5.1.2 Localization of subcategories

We will not prove the following.

Proposition 5.1.4 ([KS06, Proposition 7.2.1]). Let C be a category, D a full subcategory, $\mathcal{S}$ a right multiplicative system in C and $\mathcal{T}$ the family of morphisms in D which belong to $\mathcal{S}$.

1. If $\mathcal{T}$ is a right multiplicative system in D the functor $\mathsf{D}[\mathcal{T}^{-1}] \to \mathsf{C}[\mathcal{S}^{-1}]$ is well defined.
2. If for every $f : Y \to X$ in $\mathcal{S}$ with $Y \in \mathsf{D}$ there exists $W \in \mathsf{D}$ and $g : X \to W$ with $g \circ f \in \mathcal{S}$ then $\mathcal{T}$ is a right multiplicative system and the functor $\mathsf{D}[\mathcal{T}^{-1}] \to \mathsf{C}[\mathcal{S}^{-1}]$ is fully faithful.

Corollary 5.1.2. Let C be a category, D a full subcategory, $\mathcal{S}$ a right multiplicative system in C and $\mathcal{T}$ the family of morphisms in D which belong to $\mathcal{S}$. Suppose that for all $X \in \mathsf{C}$ there exists $s : X \to W$ with $W \in \mathsf{D}$ and $s \in \mathcal{S}$. Then $\mathcal{T}$ is a right multiplicative system and $\mathsf{D}[\mathcal{T}^{-1}]$ is equivalent to $\mathsf{C}[\mathcal{S}^{-1}]$.

Proof. By Proposition 5.1.4 $\mathcal{T}$ is a right multiplicative system and the functor $\mathsf{D}[\mathcal{T}^{-1}] \to \mathsf{C}[\mathcal{S}^{-1}]$ is fully faithful. Furthermore by assumption it is essentially surjective.

Therefore by Theorem 2.2.1 it is an equivalence of categories. □

5.2 Localization of functors

Definition 5.2.1. Let C be a category, $\mathcal{S}$ a right multiplicative system and $F : \mathsf{C} \to \mathsf{D}$ a functor.
A **right localization** of F is a functor $R_{\mathcal{S}}F : \mathsf{C}[\mathcal{S}^{-1}] \to \mathsf{D}$ together with a natural transformation $\tau : F \to R_{\mathcal{S}}F \circ Q$ such that for any functor $G : \mathsf{C}[\mathcal{S}^{-1}] \to \mathsf{D}$ the map

$$\mathrm{Hom}_{\mathrm{Fct}(\mathsf{C}[\mathcal{S}^{-1}],\mathsf{D})}(R_{\mathcal{S}}F, G) \to \mathrm{Hom}_{\mathrm{Fct}(\mathsf{C},\mathsf{D})}(F, G \circ Q)$$

is bijective.
We say that a functor is right localizable if it admits a right localization.

Notation. Since we will mainly work with right localizations, we may sometimes ommit the word "right".

Lemma 5.2.1. Localizations of functors are unique up to unique natural isomorphism.

Proof. The functor $R_{\mathcal{S}}$ is a representative of the functor

$$G \mapsto \mathrm{Hom}_{\mathrm{Fct}(\mathsf{C},\mathsf{D})}(F, G \circ Q)$$

□

Proposition 5.2.1. Let C be a category, C' be a full subcategory of C, $\mathcal{S}$ a right multiplicative system and $\mathcal{T}$ the family of morphisms in D which belong to $\mathcal{S}$. Let

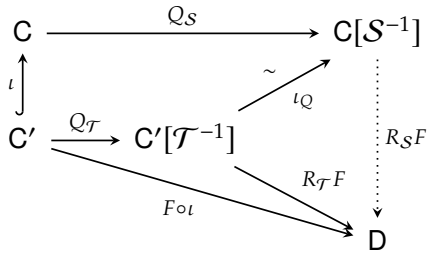
$F : \mathsf{C} \to \mathsf{D}$ a functor. Suppose that

1. For any $X \in \mathsf{C}$ there exists $s : X \to W$ with $W \in \mathsf{C}'$ and $s \in \mathcal{S}$.
2. Ft is an isomorphism for any $t \in \mathcal{T}$.

Then F is right localizable.

Proof. Let $\iota : \mathsf{C}' \hookrightarrow \mathsf{C}$ the inclussion functor. By hypothesis the localization $R_{\mathcal{T}}F$ of $F \circ \iota$ exists.

Consider the diagram



Denote by ι_Q^{-1} a quasi-inverse of ι_Q and set $R_{\mathcal{S}}F \coloneqq R_{\mathcal{T}}F \circ \iota_Q^{-1}$. Let $G : \mathsf{C}[\mathcal{S}^{-1}] \to \mathsf{D}$ be a functor. We have a chain of morphism

$$\begin{aligned}
\mathrm{Hom}_{\mathsf{Fct}(\mathsf{C},\mathsf{D})}(F, G \circ Q_{\mathcal{S}}) &\xrightarrow{\lambda} \mathrm{Hom}_{\mathsf{Fct}(\mathsf{C}',\mathsf{D})}(F \circ \iota, G \circ Q_{\mathcal{S}} \circ \iota) \\
&\cong \mathrm{Hom}_{\mathsf{Fct}(\mathsf{C}',\mathsf{D})}(R_{\mathcal{T}}F \circ Q_{\mathcal{T}}, G \circ \iota_Q \circ Q_{\mathcal{T}}) \\
&\cong \mathrm{Hom}_{\mathsf{Fct}(\mathsf{C}'[\mathcal{T}^{-1}],\mathsf{D})}(R_{\mathcal{T}}F, G \circ \iota_Q) \\
&\cong \mathrm{Hom}_{\mathsf{Fct}(\mathsf{C}[\mathcal{S}^{-1}],\mathsf{D})}(R_{\mathcal{T}}F \circ \iota_Q^{-1}, G) \\
&\cong \mathrm{Hom}_{\mathsf{Fct}(\mathsf{C}[\mathcal{S}^{-1}],\mathsf{D})}(R_{\mathcal{S}}F, G)
\end{aligned}$$

One can see that λ is an isomorphism (see [KS06, Proposition 7.3.2]). The second isomorphism comes from $Q_{\mathcal{T}}$ being a localization functor and the rest are trivial. □

PART II

Homological algebra

Chapter 6

Additive categories

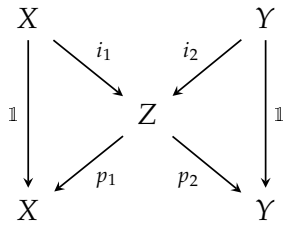
Definition 6.0.1 (Additive category). A category C is **additive** if it satisfies

- **Ad.1)** For every pair of objects $X, Y \in \mathsf{C}$ the set $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.
- **Ad.2)** There exists a zero object $0 \in \mathsf{C}$.
- **Ad.3)** C admits finite coproducts.
- **Ad.4)** C admits finite products.

Notation. If $X, Y \in \mathsf{C}$ we denote by $X \oplus Y$ its coproduct and call it their **direct sum**.

Lemma 6.0.1. Let C be a category satisfying Ad.1 and Ad.2.
Suppose that C verifies

- **Ad.5)** For any pair of objects $X, Y \in \mathsf{C}$ there exists $Z \in \mathsf{C}$ and morphisms



such that

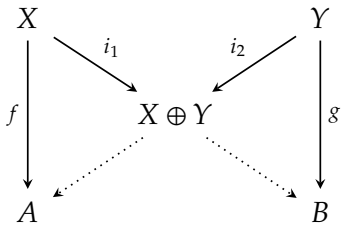
$$p_1 \circ i_1 = \mathbb{1}_X \quad p_1 \circ i_2 = 0$$
$$p_2 \circ i_2 = \mathbb{1}_Y \quad p_2 \circ i_1 = 0$$
$$i_1 \circ p_1 + i_2 \circ p_2 = \mathbb{1}_Z$$

Then conditions Ad.4, Ad.3 and Ad.5 are equivalent and the objects $X \times Y$, $X \oplus Y$ and Z are naturally isomorphic.

Proof. We will show that conditions Ad.3 and Ad.5 are equivalent and that $X \oplus Y \cong Z$. A dual argument shows that Ad.4 and Ad.5 are equivalent and that $X \times Y \cong Z$.

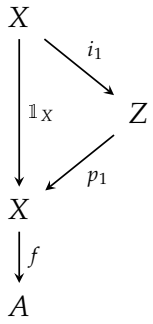
Ad.3 $\implies$ Ad.5 Let $X \oplus Y$ be the coproduct of X and Y, that comes with morphisms $i_1 : X \to X \oplus Y$ and $i_2 : Y \to X \oplus Y$.

By definition, for all morphisms $f : X \to A$, $g : Y \to B$ there exist unique morphisms $X \oplus Y \to A$, $X \oplus Y \to B$ such that



In particular if $f = \mathbb{1}_X$ and $g = \mathbb{1}_Y$

Ad.5 $\implies$ Ad.3 We are going to see that Z verifies the universal property of the coproduct. Let $f : X \to A$ be a morphism. We want to find a morphism $\widetilde{f} : Z \to A$ such that $f = \widetilde{f} \circ i_1$. We have the following



Therefore if $\widetilde{f} := f \circ p_1$ since $i_1 \circ p_1 = \mathbb{1}_X$ we have $\widetilde{f} \circ i_1 = f$.
Analogously we see that any morphism $g : Y \to B$ can can be factored through Z. Therefore since Z verifies the universal property of the coproduct, $Z \cong X \oplus Y$.

□

Lemma 6.0.2. A morphism $f : X \to Y$ in an additive category is a monomorphism if and only if for every morphism $g : Z \to X$

$$f \circ g = 0 \implies g = 0$$

Dually f is an epimorphism if and only if for every morphism $g : Y \to Z$

$$g \circ f = 0 \implies g = 0$$

Proof. We will check the monomorphism caracterization.

$\implies$ Let $f : X \rightarrowtail Y$ be a monomorphism. Consider a morphism $g : Z \to X$ such that $f \circ g = 0$. Since $f \circ 0 = 0$ applying left-cancellativity we have that $g = 0$.

$\impliedby$ Let $g_1, g_2 : Z \to X$ be two morphisms such that $f \circ g_1 = f \circ g_2$.
Since $\mathrm{Hom}(Z, Y)$ is an abelian group we have

$$0 = f \circ g_1 - f \circ g_2 = f \circ (g_1 - g_2)$$

Therefore by hypothesis $g_1 - g_2 = 0 \iff g_1 = g_2$, proving that f is a monomorphism.

□

Lemma 6.0.3. Let I be a small category. If C is additive then the functor category Fct(I, C) is additive.

Proof.

Ad.1 For any two natural transformations $\phi, \psi : \mathsf{I} \Rightarrow \mathsf{C}$ we can define their addition componentwise: $(\phi + \psi)_i = \phi_i + \psi_i$ for all $i \in \mathsf{I}$. Since $\phi_i, \psi_i \in \mathrm{Hom}_{\mathsf{C}}(-,-)$ this is well defined.

Ad.2 Consider the zero functor

$$\begin{array}{rccc} 0: & \mathsf{I} & \longrightarrow & \mathsf{C} \\ & X & & 0 \\ & f \downarrow & \longmapsto & \downarrow 0 \\ & Y & & 0 \end{array}$$

This is clearly the zero object in Fct(I, C,).

Ad.3, Ad.4 Direct from Proposition 4.3.1.

□

Definition 6.0.2 (Additive functor). Let C, D be additive categories. A functor $F : \mathsf{C} \to \mathsf{D}$ is **additive** if for all $X, Y \in \mathsf{C}$

$$\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$$

is a morphism of abelian groups.

Proposition 6.0.1. Let C, D be additive categories. A functor $F : \mathsf{C} \to \mathsf{D}$ is additive if and only if it commutes with finite direct sums, i.e: if $F(0) \cong 0$ and for all $X, Y \in \mathsf{C}$

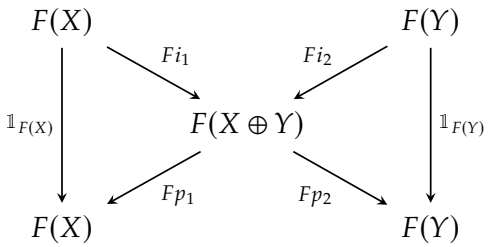
$$F(X \oplus Y) = F(X) \oplus F(Y)$$

Proof.

$\Longrightarrow$ Since F is a functor $F(\mathbb{1}_0) = \mathbb{1}_{F(0)}$. Furthermore since F is additive if $0_{X,Y}$ is the zero morphism between $X, Y \in \mathsf{C}$ then $F(0_{X,Y}) = 0_{F(X),F(Y)}$ because group homomorphisms preserve the identity. In particular taking $X = Y = 0$ since $0_{0,0} = \mathbb{1}_0$ we have

$$\mathbb{1}_{F(0)} = F(\mathbb{1}_0) = F(0_{0,0}) = 0_{F(0),F(0)}$$

which means that $F(0)$ is the zero object in D.
Let $X, Y \in \mathsf{C}$ be two objects and $X \oplus Y$ be its direct sum. Applying F to the diagram in Lemma 6.0.1 defining $Z \cong X \oplus Y$ we get



Note that

$$Fp_1 \circ Fi_1 = F(p_1 \circ i_1) = F(\mathbb{1}_X) = \mathbb{1}_{F(X)} \quad Fp_1 \circ Fi_2 = F(p_1 \circ i_2) = F(0) = 0$$
$$Fp_2 \circ Fi_2 = F(p_2 \circ i_2) = F(\mathbb{1}_Y) = \mathbb{1}_{F(Y)} \quad Fp_2 \circ Fi_1 = F(p_2 \circ i_1) = F(0) = 0$$

Furthermore since F is additive

$$Fp \circ Fi + Fj \circ Fq = F(p \circ i + j \circ q) = F\mathbb{1}_{X\oplus Y} = \mathbb{1}_{F(X\oplus Y)}$$

Therefore by Lemma 6.0.1 $F(X \oplus Y) \cong F(X) \oplus F(Y)$.

$\Longleftarrow$ Applying Lemma 6.0.1 to $X \oplus X$ and $Y \oplus Y$ we get maps $\Delta_X : X \to X \oplus X$ and $\nabla_Y : Y \oplus Y \to Y$ such that

$$p_1 \circ \Delta_X = \mathbb{1}_X = p_2 \circ \Delta_X$$
$$\nabla_Y \circ i_1 = \mathbb{1}_Y = \nabla_Y \circ i_2$$

Then for any pair of morphisms $f, g : X \to Y$ in D we have

$$f + g = \nabla_Y \circ (f \oplus g) \circ \Delta_X$$

where $f \oplus g$ is the unique morphism such that

$$p_k \circ (f \oplus g) \circ i_j = \begin{cases} f & j = k = 1 \\ g & j = k = 2 \\ 0 & \text{otherwise} \end{cases}$$

Applying F to this expression we get

$$F(f + g) = F(\nabla_Y \circ (f \oplus g) \circ \Delta_X) = F\nabla_Y \circ F(f \oplus g) \circ F\Delta_X$$

Since by hypothesis F preserves finit direct sums

$$F\nabla_Y \circ F(f \oplus g) \circ F\Delta_X = \nabla_{F(Y)} \circ (Ff \oplus Fg) \oplus \Delta_{F(X)} = Ff + Fg$$

Therefore $F(f + g) = Ff + Fg$, which means that F is additive.

□

Definition 6.0.3 (Full additive subcategory). Let C be an additive category.
A full subcategory D of C is a **full additive subcategory** if the inclussion functor $\iota : D \hookrightarrow \mathsf{C}$ is additive.

Remark 6.0.1. By Proposition 6.0.1 a full subcategory D of C is a full additive subcategory if and only if $0 \in \mathsf{D}$ and D is closed under finite direct sums.

6.1 Complexes

From now until the end of the chapter, let C be an additive category.

Definition 6.1.1 (Differential object). A **differential object** $(X^\bullet, \mathrm{d}_X^\bullet)$ in C is a sequence of objects $X^k \in \mathsf{C}$ and morphisms d_X^k for every $k \in \mathbb{Z}$:

$$\cdots \longrightarrow X^{k-1} \xrightarrow{\mathrm{d}_X^{k-1}} X^k \xrightarrow{\mathrm{d}_X^k} X^{k+1} \longrightarrow \cdots$$

A morphism of differential objects $f^\bullet : (X^\bullet, \mathrm{d}_X^\bullet) \to (Y^\bullet, \mathrm{d}_Y^\bullet)$ is a collection of morphisms $\{f^k : X^k \to Y^k\}_{k\in\mathbb{Z}}$ such that, for all $k \in \mathbb{Z}$, the following diagram commutes

$$\begin{array}{ccccccc} \cdots & \longrightarrow & X^k & \xrightarrow{\mathrm{d}_X^k} & X^{k+1} & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle f^k} & & \downarrow{\scriptstyle f^{k+1}} & & \\ \cdots & \longrightarrow & Y^k & \xrightarrow[\mathrm{d}_Y^k]{} & Y^{k+1} & \longrightarrow & \cdots \end{array}$$

Therefore we get a category $\mathsf{Diff}(\mathsf{C})$ of differential objects in C. By abuse of notation we will often denote differential objects $(X^\bullet, \mathrm{d}_X^\bullet)$ as $X^\bullet$.

Remark 6.1.1. We can view the category of differential objects $\mathsf{Diff}(\mathsf{C})$ as the functor category $\mathsf{Fct}(\mathbb{Z}, \mathsf{C})$. Therefore by Lemma 6.0.3 $\mathsf{Diff}(\mathsf{C})$ is an additive category.

Definition 6.1.2 (Complex)**.** A **complex** in C is a differential object $(X^\bullet, \mathrm{d}_X^\bullet)$ such that for all $k \in \mathbb{Z}$

$$\mathrm{d}_X^{k+1} \circ \mathrm{d}_X^k = 0$$

We denote by $\mathsf{Comp}(\mathsf{C})$ the full additive subcategory of $\mathsf{Diff}(\mathsf{C})$ consisting of complexes in C.

Definition 6.1.3. A complex $X^\bullet$ is

1. **Bounded above** if $X^n = 0$ of $n \gg 0$.
2. **Bounded below** if $X^n = 0$ of $n \ll 0$.
3. **Bounded** if it is bounded above and below.

We denote by $\mathsf{Comp}^*(\mathsf{C})$ $(* = +, -, \mathrm{b})$ the full additive subcategory of $\mathsf{Comp}(\mathsf{C})$ consisting of bounded above (resp. bounded below, bounded) complexes. To emphasize that we are working with unbounded complexes we may write $\mathsf{Comp}^{\mathrm{ub}}(\mathsf{C}) := \mathsf{Comp}(\mathsf{C})$.

Notation. For $a \in \mathbb{Z}$ we denote $\mathsf{Comp}^{\geq a}(\mathsf{C})$ the full additive subcategory of $\mathsf{Comp}(\mathsf{C})$ consisting of complexes $X^\bullet$ such that $X^j = 0$ for all $j < a$. We define the categories $\mathsf{Comp}^{\leq a}(\mathsf{C})$ and $\mathsf{Comp}^{[a,b]}(\mathsf{C})$ for $a \leq b$ analogously.

Definition 6.1.4 (Shift functor)**.** Let $p \in \mathbb{Z}$. The **shift functor** is the functor

$$[p] : \mathsf{Comp}(\mathsf{C}) \to \mathsf{Comp}(\mathsf{C})$$

as

– For all complexes $(X^\bullet, \mathrm{d}_X^\bullet) \in \mathsf{Comp}(\mathsf{C})$

$$(X[p])^n := X^{n+p} \qquad \mathrm{d}_{X[p]}^n := (-1)^p \mathrm{d}_X^{n+p}$$

– For all morphisms of complexes $f^\bullet : X^\bullet \to Y^\bullet$

$$f^\bullet[p] : X[p] \to Y[p],\ (f[p])^n := f^{n+p}$$

Remark 6.1.2. Note that, for all $p \in \mathbb{Z}$

$$\mathrm{d}_{X[p]}^{n+1} \circ \mathrm{d}_{X[p]}^n = (-1)^p \mathrm{d}_X^{n+p+1} \circ (-1)^p \mathrm{d}_X^{n+p} = (-1)^p \left(\mathrm{d}_X^{(n+p)+1} \circ \mathrm{d}_X^{n+p} \right) = (-1)^p 0 = 0$$

Therefore the shift functor is well defined.

Definition 6.1.5 (Mapping cone). Let $f^\bullet : X^\bullet \to Y^\bullet$ be a morphism of complexes. The **mapping cone** of f, denoted $\mathrm{Cone}(f)$ is the complex defined by

$$\mathrm{Cone}(f^\bullet)^n := (X[1])^n \oplus Y^n = X^{n+1} \oplus Y^n$$
$$\mathrm{d}^n_{\mathrm{Cone}(f^\bullet)} := \begin{pmatrix} \mathrm{d}^n_{X[1]} & 0 \\ f^{n+1} & \mathrm{d}^n_Y \end{pmatrix} = \begin{pmatrix} -\mathrm{d}^{n+1}_X & 0 \\ f^{n+1} & \mathrm{d}^n_Y \end{pmatrix}$$

Note that there a natural morphisms of complexes

$$\alpha f^\bullet : Y \to \mathrm{Cone}(f^\bullet) \qquad \beta f^\bullet : \mathrm{Cone}(f^\bullet) \to X[1]$$

Remark 6.1.3. Note that for any $f^\bullet : X^\bullet \to Y^\bullet$

$$\begin{aligned} \mathrm{d}^{n+1}_{\mathrm{Cone}(f^\bullet)} \circ \mathrm{d}^n_{\mathrm{Cone}(f^\bullet)} &= \begin{pmatrix} -\mathrm{d}^{n+2}_X & 0 \\ f^{n+2} & \mathrm{d}^{n+1}_Y \end{pmatrix} \circ \begin{pmatrix} -\mathrm{d}^{n+1}_X & 0 \\ f^{n+1} & \mathrm{d}^n_Y \end{pmatrix} \\ &= \begin{pmatrix} -\mathrm{d}^{n+2}_X \circ -d^{n+1}_X & 0 \\ f^{n+2} \circ -\mathrm{d}^{n+1}_X + \mathrm{d}^{n+1}_Y \circ f^{n+1} & \mathrm{d}^{n+1}_Y \circ \mathrm{d}^n_Y \end{pmatrix} \\ &= 0 \end{aligned}$$

were we have used that $\mathrm{d}^{n+1}_Y \circ f^{n+1} = f^{n+2} \circ \mathrm{d}_X$.
Therefore the mapping cone is well defined.

Proposition 6.1.1. Let $F : \mathsf{C} \to \mathsf{D}$ be a additive functor. Then F commutes with mapping cones: for all morphisms $f^\bullet : X^\bullet \to Y^\bullet$ in $\mathsf{Comp}(\mathsf{C})$

$$F(\mathrm{Cone}(f^\bullet)) \cong \mathrm{Cone}(Ff^\bullet)$$

Proof. Since F preserves finite direct sums for all $n \in \mathbb{Z}$

$$F(X^{n+1} \oplus Y^n) \cong F(X^{n+1}) \oplus F(Y^n)$$

Therefore the objects of $F(\mathrm{Cone}(f^\bullet))$ and $\mathrm{Cone}(Ff^\bullet)$ are the same.

Furthermore

$$\begin{aligned} F\mathrm{d}^n_{\mathrm{Cone}(f^\bullet)} &= F\begin{pmatrix} -\mathrm{d}^{n+1}_X & 0 \\ f^{n+1} & \mathrm{d}^n_Y \end{pmatrix} \\ &= \begin{pmatrix} F(-\mathrm{d}^{n+1}_X) & F0 \\ Ff^{n+1} & F\mathrm{d}^n_Y \end{pmatrix} \\ &= \begin{pmatrix} -Fd^{n+1}_X & 0 \\ Ff^{n+1} & F\mathrm{d}^n_Y \end{pmatrix} \\ &= \begin{pmatrix} -d^{n+1}_{F(X)} & 0 \\ Ff^{n+1} & \mathrm{d}^n_{F(Y)} \end{pmatrix} \\ &= \mathrm{d}^n_{\mathrm{Cone}(Ff^\bullet)} \end{aligned}$$

□

6.2 Double complexes

Definition 6.2.1 (Double complex). A **double complex** in C is the data of

$$\{X^{n,m}, \mathrm{d}_{\uparrow}^{n,m}, \mathrm{d}_{\rightarrow}^{n,m}\}_{n,m\in\mathbb{Z}}$$

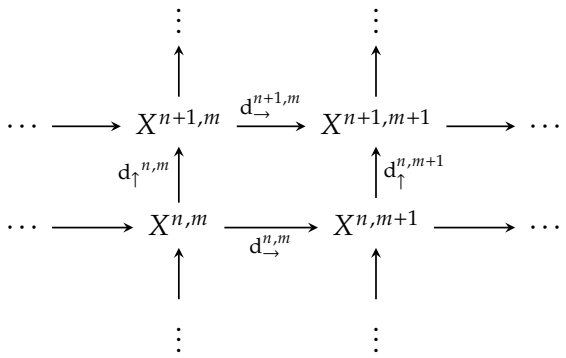
where $X^{n,m} \in \mathsf{C}$ and the *differential maps*

$$\mathrm{d}_{\uparrow}^{n,m} : X^{n,m} \to X^{n+1,m} \qquad \mathrm{d}_{\rightarrow}^{n,m} : X^{n,m} \to X^{n,m+1}$$

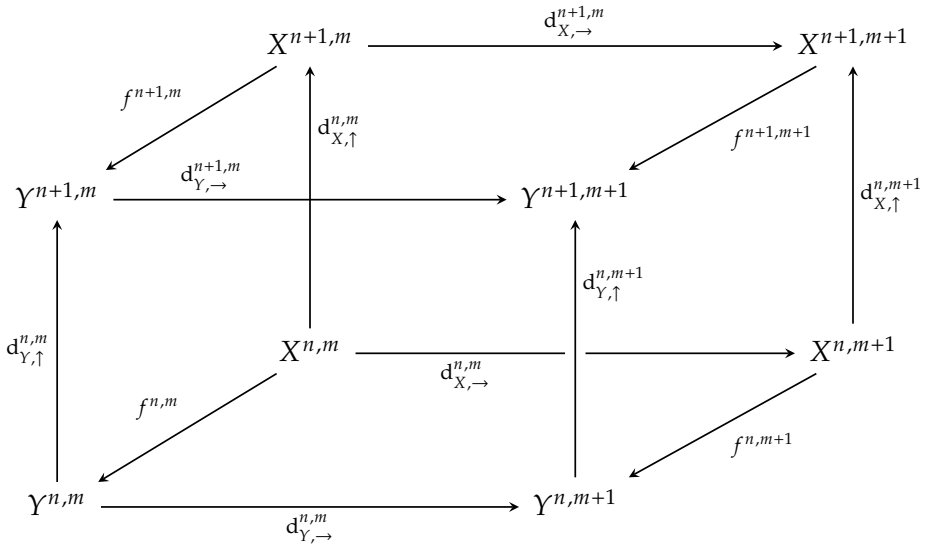
satisfy

$$\mathrm{d}_{\uparrow}^{n+1,m} \circ \mathrm{d}_{\uparrow}^{n,m} = \mathrm{d}_{\rightarrow}^{n,m+1} \circ \mathrm{d}_{\rightarrow}^{n,m} = 0 \qquad \mathrm{d}_{\rightarrow}^{n+1,m} \circ \mathrm{d}_{\uparrow}^{n,m} = \mathrm{d}_{\uparrow}^{n,m+1} \circ \mathrm{d}_{\rightarrow}^{n,m}$$

This can be represented by a commutative diagram



We denote double complexes by $X^{\bullet,\bullet}$
A morphism of double complexes $f^{\bullet,\bullet} : X^{\bullet,\bullet} \to Y^{\bullet,\bullet}$ consists of maps $\{f^{n,m} : X^{n,m} \to Y^{n,m}\}_{n,m\in\mathbb{Z}}$ such that the following cube commutes for all $n, m \in \mathbb{Z}$:



In this way we obtain the additive category $\mathsf{Comp}^2(\mathsf{C})$ of double complexes in C.

Remark 6.2.1. There is a functor $F_I : \mathsf{Comp}^2(\mathsf{C}) \to \mathsf{Comp}(\mathsf{Comp}(C))$ which to a double complex X associates the complex whose objects are the rows of X. More precisely for all $n \in \mathbb{Z}$ consider the complex

$$X_I^n := \{X^{n,m}, \mathrm{d}_{\rightarrow}^{n,m}\}_{m\in\mathbb{Z}}$$

The collection of morphisms $\{d^{n,m}_{\rightarrow}\}_{m\in\mathbb{Z}}$ defines a morphism $d^n_I : X^n_I \to X^{n+1}_I$, which trivially verifies $d^{n+1}_I \circ d^n_I = 0$. Therefore X_I is a complex in $\mathsf{Comp}(\mathsf{C})$ and we have constructed the functor F_I.
Interchainging rows and columns we get a functor $F_{II} : \mathsf{Comp}^2(\mathsf{C}) \to \mathsf{Comp}(\mathsf{Comp}(C))$.

Remark 6.2.2. To make things easier we will assume that all double complexes $X^{\bullet,\bullet} \in \mathsf{Comp}^2(\mathsf{C})$ verify the following finiteness condition:

$$\forall p \in \mathbb{Z} \quad \{(n,m) \in \mathbb{Z}^2 \mid X^{n,m} \neq 0 \text{ and } n+m=p\} \text{ is finite.}$$

Definition 6.2.2 (Total complex)**.** Let $X^{\bullet,\bullet}$ be a double complex. The **total complex** $\mathrm{tot}(X)^\bullet$ associated to $X^{\bullet,\bullet}$ is the complex defined by

$$\mathrm{tot}(X^{\bullet,\bullet})^n := \bigoplus_{p+q=n} X^{p,q} \qquad d^n_{\mathrm{tot}(X)} := \sum_{p+q=n} \left(d^{p,q}_{\rightarrow} + (-1)^p d^{p,q}_{\uparrow}\right)$$

Proposition 6.2.1. The differential object $\{\mathrm{tot}(X^{\bullet,\bullet})^p, d^p_{\mathrm{tot}(X^{\bullet,\bullet})}\}_{p\in\mathbb{Z}}$ is a complex and $\mathrm{tot} : \mathsf{Comp}^2(\mathsf{C}) \to \mathsf{Comp}(\mathsf{C})$ is an additive functor.

Proof. To see that $d^{n+1}_{\mathrm{tot}(X^{\bullet,\bullet})} \circ d^n_{\mathrm{tot}(X^{\bullet,\bullet})} = 0$ we look at the maps from a specific source $X^{p,q}$ with $p+q=n$ to tot^{n+2}. Note that in two steps the only targets possible are

$X^{p+2,q}$ Reached via $d^{p+1,q}_{\rightarrow} \circ d^{p,q}_{\rightarrow} = 0$.

$X^{p,q+2}$ Reached via $(-1)^p(d^{p,q+1}_{\uparrow} \circ d^{p,q}_{\uparrow}) = 0$.

$X^{p+1,q+1}$ There are two possible paths: $(-1)^{p+1} d^{p+1,q}_{\uparrow} \circ d^{p,q}_{\rightarrow}$ or $d^{p,q+1}_{\rightarrow} \circ (-1)^p d^{p,q}_{\uparrow}$. Summing these we have

$$\begin{aligned}(-1)^{p+1} d^{p+1,q}_{\uparrow} \circ d^{p,q}_{\rightarrow} + d^{p,q+1}_{\rightarrow} \circ (-1)^p d^{p,q}_{\uparrow} &= (-1)^p \left(-d^{p+1,q}_{\uparrow} \circ d^{p,q}_{\rightarrow} + d^{p,q+1}_{\rightarrow} \circ d^{p,q}_{\uparrow}\right) \\ &= (-1)^p \left(-d^{p+1,q}_{\uparrow} \circ d^{p,q}_{\rightarrow} + d^{p+1,q}_{\uparrow} \circ d^{p,q}_{\rightarrow}\right) \\ &= 0\end{aligned}$$

Therefore since all possibilities lead to the map being zero, $d^{n+1}_{\mathrm{tot}(X^{\bullet,\bullet})} \circ d^n_{\mathrm{tot}(X^{\bullet,\bullet})} = 0$.

To see that tot is additive note that

$$\mathrm{tot}(0^{\bullet,\bullet}) = \bigoplus_{p+q=n} 0^{p,q} = 0^\bullet$$

Furthermore for any $X^{\bullet,\bullet}, Y^{\bullet,\bullet} \in \mathsf{Comp}^2(\mathsf{C})$

$$\begin{aligned}
\mathrm{tot}(X^{\bullet,\bullet} \oplus Y^{\bullet,\bullet}) &= \bigoplus_{p+q=n} (X^{p,q} \oplus Y^{p,q}) = \left(\bigoplus_{p+q=n} X^{p,q}\right) \oplus \left(\bigoplus_{p+q=n} Y^{p,q}\right) = \mathrm{tot}(X^{\bullet,\bullet}) \oplus \mathrm{tot}(Y^{\bullet,\bullet}) \\
\mathrm{d}^n_{\mathrm{tot}(X^{\bullet,\bullet}\oplus Y^{\bullet,\bullet})} &= \sum_{p+q=n} \left(\mathrm{d}^{p,q}_{X^{\bullet,\bullet}\oplus Y^{\bullet,\bullet},\rightarrow} + (-1)^p \mathrm{d}^{p,q}_{X^{\bullet,\bullet}\oplus Y^{\bullet,\bullet},\uparrow}\right) \\
&= \sum_{p+q=n} \left(\left(\mathrm{d}^{p,q}_{X^{\bullet,\bullet},\rightarrow} \oplus \mathrm{d}^{p,q}_{Y^{\bullet,\bullet},\rightarrow}\right) + (-1)^p \left(\mathrm{d}^{p,q}_{X^{\bullet,\bullet},\uparrow} \oplus \mathrm{d}^{p,q}_{Y^{\bullet,\bullet},\uparrow}\right)\right) \\
&= \sum_{p+q=n} \left(\left(\mathrm{d}^{p,q}_{X^{\bullet,\bullet},\rightarrow} + (-1)^p \mathrm{d}^{p,q}_{X^{\bullet,\bullet},\uparrow}\right) \oplus \left(\mathrm{d}^{p,q}_{Y^{\bullet,\bullet},\rightarrow} + (-1)^p \mathrm{d}^{p,q}_{Y^{\bullet,\bullet},\uparrow}\right)\right) \\
&= \left(\sum_{p+q=n} \left(\mathrm{d}^{p,q}_{X^{\bullet,\bullet},\rightarrow} + (-1)^p \mathrm{d}^{p,q}_{X^{\bullet,\bullet},\uparrow}\right)\right) \oplus \left(\sum_{p+q=n} \left(\mathrm{d}^{p,q}_{Y^{\bullet,\bullet},\rightarrow} + (-1)^p \mathrm{d}^{p,q}_{Y^{\bullet,\bullet},\uparrow}\right)\right) \\
&= \mathrm{d}^n_{\mathrm{tot}(X^{\bullet,\bullet})} \oplus \mathrm{d}^n_{\mathrm{tot}(Y^{\bullet,\bullet})}
\end{aligned}$$

Therefore by Proposition 6.0.1 tot is additive. □

Remark 6.2.3. Since Comp(C) is additive and thus finite products and coproducts coincide we can write $\prod$ instead of $\bigoplus$ in the tot definition. We can do this because we are assuming the finiteness condition.

Remark 6.2.4 (Bifunctors and double complexes). If C, C' and C'' are additive categories and $F : \mathsf{C} \times \mathsf{C}' \to \mathsf{C}''$ is an additive bifunctor (i.e: it is additive in each component). Then it defines an additive bifunctor

$$\mathsf{Comp}^2(F) : \mathsf{Comp}(\mathsf{C}) \times \mathsf{Comp}(\mathsf{C}') \to \mathsf{Comp}^2(\mathsf{C}'')$$

In other words, if $X \in \mathsf{Comp}(\mathsf{C})$ and $X' \in \mathsf{Comp}(\mathsf{C}')$ are complexes then $\mathsf{Comp}^2(F)(X, X')$ is a double complex.

Example 6.2.1. Let A be a ring. Consider the tensor product bifunctor

$$- \otimes - : \mathsf{Mod}A^{\mathrm{op}} \times \mathsf{Mod}A \to \mathsf{Ab}$$

In what follows we will write $\otimes$ instead of $\mathsf{Comp}^2(\otimes)$. Then for any complexes $X^\bullet \in \mathsf{Comp}(\mathsf{Mod}A^{\mathrm{op}}), Y^\bullet \in \mathsf{Comp}(\mathsf{Mod}A)$ we have

$$(X^\bullet \otimes Y^\bullet)^{n,m} = X^n \otimes Y^m \quad \mathrm{d}^{n,m}_\uparrow = \mathrm{d}^n_{X^\bullet} \otimes \mathbb{1}_{Y^m} \quad \mathrm{d}^{n,m}_\rightarrow = \mathbb{1}_{X^n} \otimes \mathrm{d}^m_{Y^\bullet}$$

Therefore applying tot we have

$$\begin{aligned}
(\mathrm{tot}(X, Y))^k &= \bigoplus_{n+m=k} X^n \oplus Y^m \\
\mathrm{d}_{\mathrm{tot}(X^\bullet, Y^\bullet)}|_{X^n \oplus Y^m} &= \mathrm{d}^{n,m}_\rightarrow + (-1)^n \mathrm{d}^{n,m}_\uparrow
\end{aligned}$$

Example 6.2.2 ($\mathrm{Hom}^{\bullet,\bullet}$ complex). Consider the bifunctor

$$\mathrm{Hom}_\mathsf{C} : \mathsf{C}^{\mathrm{op}} \times \mathsf{C} \to \mathsf{Ab}$$

In what follows we will write $\mathrm{Hom}^{\bullet,\bullet}_\mathsf{C}$ instead of $\mathsf{Comp}^2(\mathrm{Hom}_\mathsf{C})$. Then for any

complexes $X^\bullet \in \mathsf{Comp}(\mathsf{C}^{\mathrm{op}})$, $Y^\bullet \in \mathsf{Comp}(\mathsf{C})$ we have

$$\mathrm{Hom}_{\mathsf{C}}^{\bullet,\bullet}(X^\bullet, Y^\bullet)^{n,m} = \mathrm{Hom}_{\mathsf{C}}(X^{-m}, Y^n)$$
$$\mathrm{d}_{\uparrow}^{n,m} = \mathrm{Hom}_{\mathsf{C}}(X^{-m}, \mathrm{d}_{Y}^{n}) \quad \mathrm{d}_{\rightarrow}^{n,m} = \mathrm{Hom}_{\mathsf{C}}((-1)^{m+1}\mathrm{d}_{X}^{m+1}, Y^n)$$

Using that Ab admits countable products we set

$$\mathrm{Hom}_{\mathsf{C}}^{\bullet}(X^\bullet, Y^\bullet) := \mathrm{tot}\,\mathrm{Hom}_{\mathsf{C}}^{\bullet,\bullet}(X^\bullet, Y^\bullet) \in \mathsf{Comp}(\mathsf{Ab})$$

Therefore we have

$$\mathrm{Hom}_{\mathsf{C}}(X^\bullet, Y^\bullet)^n = \prod_{j\in\mathbb{Z}} \mathrm{Hom}_{\mathsf{C}}(X^j, Y^{n-j})$$
$$(\mathrm{d}^n f)^j = \mathrm{d}_Y^{j+n} \circ f^n + (-1)^{n+1} f^{j+1} \circ \mathrm{d}_X^j$$

Note that for $X^\bullet, Y^\bullet, Z^\bullet \in \mathsf{Comp}(\mathsf{C})$ we have *composition* map

$$\mathrm{Hom}_{\mathsf{C}}^{\bullet}(X^\bullet, Y^\bullet) \times \mathrm{Hom}_{\mathsf{C}}(Y^\bullet, Z^\bullet) \xrightarrow{\circ} \mathrm{Hom}_{\mathsf{C}}(X^\bullet, Z^\bullet)$$

associated to the maps

$$\mathrm{Hom}_{\mathsf{C}}(X^\bullet, Y^\bullet)^m \times \mathrm{Hom}_{\mathsf{C}}(Y^\bullet, Z^\bullet)^n \to \mathrm{Hom}_{\mathsf{C}}(X^\bullet, Z^\bullet)^{n+m}$$
$$\prod_{j\in\mathbb{Z}} \mathrm{Hom}_{\mathsf{C}}(X^j, Y^{j+j}) \times \prod_{j\in\mathbb{Z}} \mathrm{Hom}_{\mathsf{C}}(Y^{j+m}, Z^{j+m+n}) \mapsto \prod_{j\in\mathbb{Z}} \mathrm{Hom}_{\mathsf{C}}(X^j, Z^{j+m+n})$$

6.3 The homotopy category

Definition 6.3.1 (Homotopy).

1. A morphism of complexes $f^\bullet : X^\bullet \to Y^\bullet$ is **homotopic** to zero if for all $p \in \mathbb{Z}$ there exists a morphism $s^p : X^p \to Y^{p-1}$ in C such that
$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$
2. Two morphisms $f^\bullet, g^\bullet : X^\bullet \to Y^\bullet$ are homotopic if $f^\bullet - g^\bullet$ is homotopic to zero. We denote this by $f^\bullet \sim g^\bullet$.
3. A complex $X^\bullet \in \mathsf{Comp}(\mathsf{C})$ is homotopic to zero if $\mathbb{1}_{X^\bullet} \sim 0$.
4. A morphism of complexes $f^\bullet : X^\bullet \to Y^\bullet$ is a **homotopy equivalence** if there exists $g^\bullet : Y^\bullet \to X^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathbb{1}_{X^\bullet}$ and $f^\bullet \circ g^\bullet \sim \mathbb{1}_{Y^\bullet}$.

Lemma 6.3.1. Let $F : \mathsf{C} \to \mathsf{D}$ be an additive functor. If $f^\bullet : X^\bullet \to Y^\bullet$ is homotopic to zero then Ff is also homotopic to zero.

Proof. Since $f^\bullet$ is homotopic to zero for all $p \in \mathbb{Z}$ there exists a morphism $s^p : X^p \to Y^{p-1}$ such that

$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$

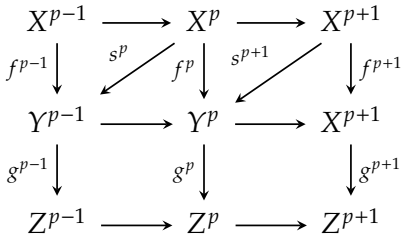
Therefore

$$Ff^p = F(s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p) = Fs^{p+1} \circ \mathrm{d}_{F(X)}^p + \mathrm{d}_{F(Y)}^{p-1} \circ Fs^p$$

□

Lemma 6.3.2. Let $f^\bullet : X^\bullet \to Y^\bullet$ and $g^\bullet : Y^\bullet \to Z^\bullet$ be two morphisms of complexes in $\mathsf{Comp}(\mathsf{C})$. If either f or g is homotopic to 0 then $g \circ f$ is homotopic to 0.

Proof. Assume that f is homotopic to 0. The proof consists on chasing the following diagram:



We have that for all $p \in \mathbb{Z}$ there exists $s^p : X^p \to Y^{p-1}$ such that

$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$

Therefore

$$\begin{aligned} g^p \circ f^p &= g^p \circ \left(s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p\right) \\ &= \left(g^p \circ s^{p+1}\right) \circ \mathrm{d}_X^p + \left(g^p \circ \mathrm{d}_Y^{p-1}\right) \circ s^p \\ &= \left(g^p \circ s^{p+1}\right) \circ \mathrm{d}_X^p + \mathrm{d}_Z^{p-1} \circ \left(g^{p-1} \circ s^p\right) \end{aligned}$$

□

Definition 6.3.2 (Homotopy category). Let C be a additive-category. For every pair of complexes $X^\bullet, Y^\bullet \in \mathsf{Comp}(\mathsf{C})$ set

$$\mathrm{Ht}(X^\bullet, Y^\bullet) := \{f^\bullet : X^\bullet \to Y^\bullet \mid f \sim 0\}$$

The **homotopy category** $\mathrm{K}(\mathsf{C})$ of C is the category defined by

$$\begin{aligned} \mathrm{Ob}(\mathrm{K}(\mathsf{C})) &:= \mathrm{Ob}(\mathsf{Comp}(\mathsf{C})) \\ \mathrm{Hom}_{\mathrm{K}(\mathsf{C})}(X^\bullet, Y^\bullet) &:= \mathrm{Hom}_{\mathsf{Comp}(\mathsf{C})}(X^\bullet, Y^\bullet) \Big/ \mathrm{Ht}(X^\bullet, Y^\bullet) \end{aligned}$$

We define analogously the categories $\mathrm{K}^*(\mathsf{C})$ $(* = +, -, \mathrm{b})$.
Note that all of these categories are endowed with an equivalence, the shift functor $[1] : X \mapsto X[1]$.

Remark 6.3.1. Let $F : \mathsf{C} \to \mathsf{C}'$ be an additive functor. This defines a additive functor $F : \mathsf{Comp}(\mathsf{C}) \to \mathsf{Comp}(\mathsf{C})$.
If $f : X \to Y$ is homotopic to 0 in C then by Lemma 6.3.1 Ff is homotopic to 0. Therefore F defines an additive functor

$$\mathrm{K}(F) : \mathrm{K}(\mathsf{C}) \to \mathrm{K}(\mathsf{C}')$$

Now let $F : \mathsf{C} \times \mathsf{C}' \to \mathsf{C}''$ be an additive bifunctor. This defines an additive bifunctor $F : \mathsf{Comp}(\mathsf{C}) \times \mathsf{Comp}(\mathsf{C}') \to \mathsf{Comp}(\mathsf{C}'')$ and by composing with the tot functor a bifunctor $\mathsf{Comp}^+(\mathsf{C}) \times \mathsf{Comp}^+(\mathsf{C}') \to \mathsf{Comp}^+(\mathsf{C}'')$.
If $f : X \to Y$ homotopic to 0 in C and $X' \in \mathsf{C}'$ then $F(f, X')$ is homotopic to 0, and

analogously interchanging C and C'. Therefore F defines an additive bifunctor

$$\mathrm{K}(F) : \mathrm{K}(\mathsf{C}) \times \mathrm{K}(\mathsf{C}') \to \mathrm{K}(\mathsf{C}'')$$

By abuse of notation when the context is clear we will write F instead of $\mathrm{K}(F)$.

Chapter 7

Abelian categories

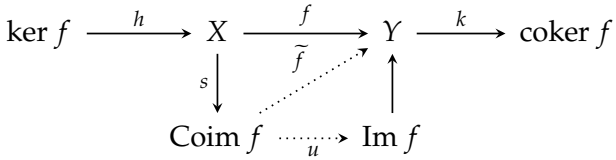
Definition 7.0.1. Let $f : X \to Y$ be a morphism in a additive category C that admits kernels and cokernels. Consider the following

$$\ker f \xrightarrow{h} X \xrightarrow{f} Y \xrightarrow{k} \operatorname{coker} f$$

We denote

$$\operatorname{Im} f := \ker k \qquad \operatorname{Coim} f := \operatorname{coker} h$$

I.e: we have the following diagram



Note that, since $f \circ h = 0$, f factors uniquely through $\operatorname{Coim} f$ producing the morphism $\widetilde{f}$. Furthermore since $k \circ f = k \circ \widetilde{f} = 0$ we have that $\widetilde{f}$ factors uniquely through $\operatorname{Im} f$. Therefore we have a canonical morphism

$$u : \operatorname{Coim} f \to \operatorname{Im} f$$

Definition 7.0.2 (Abelian category)**.** An additive category C is **abelian** if it verifies

Ab.1) Any morphism in C admits a kernel and a cokernel.

Ab.2) For any morphism f in C the canonical morphism $\operatorname{Coim} f \to \operatorname{Im} f$ is an isomorphism.

Lemma 7.0.1. An abelian category admits all finite limits and colimits.

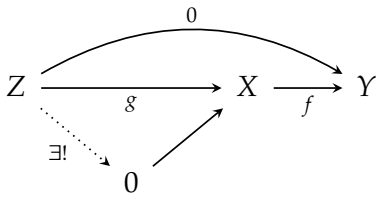
Proof. Direct from Theorem 3.2.1. □

Lemma 7.0.2. Let $f : X \to Y$ be a morphism in an abelian category. Then f is a monomorphism if and only if $0 \to X$ is its kernel.
Dually f is an epimorphism if and only if $Y \to 0$ is its cokernel.

Proof. We will do the kernel one.

$\Longrightarrow$ Suppose f is a monomorphism.
If $g : Z \to X$ is a morphism such that $f \circ g = 0$ then by Lemma 6.0.2 $g = 0$. In particular g factors uniquely through $0 \to X$, i.e: $0 \to X$ is the kernel of f.

$\Longleftarrow$ Suppose now that $0 \to X$ is the kernel of f.
Consider a morphism $g : Z \to X$ such that $f \circ g = 0$. By the kernel properties we have the following diagram



Therefore by uniqueness $g = 0$, making f a monomorphism by Lemma 6.0.2.

□

Proposition 7.0.1. Let I be a small category and C an abelian category. Then the functor category $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is abelian.

Proof. By Lemma 6.0.3 we have that $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is additive.

Furthermore since C has kernels and cokernels, which are no more than special cases of limits and colimits, by Proposition 4.3.1 $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ has kernels and cokernels and they are computed elementwise.

Finally since $\operatorname{Coim} f(i) \xrightarrow{\sim} \operatorname{Im} f(i)$ for all $i \in \mathsf{I}$ we have that $\operatorname{Coim} f \xrightarrow{\sim} \operatorname{Im} f$. □

Corollary 7.0.1. If C is abelian then the categories of complexes $\mathsf{Comp}^*(\mathsf{C})$ $(* = +, -, \mathrm{b})$ are abelian.

Proof. Direct since by Proposition 7.0.1 the category of differential objects $\mathsf{Diff}(\mathsf{C})$ is abelian. □

In what remains of the chapter we will assume that C is an abelian category.

Definition 7.0.3.

1. A complex $X \xrightarrow{f} Y \xrightarrow{g} Z$ is **exact** if $\operatorname{Im} f \xrightarrow{\sim} \ker g$. Note that this is equivalent to $\operatorname{coker} f \xrightarrow{\sim} \operatorname{Im} g$.
2. More generally, a sequence of morphisms $X^p \xrightarrow{\mathrm{d}^p} \cdots \to X^n$ with $\mathrm{d}^{i+1} \circ \mathrm{d}^i = 0$ for all $i \in \{p, \dots, n-1\}$ is **exact** if $\operatorname{Im} \mathrm{d}^i \xrightarrow{\sim} \ker \mathrm{d}^{i+1}$ for all $i \in \{p, \dots, n-1\}$.
3. A **short exact sequence** is an exact sequence of the form $0 \to X \to Y \to Z \to 0$.

Remark 7.0.1. Note that any morphism $f : X \to Y$ may be decomposed in short exact sequences

$$0 \longrightarrow \ker f \longrightarrow X \longrightarrow \operatorname{Coim} f \longrightarrow 0$$

$$0 \longrightarrow \operatorname{Im} f \longrightarrow Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

with $\operatorname{Coim} f \cong \operatorname{Im} f$.

Proposition 7.0.2. Let

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

be a short exact sequence in an abelian category C. We say that the sequence **splits** if any of the following equivalent conditions are satisfied

1. There exists $h : Z \to Y$ such that $g \circ h = \mathbb{1}_Z$.
2. There exists $k : Y \to X$ such that $k \circ f = \mathbb{1}_X$.
3. There exists $\varphi = (k, h)$ and $\psi = \binom{f}{g}$ such that $Y \xrightarrow{\varphi} X \oplus Z$ and $X \otimes Z \xrightarrow{\psi} Y$ are inverses to each other.
4. The sequence is homotopic to 0.
5. The sequence is isomorphic to $0 \to X \to X \oplus Z \to Z \to 0$.

Proof.

$1 \implies 3$ Since $g = g \circ h \circ g$ we have that $g \circ (\mathbb{1}_Y - h \circ g) = 0$, i.e: $\mathbb{1}_Y - h \circ g$ factors through $\ker g$ and so it factors through X. Therefore there exists $h : Y \to Z$ such that $\mathbb{1}_Y - h \circ g = f \circ k$.

$3 \implies 1$ Since $g \circ f = 0$ we have $g = g \circ h \circ g$, i.e: $(g \circ h - \mathbb{1}_Z) \circ g$. Therefore since g is an epimorphism we have $g \circ h - \mathbb{1}_Z = 0 \iff g \circ h = \mathbb{1}_Z$.

$2 \iff 3$ Analogous to $1 \iff 3$.

$3 \iff 4$ By definition the sequence is homopic to zero if and only if there exists

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mathbb{1}_X} & \swarrow{\scriptstyle k} & \downarrow{\scriptstyle \mathbb{1}_Y} & \swarrow{\scriptstyle h} & \downarrow{\scriptstyle \mathbb{1}_Z} & & \\ 0 & \longrightarrow & X & \xrightarrow[f]{} & Y & \xrightarrow[g]{} & Z & \longrightarrow & 0 \end{array}$$

such that $\mathbb{1}_X = k \circ f$, $\mathbb{1}_Z = h \circ g$ and $\mathbb{1}_Z = h \circ g + f \circ k$.

$3 \iff 5$ Trivial.

□

7.1 Exact functors

Definition 7.1.1. Let C, D be two abelian categories and

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be a short exact sequence in an C.
An additive functor $F : \mathsf{C} \to \mathsf{D}$ is **left-exact** if

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$

is exact in D.
Dually, it is **right-exact** if

$$F(X) \longrightarrow F(Y) \longrightarrow F(Z) \longrightarrow 0$$

is exact in D.
If F is both left and right exact we say that it is **exact**.

We state the following characterization of exact functors, which we will not prove.

Lemma 7.1.1 ([Sch, Lemma 5.2.2]). Let $F : \mathsf{C} \to \mathsf{D}$ be an additive functor.

1. The following are equivalent
   - **(a)** F is left exact (resp. right exact).
   - **(b)** F commutes with kernels: for any morphism $f : X \to Y$, $F(\ker f) \xrightarrow{\sim} \ker F f$ (resp. with cokernels).
   - **(c)** F commutes with finite limits (resp. finite colimits).
   - **(d)** For any short exact sequence $0 \to X \to Y \to Z \to 0$ in C the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is exact in D (resp. $F(X) \to F(Y) \to F(Z) \to$

0 is exact).

2. The following are equivalent
   (a) F is exact.
   (b) For any exact sequence $X \to Y \to Z$ in C the sequence $F(X) \to F(Y) \to F(Z)$ is exact in D.
   (c) F commutes with finite limits and finite colimits.

As a direct application of Lemma 7.1.1.

Example 7.1.1. The Hom bifunctor $\mathrm{Hom}_{\mathsf{C}} : \mathsf{C}^{\mathrm{op}} \times \mathsf{C} \to \mathsf{Ab}$ is left exact with respect to each of its arguments.

Example 7.1.2. Recall Theorem 4.2.1.
Right adjoints are left exact. Dually, left adjoints are right exact.

Example 7.1.3. Let I be a small category. If C admits I-indexed limits then the functor $\lim : \mathsf{Fct}(\mathsf{I}^{\mathrm{op}}, \mathsf{C}) \to \mathsf{C}$ is left exact. Dually if C admits I-indexed colimits then the functor $\mathrm{colim} : \mathsf{Fct}(\mathsf{I}, \mathsf{C}) \to \mathsf{C}$ is right exact.

Example 7.1.4. Let R be a ring and $\mathcal{I}$ an index set. The two functors

$$\prod_{i \in \mathcal{I}} : \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R) \qquad \bigoplus_{i \in \mathcal{I}} : \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R)$$

are exact.

Example 7.1.5. Let R be a ring and $\mathcal{I}$ a small directed category. The functor $\varinjlim_{i \in \mathcal{I}} : \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R)$ is exact.

7.1.1 Diagram lemmas

Theorem 7.1.1 (Freyd-Mitchell embedding - [Wei94, Section 1.6]). Let C be a small abelian category. Then there exists a ring R (with unit, not necessarily commutative) and a fully faithful exact functor $F : \mathsf{C} \to \mathsf{Mod}(R)$ which embeds C as a full subcategory of $\mathsf{Mod}(R)$.

Lemma 7.1.2 (Five lemma). Consider a commutative diagram

$$\begin{array}{ccccccc} X^0 & \xrightarrow{\alpha^0} & X^1 & \xrightarrow{\alpha^1} & X^2 & \xrightarrow{\alpha^2} & X^3 \\ \downarrow{\scriptstyle f^0} & & \downarrow{\scriptstyle f^1} & & \downarrow{\scriptstyle f^2} & & \downarrow{\scriptstyle f^3} \\ Y^0 & \xrightarrow[\beta^0]{} & Y^1 & \xrightarrow[\beta^1]{} & Y^2 & \xrightarrow[\beta^2]{} & Y^3 \end{array}$$

where the rows are exact.
If f^0 is an epimorphism and f^1, f^3 are monomorphisms then f^2 is a monomorphism. Dually if f^3 is an monomorphism and f^0, f^2 are epimorphisms then f^1 is a epimorphism.

Proof. Let $x_2 \in X^2$ such that $f^2(x_2) = 0$. We want to see that $x_2 = 0$.

We have that $(f^3 \circ \alpha^2)(x_2) = 0$ which, since f^3 is a monomorphism, implies that $\alpha^2(x_2) = 0$. Since the first row is exact there exists $x_1 \in X^1$ such that $\alpha(x_1) = x_2$.

Let $y_1 := f^1(x_1)$. Since $(\beta^1 \circ f^1)(x_1) = 0$ and the second row is exact there exists $y_0 \in X^0$ such that $\beta^0(y_0) = f^1(x_1)$.

Since f^0 is an epimorphism there exists $x_0 \in X^0$ such that $y_0 = f^0(x_0)$.

Since $(f^1 \circ \alpha^0)(x_0) = f^1(x_1)$ and f^1 is a monomorphism we have $\alpha^0(x_0) = x_1$. Therefore $x_2 = \alpha^1(x_1) = 0$. □

Lemma 7.1.3 (Snake lemma). Consider a commutative diagram

$$\begin{array}{ccccccc} X' & \xrightarrow{f} & X & \xrightarrow{g} & X'' & \longrightarrow & 0 \\ \downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\gamma} & & \\ 0 & \longrightarrow & Y' & \xrightarrow[f']{} & Y & \xrightarrow[g']{} & Y'' \end{array}$$

where the rows are exact.
Then there exists a morphism $\delta : \ker\gamma \to \operatorname{coker}\alpha$ giving rise to an exact sequence

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & \ker\alpha & \dashrightarrow & \ker\beta & \dashrightarrow & \ker\gamma & \dashrightarrow & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & X' & \xrightarrow{f} & X & \xrightarrow{g} & X'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\gamma} & & \\ 0 & \longrightarrow & Y' & \xrightarrow[f']{} & Y & \xrightarrow[g']{} & Y'' & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & \operatorname{coker}\alpha & \dashrightarrow & \operatorname{coker}\beta & \dashrightarrow & \operatorname{coker}\gamma & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}$$

Proof. We will first check that $\ker\alpha \to \ker\beta \to \ker\gamma$ is exact. Let $x \in \ker\beta$ with $g(x) = 0$. Since the first row is exact there exists $x' \in X'$ with $f(x') = x$. Then $(f' \circ \alpha)(x') = (\beta \circ f)(x') = 0$. Since f' is a monomorphism we get $\alpha(x') = 0$ and thus $x' \in \ker\alpha$.

A dual argument shows that $\operatorname{coker}\alpha \to \operatorname{coker}\beta \to \operatorname{coker}\gamma$ is exact.

It only remains to construct δ. Let $x'' \in \ker\gamma$ and choose $x \in X$ with $g(x) = x''$. Set $y := \beta(y)$. Since $g'(y) = 0$ there exists $y' \in Y'$ such that $f'(y') = y$. Therefore we can define $\delta(x'')$ to be the image of y' in $\operatorname{coker}\alpha$. □

7.2 Injective and projective objects

Definition 7.2.1.

– An object $I \in \mathsf{A}$ is **injective** if all the diagrams of the form

$$\begin{array}{ccc} X & \xrightarrow{k} & I \\ {\scriptstyle f}\downarrow & & \\ Z & & \end{array}$$

where $X \rightarrowtail Z$ is a monomorphism admit an extension

$$\begin{array}{ccc} X & \xrightarrow{k} & I \\ {\scriptstyle f}\downarrow & \nearrow_{h} & \\ Z & & \end{array}$$

– An object $P \in \mathsf{A}$ is **projective** if all the diagrams of the form

$$\begin{array}{ccc} & & X \\ & & \downarrow{\scriptstyle f} \\ P & \xrightarrow[k]{} & Z \end{array}$$

where $X \twoheadrightarrow Z$ is an epimorphism admit an extension

$$\begin{array}{ccc} & & X \\ & \nearrow_{h} & \downarrow{\scriptstyle f} \\ P & \xrightarrow[k]{} & Z \end{array}$$

Proposition 7.2.1.

– An object $I \in \mathsf{A}$ is injective if and only if the functor $\mathrm{Hom}_{\mathsf{A}}(-, I)$ is exact.
– An object $P \in \mathsf{A}$ is projective if and only if the functor $\mathrm{Hom}_{\mathsf{A}}(P, -)$ is exact.

Proof. By Example 7.1.1 we only need to check right exactness.

We will do the injective version this time:

$\Longrightarrow$ Suppose $I \in A$ is injective and consider a short exact sequence

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

Using Lemma 7.1.1, we want to see that

$$0 \longrightarrow \mathrm{Hom}(Z, I) \xrightarrow{g^*} \mathrm{Hom}(Y, I) \xrightarrow{f^*} \mathrm{Hom}(X, I)$$

is exact.
Since by Example 7.1.1 this sequence is exact at $\mathrm{Hom}(Z, I)$ and $\mathrm{Hom}(Y, I)$, we only need to check exactness at $\mathrm{Hom}(X, I)$, i.e: we want to see that f^* is surjective.
Consider a morphism $\alpha : X \to I$. Since f is a monomorphism and I is injective there exists a morphism $\beta : Y \to I$ such that $\beta \circ f = f^*(\beta) = \alpha$, making f^* surjective as wanted.

$\Longleftarrow$ Suppose $\mathrm{Hom}(-, I)$ is exact and consider a monomorphism $f : A \rightarrowtail B$ in A.
We have the following short exact sequence

$$0 \longrightarrow A \xrightarrow{f} B \longrightarrow \mathrm{coker}\, f \longrightarrow 0$$

Applying $\mathrm{Hom}(-, I)$ to this sequence we get

$$0 \longrightarrow \mathrm{Hom}(\mathrm{coker}\, f, I) \longrightarrow \mathrm{Hom}(B, I) \xrightarrow{f^*} \mathrm{Hom}(A, I) \longrightarrow 0$$

which is exact since $\mathrm{Hom}(-, I)$ is exact. In particular we have that f^* is surjective.
Therefore if $\gamma : A \to I$ is any morphism there exists $\beta \in \mathrm{Hom}(B, I)$ such that $\gamma = \beta \circ f$, which is exactly the universal property of injective objects.

□

Definition 7.2.2. Let A be an abelian category. We say that:

- A has **enough injectives** if for every object $A \in \mathsf{A}$ there exists an injective object $I \in \mathsf{A}$ and a monomorphism $A \rightarrowtail I$.
- A has **enough projectives** if for every object $A \in \mathsf{A}$ there exists a projective object $P \in \mathsf{A}$ and an epimorphism $P \twoheadrightarrow A$.

Lemma 7.2.1. Let

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

be a short exact sequence in C with X injective. Then the sequence splits.

Proof. By definition taking $k = \mathbb{1}_X$ there exists $h : Y \to X$ such that $k \circ f = \mathbb{1}_X$. Then we just need to apply Proposition 7.0.2. □

Remark 7.2.1. If $F : \mathsf{C} \to \mathsf{D}$ is an additive functor between abelian categories and the hypothesis of Lemma 7.2.1 are satisfied the sequence

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z) \longrightarrow 0$$

splits and in particular it is exact.

Lemma 7.2.2. Let $X, Z \in \mathsf{C}$. Then $X \oplus Z$ is injective if and only if X, Z are injective.

Proof. This follows from the fact that if F, G are two additive functors between abelian categories then $F \oplus G$ is exact if and only if F and G are exact. □

Proposition 7.2.2. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be a short exact sequence in C with X and Z injective. Then Y is injective.

Proof. By Lemma 7.2.1 since X is injective the sequence splits. Therefore the sequence is isomorphic to

$$0 \longrightarrow X \longrightarrow X \oplus Z \longrightarrow Z \longrightarrow 0$$

Finally since X and Z are injective by Lemma 7.2.2 $X \oplus Z \cong Y$ is injective. □

7.3 Complexes and cohomology

Recall that the categories $\mathsf{Comp}^*(\mathsf{C})$ are abelian for $* = +, -, \mathrm{b}, \mathrm{ub}$.

Definition 7.3.1 (Cohomology). Let

$$X^\bullet := \cdots \longrightarrow X^{n-1} \xrightarrow{\mathrm{d}_X^{n-1}} X^n \xrightarrow{\mathrm{d}_X^n} X^{n+1} \longrightarrow \cdots$$

be a complex in C (i.e: $X^\bullet \in \mathsf{Comp}(\mathsf{C})$). We define

$$\begin{aligned} Z^n(X^\bullet) &:= \ker \mathrm{d}_X^n \\ B^n(X^\bullet) &= \operatorname{Im} \mathrm{d}_X^{n-1} \\ \mathrm{H}^n(X^\bullet) &:= {}^{Z^n(X^\bullet)}\!/_{B^n(X^\bullet)} = \operatorname{coker}(B^n(X^\bullet) \to Z^n(X^\bullet)) \end{aligned}$$

We call $\mathrm{H}^n(X^\bullet)$ the n-th **cohomology** of $X^\bullet$.

Lemma 7.3.1. Let C be an abelian category and $f^\bullet : X^\bullet \to Y^\bullet$ a morphism in $\mathsf{Comp}(\mathsf{C})$ homotopic to zero. Then $\mathrm{H}^n f^\bullet : \mathrm{H}^n(X^\bullet) \to \mathrm{H}^n(Y^\bullet)$ is the zero morphism.
In particular the functor $\mathrm{H}^0 : \mathsf{Comp}(\mathsf{C}) \to \mathsf{C}$ extends to a functor $\mathrm{H}^0 : \mathrm{K}(\mathsf{C}) \to \mathsf{C}$

Proof. Let

$$f^n = s^{n+1} \circ \mathrm{d}_X^n + \mathrm{d}_Y^{n-1} \circ s^n$$

Then $\mathrm{d}_X^n = 0$ in $\ker \mathrm{d}_X^n$ and the image of d_Y^{n-1} is zero on $\ker \mathrm{d}_Y^n \big/ \operatorname{Im} \mathrm{d}_Y^{n-1}$.

Therefore $\mathrm{H}^n f : Z^n(X^\bullet) \big/ B^n(X^\bullet) \to Z^n(Y^\bullet) \big/ B^n(Y^\bullet)$ is the zero morphism. □

Definition 7.3.2 (Quasi-isomorphism)**.** Let C be an abelian category. A morphism $f^\bullet : X^\bullet \to Y^\bullet$ is a **quasi-isomorphism** (abbreviated q.i.s) if $\mathrm{H}^n f^\bullet$ is an isomorphism for all $n \in \mathbb{Z}$. In this case we say that $X^\bullet$ and $Y^\bullet$ are **quasi-isomorphic**.
In particular $X^\bullet$ is quasi-isomorphic to zero if and only if $X^\bullet$ is exact.

Remark 7.3.1. We have short exact sequences

$$X^{n-1} \xrightarrow{\mathrm{d}_X^{n-1}} \ker \mathrm{d}_X^n \longrightarrow \mathrm{H}^n(X^\bullet) \longrightarrow 0$$

$$0 \longrightarrow \mathrm{H}^n(X^\bullet) \longrightarrow \operatorname{coker} \mathrm{d}_X^{n-1} \xrightarrow{\mathrm{d}_X^n} X^{n+1}$$

We can glue this into a exact sequence

$$0 \longrightarrow \mathrm{H}^n(X^\bullet) \longrightarrow \operatorname{coker} \mathrm{d}_X^{n-1} \xrightarrow{\mathrm{d}_X^n} \ker \mathrm{d}_X^{n+1} \longrightarrow \mathrm{H}^{n+1}(X^\bullet) \longrightarrow 0$$

Theorem 7.3.1. Let

$$0 \longrightarrow X^\bullet \longrightarrow Y^\bullet \longrightarrow Z^\bullet \longrightarrow 0$$

be a short exact sequence in an abelian category $\mathsf{Comp}(\mathsf{C})$.

1. For all $k \in \mathbb{Z}$ the sequence $\mathrm{H}^k(X^\bullet) \to \mathrm{H}^k(Y^\bullet) \to \mathrm{H}^k(Z^\bullet)$ is exact.
2. For all $k \in \mathbb{Z}$ there exists $\delta^k : \mathrm{H}^k(Z^\bullet) \to \mathrm{H}^{k+1}(X^\bullet)$ making the long sequence

$$\cdots \longrightarrow \mathrm{H}^{k-1}(Y^\bullet) \longrightarrow \mathrm{H}^{k-1}(Z^\bullet) \xrightarrow{\delta^{k-1}} \mathrm{H}^k(X^\bullet) \longrightarrow \mathrm{H}^k(Y^\bullet) \longrightarrow \mathrm{H}^k(Z) \xrightarrow{\delta^k} \mathrm{H}^{k+1}(X^\bullet) \longrightarrow \mathrm{H}^{k+1}(Y^\bullet) \longrightarrow \cdots$$

exact. Furthermore δ^k can be constructed to be functorial with respect to short exact sequences of $\mathsf{Comp}(\mathsf{C})$.

Proof. Consider the diagram

$$\begin{array}{ccccccccc}
0 & & 0 & & 0 & & & & \\
\downarrow & & \downarrow & & \downarrow & & & & \\
\mathrm{H}^k(X^\bullet) & & \mathrm{H}^k(Y^\bullet) & & \mathrm{H}^k(Z^\bullet) & & & & \\
\downarrow & & \downarrow & & \downarrow & & & & \\
\operatorname{coker} \mathrm{d}_X^{k-1} & \longrightarrow & \operatorname{coker} \mathrm{d}_Y^{k-1} & \longrightarrow & \operatorname{coker} \mathrm{d}_Z^{k-1} & \longrightarrow & 0 & & \\
\downarrow & & \downarrow & & \downarrow & & & & \\
0 \longrightarrow \ker \mathrm{d}_X^{k+1} & \longrightarrow & \ker \mathrm{d}_Y^{k+1} & \longrightarrow & \ker \mathrm{d}_Z^{k+1} & & & & \\
\downarrow & & \downarrow & & \downarrow & & & & \\
\mathrm{H}^k(X^\bullet) & & \mathrm{H}^k(Y^\bullet) & & \mathrm{H}^k(Z^\bullet) & & & & \\
\downarrow & & \downarrow & & \downarrow & & & & \\
0 & & 0 & & 0 & & & &
\end{array}$$

where the columns are exact by Remark 7.3.1 and the rows are exact by hypothesis. Therefore the result follows from the Snake lemma. □

Corollary 7.3.1. For any morphism $f^\bullet : X^\bullet \to Y^\bullet$ in $\mathsf{Comp}(\mathsf{C})$ there exists a long exact sequence

$$\begin{array}{l}
\cdots \longrightarrow \mathrm{H}^{k-1}(Y^\bullet) \longrightarrow \mathrm{H}^{k-1}(\mathrm{Cone}(f^\bullet)) \\
\qquad\qquad \xrightarrow{\delta^{k-1}} \\
\mathrm{H}^k(X^\bullet) \longrightarrow \mathrm{H}^k(Y^\bullet) \longrightarrow \mathrm{H}^k(\mathrm{Cone}(f^\bullet)) \\
\qquad\qquad \xrightarrow{\delta^{k}} \\
\mathrm{H}^{k+1}(X^\bullet) \longrightarrow \mathrm{H}^{k+1}(Y^\bullet) \longrightarrow \cdots
\end{array}$$

Proof. We have a complex

$$0 \longrightarrow Y^\bullet \longrightarrow \mathrm{Cone}(f^\bullet) \longrightarrow X[1]^\bullet$$

Since in degree n this gives the split sequence

$$0 \longrightarrow Y^n \longrightarrow Y^n \oplus X^{n+1} \longrightarrow X^{n+1}$$

the complex is exact. Therefore the result follows from Theorem 7.3.1. □

7.3.1 Truncation functors

Definition 7.3.3 (Truncation functor)**.** The **truncation functors**

$$\begin{aligned}
\tau^{\leq n}, \widetilde{\tau}^{\leq n} &: \mathsf{Comp}(\mathsf{C}) \to \mathsf{Comp}^-(\mathsf{C}) \\
\tau^{\geq n}, \widetilde{\tau}^{\geq n} &: \mathsf{Comp}(\mathsf{C}) \to \mathsf{Comp}^+(\mathsf{C})
\end{aligned}$$

are defined as, for any complex $X^\bullet \in \mathsf{Comp}(\mathsf{C})$

$$\tau^{\leq n}(X^\bullet) := \quad \cdots \longrightarrow X^{n-2} \longrightarrow X^{n-1} \longrightarrow \ker \mathrm{d}_X^n \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots$$

$$\widetilde{\tau}^{\leq n}(X^\bullet) := \quad \cdots \longrightarrow X^{n-2} \longrightarrow X^{n-1} \longrightarrow X^n \longrightarrow \operatorname{Im} \mathrm{d}_X^n \longrightarrow 0 \longrightarrow \cdots$$

$$\tau^{\geq n}(X^\bullet) := \quad \cdots \longrightarrow 0 \longrightarrow 0 \longrightarrow \operatorname{coker} \mathrm{d}_X^{n-1} \longrightarrow X^{n+1} \longrightarrow X^{n+2} \longrightarrow \cdots$$

$$\widetilde{\tau}^{\geq n}(X^\bullet) := \quad \cdots \longrightarrow 0 \longrightarrow \operatorname{Im} \mathrm{d}_X^{n-1} \longrightarrow X^n \longrightarrow X^{n+1} \longrightarrow X^{n+2} \longrightarrow \cdots$$

Remark 7.3.2. For any complex $X^\bullet \in \mathsf{Comp}(\mathsf{C})$ we have a chain of morphisms

$$\tau^{\leq n}(X^\bullet) \to \widetilde{\tau}^{\leq n}(X^\bullet) \to X^\bullet \to \widetilde{\tau}^{\geq n}(X^\bullet) \to \tau^{\geq n}(X^\bullet)$$

in $\mathsf{Comp}(\mathsf{C})$ given by

$$\begin{array}{lccccccccc}
\tau^{\leq n}(X^\bullet) = & \cdots \to & X^{n-2} & \to & X^{n-1} & \to & \ker \mathrm{d}_X^n & \to & 0 & \to & 0 & \to \cdots \\
\downarrow & & \| & & \| & & \downarrow & & \downarrow & & \| & \\
\widetilde{\tau}^{\leq n}(X^\bullet) = & \cdots \to & X^{n-2} & \to & X^{n-1} & \to & X^n & \to & \operatorname{Im} \mathrm{d}_X^n & \to & 0 & \to \cdots \\
\downarrow & & \| & & \| & & \| & & \downarrow & & \downarrow & \\
X^\bullet = & \cdots \to & X^{n-2} & \to & X^{n-1} & \to & X^n & \to & X^{n+1} & \to & X^{n+2} & \to \cdots \\
\downarrow & & \downarrow & & \downarrow & & \| & & \| & & \| & \\
\widetilde{\tau}^{\geq n}(X^\bullet) = & \cdots \to & 0 & \to & \operatorname{Im} \mathrm{d}_X^{n-1} & \to & X^n & \to & X^{n+1} & \to & X^{n+2} & \to \cdots \\
\downarrow & & \| & & \downarrow & & \downarrow & & \| & & \| & \\
\tau^{\geq n}(X^\bullet) = & \cdots \to & 0 & \to & 0 & \to & \operatorname{coker} \mathrm{d}_X^{n-1} & \to & X^{n+1} & \to & X^{n+2} & \to \cdots
\end{array}$$

Therefore we get exact sequences

$$0 \longrightarrow \widetilde{\tau}^{\leq n-1}(X^\bullet) \longrightarrow \tau^{\leq n}(X^\bullet) \longrightarrow \mathrm{H}^n(X^\bullet)[-n] \longrightarrow 0$$

$$0 \longrightarrow \mathrm{H}^n(X^\bullet)[-n] \longrightarrow \tau^{\geq n}(X^\bullet) \longrightarrow \widetilde{\tau}^{\geq n+1}(X^\bullet) \longrightarrow 0$$

$$0 \longrightarrow \tau^{\leq n}(X^\bullet) \longrightarrow X^\bullet \longrightarrow \widetilde{\tau}^{n+1}(X^\bullet) \longrightarrow 0$$

$$0 \longrightarrow \widetilde{\tau}^{\leq n-1}(X^\bullet) \longrightarrow X^\bullet \longrightarrow \tau^n(X^\bullet) \longrightarrow 0$$

Furthermore we have

$$\mathrm{H}^j(\tau^{\leq n}(X^\bullet)) \cong \mathrm{H}^j(\widetilde{\tau}^{\leq n}(X^\bullet)) = \begin{cases} \mathrm{H}^j(X^\bullet) & j \leq n \\ 0 & j > n \end{cases}$$

$$\mathrm{H}^j(\tau^{\geq n}(X^\bullet)) \cong \mathrm{H}^j(\widetilde{\tau}^{\geq n}(X^\bullet)) = \begin{cases} \mathrm{H}^j(X^\bullet) & j \geq n \\ 0 & j < n \end{cases}$$

7.4 Double complexes

Recall the finiteness condition we are assuming for all double complexes.

Lemma 7.4.1. The functor $\mathrm{tot} : \mathsf{Comp}^2(\mathsf{C}) \to \mathsf{Comp}(\mathsf{C})$ is exact.

Proof. Since direct products are exact and $\mathrm{tot}(X^{\bullet,\bullet})^n = \bigoplus_{p+q=n} X^{p,q}$, the result follows. □

Recall that we have defined functors $F_I, F_{II} : \mathsf{Comp}^2(\mathsf{C}) \to \mathsf{Comp}(\mathsf{Comp}(\mathsf{C}))$. Since C is category we can combine these with the truncation functors. For example we define $\tau_I^{\leq n} := F_I^{-1} \circ \tau^{\leq n} \circ F_I$, i.e:

$$\tau_I^{\leq n}(X^{\bullet,\bullet}) := \cdots \to F_I(X^{\bullet,\bullet})^n \to \ker \mathrm{d}^n_{F_I(X^{\bullet,\bullet})} \to 0$$

and similary for $\widetilde{\tau}_I^{\leq n}, \tau_I^{\geq n}, \widetilde{\tau}_I^{\geq n}$.

For $n \in \mathbb{Z}$ we define the complex

$$\mathrm{H}_I^n(X^{\bullet,\bullet}) := \mathrm{H}^n(F_I(X^{\bullet,\bullet}))$$

and the double complex

$$\mathrm{H}_I(X^{\bullet,\bullet}) := \cdots \to \mathrm{H}^n(F_I(X^{\bullet,\bullet})) \xrightarrow{0} \mathrm{H}^{n+1}(F_I(X^{\bullet,\bullet})) \to \cdots$$

We get analogous definitions using F_{II} instead of F_I.

Remark 7.4.1. By Remark 7.3.2 and Lemma 7.4.1 we get that the natural morphism

$$\mathrm{tot}(\tau_I^{\leq n}(X^{\bullet,\bullet})) \to \mathrm{tot}(\widetilde{\tau}_I^{\leq n}(X^{\bullet,\bullet}))$$

is an isomorphism for all $n \in \mathbb{N}$.
Furthermore we get the exact sequence in $\mathsf{Comp}(\mathsf{C})$

$$0 \longrightarrow \mathrm{tot}(\widetilde{\tau}_I^{\leq n-1}(X^{\bullet,\bullet})) \longrightarrow \mathrm{tot}(\tau_I^{\leq n}(X^{\bullet,\bullet})) \longrightarrow \mathrm{H}_I^n(X^{\bullet,\bullet})[-n] \longrightarrow 0$$

which is functorial with respect to $X^{\bullet,\bullet}$.

Theorem 7.4.1. Let $f^{\bullet,\bullet} : X^{\bullet,\bullet} \to Y^{\bullet,\bullet}$ be a morphism of double complexes. If $f^{\bullet,\bullet}$ induces an isomorphism $\mathrm{H}_{II}\,\mathrm{H}_I(X^{\bullet,\bullet}) \cong \mathrm{H}_{II}\,\mathrm{H}_I(Y^{\bullet,\bullet})$ then $\mathrm{tot}\, f : \mathrm{tot}(X^{\bullet,\bullet}) \to \mathrm{tot}(Y^{\bullet,\bullet})$ is a quasi-isomorphism.

Proof. Note that the hypothesis is equivalent to $\mathrm{H}_I^n f : \mathrm{H}_I^n(X^{\bullet,\bullet}) \to \mathrm{H}_I^n(Y^{\bullet,\bullet})$ being a quasi-isomorphism for all $n \in \mathbb{N}$.

Since by Remark 7.3.2 we have

$$\mathrm{H}_I^n(\tau_I^{\geq p}(X^{\bullet,\bullet})) \cong \begin{cases} \mathrm{H}_I^n(X^{\bullet,\bullet}) & n \geq p \\ 0 & n < p \end{cases}$$

for all $p \in \mathbb{Z}$ we get the isomorphism

$$\mathrm{H}_{II}\,\mathrm{H}_I(\tau_I^{\geq p}(X^{\bullet,\bullet})) \xrightarrow{\sim} \mathrm{H}_{II}\,\mathrm{H}_I(\tau_I^{\geq p}(Y^{\bullet,\bullet}))$$

For a fixed $n \in \mathbb{Z}$ we have, for $p \ll 0$

$$\mathrm{H}^n(\mathrm{tot}(X^{\bullet,\bullet})) \cong \mathrm{H}^n(\mathrm{tot}(\tau_I^{\geq p}(X^{\bullet,\bullet})))$$

and analogously for $Y^{\bullet,\bullet}$. Therefore by replacing $X^{\bullet,\bullet}$ and $Y^{\bullet,\bullet}$ by $\tau_I^{\geq p}(X^{\bullet,\bullet})$ and $\tau_I^{\geq p}(Y^{\bullet,\bullet})$ we may assume that $X_I^n = Y_I^n = 0$ for $n \ll 0$.

Consider now the commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathrm{tot}(\widetilde{\tau}_I^{\leq n-1}(X^{\bullet,\bullet})) & \longrightarrow & \mathrm{tot}(\tau_I^{\leq n}(X^{\bullet,\bullet})) & \longrightarrow & \mathrm{H}_I^n(X^{\bullet,\bullet})[-n] & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \mathrm{tot}(\widetilde{\tau}_I^{\leq n-1} f)} & & \downarrow{\scriptstyle \mathrm{tot}(\tau_I^{\leq n} f)} & & \downarrow{\scriptstyle (\mathrm{H}_I^n f)[-n]} & & \\
0 & \longrightarrow & \mathrm{tot}(\widetilde{\tau}_I^{\leq n-1}(Y^{\bullet,\bullet})) & \longrightarrow & \mathrm{tot}(\tau_I^{\leq n}(Y^{\bullet,\bullet})) & \longrightarrow & \mathrm{H}_I^n(Y^{\bullet,\bullet})[-n] & \longrightarrow & 0
\end{array}$$

Note that by hypothesis the right vertical arrow is a quasi-isomorphism and for $n \ll 0$ the left arrow is also a quasi-isomorphism. Therefore by induction and using $\mathrm{tot}(\tau_I^{\leq n}(X^{\bullet,\bullet})) \cong \mathrm{tot}(\widetilde{\tau}_I^{\leq n}(X^{\bullet,\bullet}))$ we get that all vertical arrows are quasi-isomorphisms.

Finally the results follows since for a fixed $n \in \mathbb{Z}$ and $p \gg 0$ we have

$$\mathrm{H}^n(\mathrm{tot}(X^{\bullet,\bullet})) \cong \mathrm{H}^n(\mathrm{tot}(\tau_I^{\leq p}(X^{\bullet,\bullet})))$$

□

Corollary 7.4.1. Let $X^{\bullet,\bullet}$ be a double complex in C. If $\mathrm{H}_I(X^{\bullet,\bullet}) = 0$ then $\mathrm{tot}(X^{\bullet,\bullet})$ is quasi-isomorphic to 0.

Proof. Take $Y^{\bullet,\bullet} = 0$ in Theorem 7.4.1. □

Corollary 7.4.2. Let $X^{\bullet,\bullet}$ be a double complex in C. If all rows $X^{j,\bullet}$ are exact for $j \neq n$ then $\mathrm{tot}(X^{\bullet,\bullet})$ is quasi-isomorphic to $X^{n,\bullet}[-n]$.

Proof. Apply Theorem 7.4.1 first to $\sigma_I^{\geq n}(X^{\bullet,\bullet}) \to X^{\bullet,\bullet}$ and then to $\sigma_I^{\geq n}(X^{\bullet,\bullet}) \to \sigma_I^{\leq n}\sigma_I^{\geq n}(X^{\bullet,\bullet}) \cong X^{n,\bullet}[-n]$. □

Corollary 7.4.3. Let $X^{\bullet,\bullet}$ be a double complex in C. If all rows $X^{j,\bullet}$ and columns $X^{\bullet,j}$ are zero for $j < 0$ and exact for $j > 0$ then $\mathrm{H}^p(X^{0,\bullet}) \cong \mathrm{H}^p(X^{\bullet,0})$ for all $p \in \mathbb{Z}$.

Proof. By hypothesis $X^{0,\bullet}$ and $X^{\bullet,0}$ are quasi-isomorphic to 0. □

Chapter 8

Triangulated categories

Definition 8.0.1 (Triangle). Let C be an additive category and $T : \mathsf{C} \to \mathsf{C}$ a equivalence. A **triangle** in (C, T) is a sequence of morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} T(X)$$

A morphism of triangles is a commutative diagram

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \xrightarrow{h} & T(X) \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle T\alpha} \\ X' & \xrightarrow[f']{} & Y' & \xrightarrow[g']{} & Z' & \xrightarrow[h']{} & T(X') \end{array}$$

Definition 8.0.2 (Triangulated category). A **triangulated category** is an additive category C endowed with an equivalence $T : \mathsf{C} \to \mathsf{C}$ called the *shift* or *translation* functor and a collection of triangles called *distinguished triangles* (d.t. for short) that satisfy

Tr.0) A triangle isomorphic to a d.t. is a d.t.

Tr.1) The triangle $X \xrightarrow{\mathbb{1}_X} X \to 0 \to T(X)$ is a d.t.

Tr.2) For all $f : X \to Y$ there exists a d.t $X \xrightarrow{f} Y \xrightarrow{g} Z \to T(X)$.

Tr.3) A triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} T(X)$ is a d.t. if and only if $Y \xrightarrow{g} Z \xrightarrow{h} T(X) \xrightarrow{-Tf} T(Y)$ is a d.t.

Tr.4) Given two d.t. $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} T(X)$ and $X' \xrightarrow{f'} Y' \xrightarrow{g'} Z' \xrightarrow{h'} T(X')$ and morphisms $\alpha : X \to X', \beta : Y \to Y'$ with $f' \circ \alpha = \beta \circ f$ there exists a morphism $\gamma : Z \to Z'$ giving rise to a morphism of d.t.

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \xrightarrow{h} & T(X) \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \vdots{\scriptstyle \gamma} & & \downarrow{\scriptstyle T\alpha} \\ X' & \xrightarrow[f']{} & Y' & \xrightarrow[g']{} & Z' & \xrightarrow[h']{} & T(X') \end{array}$$

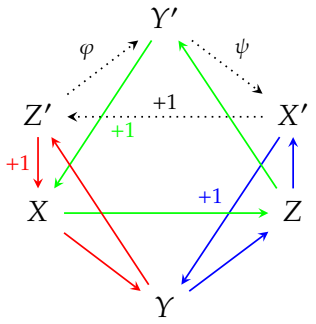
Tr.5) Given three d.t.

$$X \xrightarrow{f} Y \xrightarrow{g} Z' \longrightarrow T(X)$$

$$Y \xrightarrow{g} Z \xrightarrow{k} X' \longrightarrow T(Y)$$

$$X \xrightarrow{g\circ f} Z \xrightarrow{\ell} Y' \longrightarrow T(X)$$

There exists a d.t. $Z' \xrightarrow{\varphi} Y' \xrightarrow{\psi} X' \to T(Z)$ making the diagram below commute



where $A \xrightarrow{+1} B$ means $A \to T(B)$.

8.1 Triangulated and cohomological functors

Definition 8.1.1 (Triangulated functor). Let $(\mathsf{C}, T), (\mathsf{C}', T')$ be two triangulated categories. A **triangulated functor** $F : (\mathsf{C}, T) \to (\mathsf{C}', T')$ is an additive functor $F : \mathsf{C} \to \mathsf{C}'$ such that $F \circ T \cong T' \circ F$ and which sends distinguished triangles to distinguished triangles.

Definition 8.1.2 (Triangulated subcategory). A **triangulated subcategory** C' of a triangulated category C is a subcategory C' of C that is triangulated and such that the inclussion functor $\mathsf{C}' \hookrightarrow \mathsf{C}$ is triangulated.

Definition 8.1.3 (Cohomological functor). Let (C, T) be a triangulated category and D an abelian category. An additive functor $F : \mathsf{C} \to \mathsf{D}$ is **cohomological** if for any d.t. $X \to Y \to Z \to T(X)$ in C the sequence $F(X) \to F(Y) \to F(Z)$ is exact in D.

Remark 8.1.1. By Tr.3 any cohomological functor F gives rise to a long exact sequence

$$\cdots \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z) \longrightarrow FT(X) \longrightarrow FT(Y) \longrightarrow FT(Z) \longrightarrow \cdots$$

Proposition 8.1.1. Let (C, T) be a triangulated category.

1. If $X \xrightarrow{f} Y \xrightarrow{g} Z \to T(X)$ is a d.t. then $g \circ f = 0$.
2. For any $W \in \mathsf{C}$ the Hom functors $\mathrm{Hom}_{\mathsf{C}}(W, -)$ and $\mathrm{Hom}_{\mathsf{C}}(-, W)$ are cohomological.
   This means that if $\varphi : W \to Y$ (resp. $\varphi : Y \to W$) satisfies $g \circ \varphi = 0$ (resp. $\varphi \circ f = 0$) then φ factorizes through f (resp. through g).

Proof.

1. By Tr.1 and Tr.4 we have a commutative diagram
$$\begin{array}{ccccccc} X & \xrightarrow{\mathbb{1}_X} & X & \longrightarrow & 0 & \longrightarrow & T(X) \\ \downarrow{\scriptstyle \mathbb{1}_X} & & \downarrow{\scriptstyle f} & & \downarrow & & \downarrow{\scriptstyle \mathbb{1}_{T(X)}} \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & T(X) \end{array}$$
Therefore by the commutativity of the middle square $g \circ f = 0$.
2. We will check $\mathrm{Hom}(W, -)$, the other case is analogous.
Let $X \xrightarrow{f} Y \xrightarrow{g} Z \to T(X)$ be a d.t. and $W \in \mathsf{D}$. We want to show that
$$\mathrm{Hom}(W, X) \xrightarrow{f\circ} \mathrm{Hom}(W, Y) \xrightarrow{g\circ} \mathrm{Hom}(W, Z)$$
is exact, i.e: for all $\varphi : W \to Y$ such that $g \circ \varphi = 0$ there exists $\psi : W \to X$ such that $\varphi = f \circ \psi$. In other words, that the dotted arrow below may be completed
$$\begin{array}{ccccccc} W & \xrightarrow{\mathbb{1}_X} & W & \longrightarrow & 0 & \longrightarrow & T(W) \\ \vdots & & \downarrow{\scriptstyle \varphi} & & \downarrow & & \vdots \\ X & \xrightarrow[f]{} & Y & \xrightarrow[g]{} & Z & \longrightarrow & T(X) \end{array}$$
This follows directly from Tr.1 and Tr.4.

□

Proposition 8.1.2. Consider a morphism of d.t. in a triangulated category (C, T)
$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \xrightarrow{h} & T(X) \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle T\alpha} \\ X' & \xrightarrow[f']{} & Y' & \xrightarrow[g']{} & Z' & \xrightarrow[h']{} & T(X') \end{array}$$
If α and β are isomorphisms then so is γ.

Proof. Let $W \in \mathsf{C}$. If we apply $\mathrm{Hom}(W, -)$ to the above diagram, denoting by $\widetilde{X} := \mathrm{Hom}(W, X), \widetilde{\alpha} := \mathrm{Hom}(W, \alpha), \ldots$, we get a commutative diagram
$$\begin{array}{ccccccccc} \widetilde{X} & \xrightarrow{\widetilde{f}} & \widetilde{Y} & \xrightarrow{\widetilde{g}} & \widetilde{Z} & \xrightarrow{\widetilde{h}} & \widetilde{T(X)} & \xrightarrow{\widetilde{Tf}} & \widetilde{T(Y)} \\ \downarrow{\scriptstyle \widetilde{\alpha}} & & \downarrow{\scriptstyle \widetilde{\beta}} & & \downarrow{\scriptstyle \widetilde{\gamma}} & & \downarrow{\scriptstyle \widetilde{T\alpha}} & & \downarrow{\scriptstyle \widetilde{T\beta}} \\ \widetilde{X'} & \xrightarrow[\widetilde{f'}]{} & \widetilde{Y'} & \xrightarrow[\widetilde{g'}]{} & \widetilde{Z'} & \xrightarrow[\widetilde{h'}]{} & \widetilde{T(X')} & \xrightarrow[\widetilde{Tf'}]{} & \widetilde{T(Y')} \end{array}$$
By Proposition 8.1.1 the rows are exact and $\widetilde{\alpha}, \widetilde{\beta}, \widetilde{T(\alpha)}, \widetilde{T(\beta)}$ are isomorphisms. Therefore by the lemma $\widetilde{\gamma} = \mathrm{Hom}(W, \gamma) : \mathrm{Hom}(W, Z) \to \mathrm{Hom}(W, Z')$ is an isomorphism. Applying the Yoneda lemma we get that γ is an isomorphism. □

Corollary 8.1.1. Let C' be a full triangulated subcategory of a triangulated category C.

1. Consider a triangle $X \xrightarrow{f} Y \to Z \to T(X)$ in C'. If this triangle is distinguished in C then it is distinguished in C'.
2. Consider a d.t. $X \xrightarrow{f} Y \to Z \to T(X)$ in C with $X, Y \in \mathsf{C}'$. Then there exists $Z' \in \mathsf{C}'$ and an isomorphism $Z \cong Z'$.

Proof. By Tr.2 there exists a d.t. $X \xrightarrow{f} Y \to Z' \to T(X)$ in C'. By Tr.4 there exists a morphism of d.t. in C

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \longrightarrow & Z' & \longrightarrow & T(X) \\ \downarrow{\scriptstyle \mathbb{1}_X} & & \downarrow{\scriptstyle \mathbb{1}_Y} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \mathbb{1}_{T(X)}} \\ X & \xrightarrow[f]{} & Y & \longrightarrow & Z & \longrightarrow & T(X) \end{array}$$

By Proposition 8.1.2 this morphism is an isomorphism. Therefore $Z \cong Z'$ and the original triangle is distinguished. □

8.2 Localization of triangulated categories

Recall the definitions of chapter 5.

Definition 8.2.1. A full subcategory C' of a category C is **saturated** if $X \in \mathsf{C}'$ and $Y \cong X$ implies $Y \in \mathsf{C}'$.

Definition 8.2.2 (Null system)**.** A **null system** $\mathcal{N}$ in a triangulated category C is a full triangulated subcategory of C that satisies

- **N.1)** $0 \in \mathcal{N}$.
- **N.2)** $X \in \mathcal{N}$ if and only if $T(X) \in \mathcal{N}$.
- **N.3)** If $X \to Y \to Z \to T(X)$ is a d.t. in D and $X, Y \in \mathcal{N}$ then $Z \in \mathcal{N}$.

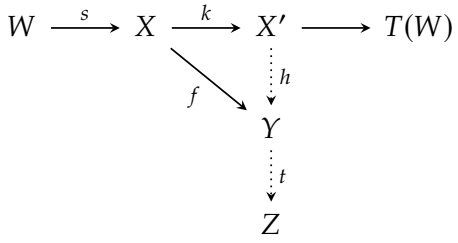
To every null system we associate the following family of morphisms

$$\mathcal{S} := \{f : X \to Y \mid \exists X \xrightarrow{f} Y \to Z \to T(X) \text{ d.t. with } Z \in \mathcal{N}\}$$

Lemma 8.2.1. $\mathcal{S}$ is a right and left multiplicative system.

Proof. We will show that $\mathcal{S}$ is a right multiplicative system. The other one is analogous.

- **S.1** Since $X \xrightarrow{\mathbb{1}_X} X \to 0 \to T(X)$ is a d.t., $\mathbb{1}_X \in \mathcal{S}$.
- **S.2** Direct from the octahedral axiom Tr.5.
- **S.3** There exists a d.t.
$$W \xrightarrow{h} X \xrightarrow{s} X' \longrightarrow T(W)$$
with $W \in \mathcal{N}$. The morphism $h \circ f : W \to Y$ gives rise to a d.t.
$$W \longrightarrow Y \xrightarrow{t} Y' \longrightarrow T(W)$$
so $t \in \mathcal{S}$. Therefore by Tr.4 there exists a morphism of triangles
$$\begin{array}{ccccccc} W & \xrightarrow{h} & X & \xrightarrow{s} & X' & \longrightarrow & T(W) \\ \| & & \downarrow{\scriptstyle f} & & \vdots{\scriptstyle g} & & \| \\ W & \longrightarrow & Y & \xrightarrow[t]{} & Y' & \longrightarrow & T(W) \end{array}$$
- **S.4** By replacing f with $f - g$ it is enough to check that there exists $s : W \to X \in \mathcal{S}$ such that if $f \circ s = 0$ then there exists $t : Y \to Z \in \mathcal{S}$ such that $t \circ f = 0$. Consider the diagram



where the top row is a d.t. Since the Hom functors are cohomological the sequence

$$\mathrm{Hom}(X', Y) \xrightarrow{\circ k} \mathrm{Hom}(X, Y) \xrightarrow{\circ s} \mathrm{Hom}(W, Y)$$

is exact. Since $f \circ s = 0$ the dotted arrow h may be completed making the diagram commutative. If we then embed h into a d.t. $X' \xrightarrow{h} Y \xrightarrow{t} Z \to T(X')$ we get an arrow t such that $t \circ h = 0$. Finally we get $t \circ f = 0$ and, since $X' \in \mathcal{N}$, $t \in \mathcal{S}$.

□

The proof of the following theorem is straightforward but rather tedious, so we refer to [KS06] for a full proof.

Theorem 8.2.1. Let C be a triangulated category, $\mathcal{N}$ a null system in C and $\mathcal{S}$ its associated right multiplicative system. Let $\mathsf{C}[\mathcal{S}^{-1}]$ be the localization of C by $\mathcal{S}$ and Q the localization functor.

1. $\mathsf{C}[\mathcal{S}^{-1}]$ is an additive category endowed with an equivalence (the image of T), which we still denote by T.
2. If we define a d.t. in $\mathsf{C}[\mathcal{S}^{-1}]$ as being isomorphic to the image of a d.t. by Q, then $\mathsf{C}[\mathcal{S}^{-1}]$ is triangulated.
3. If $X \in \mathcal{N}$ then $Q(X) \cong 0$.
4. Let $F : \mathsf{C} \to \mathsf{C}'$ be a functor of triangulated categories such that $F(X) \cong 0$ for all $X \in \mathcal{N}$. Then F factors uniquely through Q.

Proposition 8.2.1. Let C be a triangulated category, $\mathcal{N}$ a null system in C and C' a full triangulated subcategory of C. Assume that either of this conditions hold

1. Any morphism $Y \to Z$ with $Y \in \mathsf{C}'$ and $Z \in \mathcal{N}$ factorizes as $Y \to Z' \to Z$ with $Z' \in \mathcal{N} \cap \mathsf{C}'$.
2. Any morphism $Z \to Y$ with $Y \in \mathsf{C}'$ and $Z \in \mathcal{N}$ factorizes as $Z \to Z' \to Y$ with $Z' \in \mathcal{N} \cap \mathsf{C}'$.

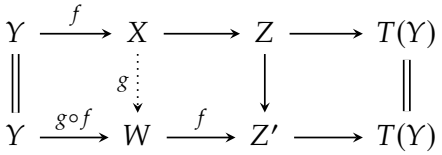
Then the functor $\mathsf{C}'[(\mathcal{N} \cap \mathsf{C}')^{-1}] \to \mathsf{C}[\mathcal{N}^{-1}]$ is fully faithful.

Proof. We will suppose 2. The proof assuming 1 is analogous.

Let $f : Y \to X$ be a morphism in C' with $Y \in \mathsf{C}'$. We want to show that there exists $g : X \to W$ with $W \in \mathsf{C}'$ and $g \circ f \in \mathsf{C}'$. We embed f into a d.t.

$$Y \xrightarrow{f} X \longrightarrow Z \longrightarrow T(Y)$$

with $Z \in \mathcal{N}$. By hypothesis the morphism $Z \to T(Y)$ factorizes through an object $Z' \in \mathcal{N} \cap \mathsf{C}'$. If we embed $Z' \to T(Y)$ into a d.t. we obtain morphism of d.t.



By Tr.4 the dotted arrow g may be completed and since $Z' \in \mathcal{N}$ we have $g \circ f \in \mathsf{C}'$. □

Proposition 8.2.2. Let C be triangulated, $\mathcal{N}$ a null system and C' a full triangulated subcategory of C. Assume that

1. For any $X \in \mathsf{C}$ there exists a d.t. $X \to Y \to Z \to T(X)$ with $Z \in \mathcal{N}$ and $Y \in \mathsf{C}'$.
2. For any $X \in \mathsf{C}$ there exists a d.t. $Y \to X \to Z \to T(Y)$ with $Z \in \mathcal{N}$ and $Y \in \mathsf{C}'$.

Then $\mathsf{C}'[(\mathcal{N} \cap \mathsf{C}')^{-1}] \to \mathsf{C}[\mathcal{N}^{-1}]$ is an equivalence of categories.

Proof. Direct from Corollary 5.1.2. □

Theorem 8.2.2. Let $F : \mathsf{C} \to \mathsf{D}$ be a functor between triangulated categories. Let $\mathcal{N}$ be a null system in C and C' a full triangulated subcategory of C. Suppose

1. For all $X \in \mathsf{C}$ there exists a d.t. $X \to Y \to Z \to T(X)$ with $Z \in \mathcal{N}$ and $Y \in \mathsf{C}'$.
2. $F(Y) \cong 0$ for all $Y \in \mathcal{N} \cap \mathsf{C}'$.

Then F is right localizable and its localizable is a triangulated functor.

Proof. Direct from Proposition 5.2.1. □

Theorem 8.2.3. Let $F : \mathsf{C} \times \mathsf{C}' \to \mathsf{C}''$ be a triangulated bifunctor. Let $\mathcal{N}, \mathcal{N}$ be null systems of C and C' respectively and let D, D' be full triangulated subcategories of C and C' respectively. Suppose

1. For any $X \in \mathsf{C}$ there exists a d.t. $X \to Y \to Z \to T(X)$ with $Z \in \mathcal{N}$ and $Y \in \mathsf{D}$.
2. For any $X' \in \mathsf{C}'$ there exists a d.t. $X' \to Y' \to Z' \to T(X')$ with $Z' \in \mathcal{N}$ and $Y' \in \mathsf{D}$.
3. For any $Y \in \mathsf{D}$ and $Y' \in \mathsf{D}' \cap \mathcal{N}'$, $F(Y, Y') = 0$.
4. For any $Y' \in \mathsf{D}'$ and $Y \in \mathsf{D} \cap \mathcal{N}$, $F(Y, Y') = 0$.

Then F is right localizable.

Proof. Direct from Proposition 5.2.1. □

To finish this chapter, we state how localization interacts with direct sums.

Proposition 8.2.3 ([KS06, Propositions 10.1.19, 10.2.8])**.** Let C be a triangulated category admitting small direct sums.

1. The localization functor commutes with small direct sums and a small direct sums of a d.t. is a d.t.
2. Let $\mathcal{N}$ be a null system in C stable under small direct sums. Then $\mathsf{C}[\mathcal{N}^{-1}]$ admits small direct sums and the localization functor commutes with them.

Chapter 9

Derived categories

Theorem 9.0.1 ([KS06, Theorem 11.2.6]). Let C be an additive category. Consider homotopy category $\mathrm{K}(\mathsf{C})$ together with the shift functor $[1] : X \mapsto X[1]$ and a triangle $X \to Y \to Z \to X[1]$ being distinguished if and only if it is isomorphic to a triangle of the form $X \to Y \to \mathrm{Cone}(f) \to X[1]$.
Then $\mathrm{K}(\mathsf{C})$ is a triangulated category.

In the rest of this chapter C will be an abelian category.

Proposition 9.0.1. The cohomology functor $\mathrm{H}^k : \mathrm{K}(\mathsf{C}) \to \mathsf{C}$ is cohomological for all $k \in \mathbb{Z}$.

Proof. Let $X \xrightarrow{f} Y \to Z \xrightarrow{+1}$ be a d.t. By definition it is isomorphic to

$$X \longrightarrow Y \xrightarrow{\alpha f} \mathrm{Cone}(f) \xrightarrow{\beta f} X[1] \xrightarrow{+1}$$

Since

$$0 \longrightarrow Y \xrightarrow{\alpha f} \mathrm{Cone}(f) \xrightarrow{\beta f} X[1] \longrightarrow 0$$

is exact, by Theorem 7.3.1 the sequence

$$\mathrm{H}^k(Y) \longrightarrow \mathrm{H}^k \mathrm{Cone}(f) \longrightarrow \mathrm{H}^{k+1}(X)$$

is exact. Therefore $\mathrm{H}^k(Y) \to \mathrm{H}^k(Z) \to \mathrm{H}^{k+1}(x)$ is exact. □

Corollary 9.0.1. Let $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ be a short exact sequence in $\mathsf{Comp}(\mathsf{C})$.
The morphism $\varphi : \mathrm{Cone}(f) \to Z, \varphi^n := (0, g)$ is a quasi-isomorphism.

Proof. Consider the exact sequence in $\mathsf{Comp}(\mathsf{C})$

$$0 \longrightarrow \mathrm{Cone}(\mathbb{1}_X) \xrightarrow{\gamma} \mathrm{Cone}(f) \xrightarrow{\varphi} Z \longrightarrow 0$$

where $\gamma : \mathrm{Cone}(\mathbb{1}_X) \to \mathrm{Cone}(f)$, $\gamma^n := \begin{pmatrix} \mathbb{1}_{X^{n+1}} & 0 \\ 0 & f^n \end{pmatrix}$. Since $\mathrm{H}^H(\mathrm{Cone}(\mathbb{1}_X)) = 0$ for all $k \in \mathbb{Z}$ we get the result from Proposition 9.0.1. □

We want to localize $\mathrm{K}(\mathsf{C})$ *inverting* the quasi-isomorphisms. Define

$$N(\mathsf{C}) := \{X \in \mathrm{K}(\mathsf{C}) \mid \mathrm{H}^k(X) = 0 \quad \forall k \in \mathbb{Z}\}$$

We can also define $N^*(\mathsf{C}) := N(\mathsf{C}) \cap \mathrm{K}^*(\mathsf{C})$ for $* = +, -, \mathrm{b}$.

Lemma 9.0.1. $N^*(\mathsf{C})$ is a null system.

Proof.

N.1 Since $\mathrm{H}^k(0) = 0$ for all $k \in \mathbb{Z}$, $0 \in N^*(\mathsf{C})$.

N.2 We have

$$\begin{aligned} X \in N^*(\mathsf{C}) &\iff \mathrm{H}^k(X) = 0 \quad \forall k \in \mathbb{Z} \\ &\iff \mathrm{H}^{k+1}(X) = 0 \quad \forall k \in \mathbb{Z} \\ &\iff \mathrm{H}^k(X[1]) = 0 \quad \forall k \in \mathbb{Z} \\ &\iff X[1] \in N^*(\mathsf{C}) \end{aligned}$$

N.3 Let $X \to Y \to Z \to X[1]$ be a d.t. with $X, Y \in N^*(\mathsf{C})$. Since $\mathrm{H}^k(X) = \mathrm{H}^k(Y) = 0$ for all $k \in \mathbb{Z}$ the long exact sequence in cohomology is

$$\cdots \longrightarrow 0 \longrightarrow \mathrm{H}^k(Z) \longrightarrow 0 \longrightarrow \cdots$$

Therefore $\mathrm{H}^k(Z) = 0$ for all $k \in \mathbb{Z}$, i.e: $Z \in N^*(\mathsf{C})$.

□

Let

$$\mathcal{S}^*(\mathsf{C}) := \{f : X \to Y \mid \exists X \xrightarrow{f} Y \to Z \to X[1] \text{d.t. with } Z \in N^*(\mathsf{C})\}$$

be the multiplicative system associated to $N^*(\mathsf{C})$.

Lemma 9.0.2. For $* = +, -, \mathrm{b}, \mathrm{ub}$, the multiplicative system $\mathcal{S}^*(\mathsf{C})$ is saturated.

Proof. It is enough to see the case $* = \mathrm{ub}$.

Let $f : X \to Y$, $g : Y \to Z$ and $h : Z \to W$ be morphisms in $\mathrm{K}(\mathsf{C})$ and suppose that $g \circ f$ and $h \circ g$ are quasi-isomorphisms. This means that for all $k \in \mathbb{Z}$

$$\mathrm{H}^k(g \circ f) : \mathrm{H}^k(X) \xrightarrow{\sim} \mathrm{H}^k(Z) \qquad \mathrm{H}^k(h \circ g) : \mathrm{H}^k(Y) \xrightarrow{\sim} \mathrm{H}^k(W)$$

Since $\mathrm{H}^k(g \circ f) = \mathrm{H}^k\, g \circ \mathrm{H}^k\, f$ and $\mathrm{H}^k(h \circ g) = \mathrm{H}^k\, h \circ \mathrm{H}^k\, g$ we have that $\mathrm{H}^k\, f$ is an isomorphism and thus f is a quasi-isomorphism. □

Definition 9.0.1 (Derived category). Let C be an abelian category. We define the **derived category** $\mathrm{D}^*(\mathsf{C})$ of C as the localization $\mathrm{K}^*(\mathsf{C})[N^*(\mathsf{C})^{-1}]$ for $* = +, -, \mathrm{b}, \mathrm{ub}$. The localization functor is denoted by $Q : \mathrm{K}^*(\mathsf{C}) \to \mathrm{D}^*(\mathsf{C})$.

Proposition 9.0.2. Let $X \in \mathrm{K}(\mathsf{C})$ and let $Q(X)$ be its image in $\mathrm{D}(X)$. Then $Q(X) = 0$ in $\mathrm{D}(\mathsf{C})$ if and only if X is q.i.s to 0 in $\mathrm{K}(\mathsf{C})$.

Remark 9.0.1. We know that $\mathrm{H}^k : \mathsf{Comp}(\mathsf{C}) \to \mathsf{C}$ can be extended to a functor $\mathrm{H}^k : \mathrm{K}(\mathsf{C}) \to \mathsf{C}$. Since this functor sends 0 to an object q.i.s to 0, it defines a functor

$$\mathrm{H}^k : \mathrm{D}(\mathsf{C}) \to \mathsf{C}$$

Moreover all of this functors are cohomological by Proposition 9.0.1.
Therefore for all d.t. $X \to Y \to Z \to X[1]$ in $\mathrm{D}(\mathsf{C})$ we get a long exact sequence

$$\cdots \longrightarrow \mathrm{H}^{i-1}(Y) \longrightarrow \mathrm{H}^{i-1}(Z) \longrightarrow \mathrm{H}^i(X) \longrightarrow \mathrm{H}^i(Y) \longrightarrow \mathrm{H}^i(Z) \longrightarrow \mathrm{H}^{i+1}(X) \longrightarrow \mathrm{H}^{i+1}(Y) \longrightarrow \cdots$$

Proposition 9.0.3. The triangulated category $\mathrm{D}^+(\mathsf{C})$ is equivalent to the full subcategory of $\mathrm{D}(\mathsf{C})$ consisting of objects X satisfying $\mathrm{H}^j(X) = 0$ for all $j \ll 0$.
There is an analogous result for $\mathrm{D}^-(\mathsf{C})$ and $\mathrm{D}^b(\mathsf{C})$.
In particular, C is equivalent to the full subcategory of $\mathrm{D}(\mathsf{C})$ consisting of objects X satisfying $\mathrm{H}^j(X) = 0$ for all $j \neq 0$.

Proof. Direct from Proposition 8.2.2. □

9.1 Resolutions

Definition 9.1.1. Let D be a full additive subcategory of C. We say that D is **cogenerating** if for all $X \in \mathsf{C}$ there exists $Y \in \mathsf{D}$ and a monomorphism $X \rightarrowtail Y$.

Theorem 9.1.1 ([KSO6, Lemma 13.2.1]). Let D be a cogenerating category of C. Then for all $a \in \mathbb{Z}$ and $X \in \mathsf{Comp}^{\geq a}(()\mathsf{C})$ there exists $Y \in \mathsf{Comp}^{\geq a}(\mathsf{C})$ and a quasi-isomorphism $X \xrightarrow{\text{q.i.s}} Y$.

Notation. If D is an additive subcategory of C we set

$$N^+(\mathsf{D}) := N(\mathsf{C}) \cap \mathrm{K}^+(\mathsf{D})$$

which is a null system in $\mathrm{K}^+(\mathsf{D})$.

Corollary 9.1.1. Let D be an additive subcategory of C and assume that D is cogenerating. For any $X \in \mathrm{K}^+(\mathsf{C})$ there exists $Y \in \mathrm{K}^+(\mathsf{D})$ and a quasi-isomorphism $X \to Y$. Furthermore the natural functor $\theta : \mathrm{K}^+(\mathsf{D})[(N^+(\mathsf{D}))^{-1}] \to \mathrm{D}^+(\mathsf{D})$ is an equivalence of categories.

Proof. Direct from Proposition 8.2.2. □

9.1.1 Injective resolutions

In this subsection C denotes an abelian category and $\mathcal{I}_{\mathsf{C}}$ is full additive subcategory of injective objects. We will assume that C admits enough injectives or, in other words, that $\mathcal{I}_{\mathsf{C}}$ is cogenerating.

Proposition 9.1.1.

1. Let $f : X \to I$ be a morphism in $\mathsf{Comp}^+(\mathsf{C})$ with $I \in \mathcal{I}_{\mathsf{C}}$. Then f is homotopic to zero.
2. Let $I \in \mathsf{Comp}^+(\mathsf{C})$ be an exact complex. Then I is homotopic to zero.

Proof.

1. We want to complete the following diagram

$$\begin{array}{ccccccccc} X^{k-2} & \xrightarrow{\mathrm{d}_X^{k-2}} & X^{k-1} & \xrightarrow{\mathrm{d}_X^{k-1}} & X^k & \xrightarrow{\mathrm{d}_X^{k}} & X^{k+1} \\ & \overset{s^{k-1}}{\swarrow} & & \overset{s^{k}}{\swarrow} & & \overset{s^{k+1}}{\swarrow} & \\ I^{k-2} & \xrightarrow[\mathrm{d}_I^{k-2}]{} & I^{k-1} & \xrightarrow[\mathrm{d}_I^{k-1}]{} & I^k & \xrightarrow[\mathrm{d}_I^{k}]{} & I^{k+1} \end{array}$$

   with morphisms s^k satisfying

$$f^k = s^{k+1} \circ \mathrm{d}_X^k + \mathrm{d}_I^{k-1} \circ s^k$$

   We will do it by induction.
   For $j \ll 0$ it suffices to set $s^j := 0$. Suppose that we have constructe s^j for $j \leq k$. If we define $g^k := f^k - \mathrm{d}_I^{k-1} \circ s^k$ we have

$$\begin{aligned} g^k \circ \mathrm{d}_X^{k-1} &= (f^k - \mathrm{d}_I^{k-1} \circ s^k) \circ \mathrm{d}_X^{k-1} \\ &= f^k \circ \mathrm{d}_X^{k-1} - \mathrm{d}_I^{k-1} \circ s^k \circ \mathrm{d}_X^{k-1} \\ &= f^k \circ \mathrm{d}_X^{k-1} - \mathrm{d}_I^{k-1} \circ (f^{k-1} - \mathrm{d}_I^{k-2} \circ s^{k-1}) \\ &= f^k \circ \mathrm{d}_X^{k-1} - \mathrm{d}_I^{k-1} \circ f^{k-1} + \mathrm{d}_I^{k-1} \circ \mathrm{d}_I^{k-2} \circ s^{k-1} \\ &= 0 \end{aligned}$$

   Therefore g^k factorizes through $X^k / \operatorname{Im} \mathrm{d}_X^{k-1}$.
   Since X is exact, the sequence $0 \to X / \operatorname{Im} \mathrm{d}_X^{k-1} \to X^{k+1}$ is exact. Therefore by definition of injective object the following diagram may be completed

$$\begin{array}{ccccc} 0 & \longrightarrow & X^k / \operatorname{Im} \mathrm{d}_X^{k-1} & \xrightarrow{\mathrm{d}_X^k} & X^{k+1} \\ & & {\scriptstyle g^k}\downarrow & \swarrow_{s^{k+1}} & \\ & & I^k & & \end{array}$$

2. Apply the result of part 1 to $X = I$ and $f = \mathbb{1}_I$.

□

Corollary 9.1.2. Suppose C admits enough injectives. Then $\mathrm{K}^+(\mathcal{I}_{\mathsf{C}}) \to \mathrm{D}^+(\mathsf{C})$ is an equivalence of categories.

Proof. Recall that $N^+(\mathcal{I}_{\mathsf{C}})$ is the subcategory of $\mathrm{K}^+(\mathcal{I}_{\mathsf{C}})$ consisting of complexes quasi-isomorphic to 0. By Corollary 9.1.1 the natural functor

$$\mathrm{K}^+(\mathcal{I}_{\mathsf{C}})[N^+(\mathcal{I}_{\mathsf{C}})^{-1}] \to \mathrm{D}^+(\mathcal{I}_{\mathsf{C}})$$

is an equivalence of categories.

Finally, since the objects of $N^+(\mathcal{I}_{\mathsf{C}})$ are isomorphic to 0 in $\mathrm{K}^+(\mathcal{I}_{\mathsf{C}})$ we have that $\mathrm{K}^+(\mathcal{I}_{\mathsf{C}})[N^+(\mathcal{I}_{\mathsf{C}})^{-1}]$ is equivalent to $\mathrm{K}^+(\mathcal{I}_{\mathsf{C}})$. □

9.2 Derived functors

In this section C and C' are abelian categories and $F : \mathsf{C} \to \mathsf{C}'$ is a left-exact functor.

There are dual definitions and statements for right-exact functors of all of the given in this section.

Note that by Remark 6.3.1 F defines a functor

$$\mathrm{K}^+(F) : \mathrm{K}^+(\mathsf{C}) \to \mathrm{K}^+(\mathsf{C}')$$

Definition 9.2.1. If the functor $\mathrm{K}^+(F) : \mathrm{K}^+(\mathsf{C}) \to \mathrm{K}^+(\mathsf{C}')$ admits a right localization (with respect to quasi-isomorphisms in $\mathrm{K}^+(\mathsf{C})$) we say that F admits a **right derived functor** and denote this localization by

$$\mathrm{R}F : \mathrm{D}^+(\mathsf{C}) \to \mathrm{D}^+(\mathsf{C}')$$

For every $X \in \mathsf{C}$ and $j \geq 0$ we set

$$\mathrm{R}^j F(X) := \mathrm{H}^j(\mathrm{R}F(X))$$

There are dual definitions of **left derived functors** $\mathrm{L}F : \mathrm{D}^-(\mathsf{C}) \to \mathrm{D}^-(\mathsf{C})$ for right-exact functors.

Notation. If $F : \mathsf{C} \to \mathsf{C}'$ is exact then it admits both a right and left derived functor and both they coincide. In this case we denote by F its localization.

Definition 9.2.2. An object $X \in \mathsf{C}$ is F-**acyclic** if $\mathrm{R}^j F(X) = 0$ for all $j \neq 0$.

Proposition 9.2.1. For any short exact sequence

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

in C there exists a long exact sequence

$$\begin{array}{l} 0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z) \\ \quad \longrightarrow \mathrm{R}^1 F(X) \longrightarrow \mathrm{R}^1 F(Y) \longrightarrow \mathrm{R}^1 F(Z) \\ \quad \longrightarrow \mathrm{R}^2 F(X) \longrightarrow \mathrm{R}^2 F(Y) \longrightarrow \cdots \end{array}$$

Proof. $\mathrm{R}F$ sends distinguished triangles to distinguished triangles. □

Definition 9.2.3. Let D be a full additive subcategory of C. We say that D is F-**injective** if

1. D is cogenerating.
2. For any short exact sequence $0 \to X \to Y \to Z \to 0$ in C if X and Y are in D then so is D.
3. For any short exact sequence $0 \to X \to Y \to Z \to 0$ in C with X in D the sequence $0 \to F(X) \to F(Y) \to F(Z) \to 0$ is exact.

The dual concept is of a F-projective subcategory.

Example 9.2.1. If C has enough injectives then $\mathcal{I}_{\mathsf{C}}$ is F-injective for any left-exact functor F.

Example 9.2.2. Let $\mathcal{I}_F$ be the full additive subcategory of C made of F-acyclic objects. If $\mathcal{I}_F$ is cogenerating then it is F-injective, since conditions 2 and 3 are satisfied by Proposition 9.2.1.

Lemma 9.2.1. Let D be F-injective and let $X^\bullet \in \mathsf{Comp}^+(\mathsf{C})$ be a complex quasi-isomorphic to 0. Then $F(X^\bullet)$ is quasi-isomorphic to 0.

Proof. Let's assume that $X^\bullet$ starts at zero to ease the notation.

We can decompose $X^\bullet$ into short exact sequences

$$0 \to X^0 \to X^1 \to Z^1 \to 0$$
$$0 \to Z^{k-1} \to X^k \to Z^k \to 0 \quad \forall k \in \{2, \dots, n\}$$

Applying the second axiom of being F-injective by induction we see that all of the Z^k are in D. Therefore the sequences

$$0 \to F(X^0) \to F(X^1) \to F(Z^1) \to 0$$
$$0 \to F(Z^{k-1}) \to F(X^k) \to F(Z^k) \to 0 \quad \forall k \in \{2, \dots, n\}$$

are exact. *Glueing* them we get that

$$0 \to F(X^0) \to F(X^1) \to F(X^2) \to \cdots$$

is exact. □

Theorem 9.2.1. Let $F : \mathsf{C} \to \mathsf{C}'$ be a left-exact functor between abelian categories. Let D be an F-injective subcategory of C. Then F admits a right derived functor $\mathrm{R}F : \mathrm{D}^+(\mathsf{C}) \to \mathrm{D}^+(\mathsf{C}')$.

Proof. We want to apply Theorem 8.2.2. The first condition is satisfied by Corollary 9.1.1 and the second thanks to Lemma 9.2.1. □

Remark 9.2.1. The construction of $\mathrm{R}F$ may be visualized by the diagram

$$\begin{array}{ccc}
\mathrm{K}^+(\mathsf{D}) & \xrightarrow{\mathrm{K}^+F} & \mathrm{K}^+(\mathsf{C}') \\
{\scriptstyle Q}\downarrow & & \\
\mathrm{K}^+(\mathsf{D})[N^+(\mathsf{D})^{-1}] & & \downarrow{\scriptstyle Q} \\
{\scriptstyle \sim}\downarrow & \searrow & \\
\mathrm{D}^+(\mathsf{C}) & \xrightarrow[\mathrm{R}F]{} & \mathrm{D}^+(\mathsf{C}')
\end{array}$$

Furthermore $\mathrm{R}F$ is triangulated and does not depend on D. Therefore if $X \to Y \to Z \xrightarrow{+1}$ is a d.t. in $\mathrm{D}^+(\mathsf{C})$ then $\mathrm{R}F(X) \to \mathrm{R}F(Y) \to \mathrm{R}F(Z) \xrightarrow{+1}$ is a d.t. in $\mathrm{D}^+(\mathsf{C}')$.

The following result gives an explicit construction of the derived functor.

Proposition 9.2.2. Let D be F-injective. Then for any $X \in \mathsf{C}$ and resolution $0 \to X \to Y^\bullet$ with $Y^\bullet \in \mathsf{Comp}^+(\mathsf{D})$ there is an isomorphism

$$\mathrm{R}^n F(X) \cong \mathrm{H}^n(F(Y^\bullet))$$

In other words, to compute the functors $\mathrm{R}^n F(X)$ it is enough to replace X with a

right D-resolution, apply F to this complex and take its cohomology.

Proof. Note that $0 \to X \to Y^\bullet$ being a resolution is equivalent to saying that $X \to Y^\bullet$ is a quasi-isomorphism in $\mathrm{K}^+(\mathsf{C})$. Therefore since Q *inverts* quasi-isomorphisms we get that $X \cong Y^\bullet$ in $\mathrm{D}^+(\mathsf{C})$. Applying $\mathrm{R}F$ we get $\mathrm{R}F(X) \cong \mathrm{R}F(Y^\bullet)$ in $\mathrm{D}^+(\mathsf{C}')$.

Since we can view $Y^\bullet \in \mathrm{K}^+(\mathsf{D})$ by the construction of $\mathrm{R}F$ we have $\mathrm{R}F(Y^\bullet) \cong Q(F(Y^\bullet))$, which gives

$$\mathrm{R}F(X) \cong Q(F(Y^\bullet))$$

Therefore for all $n \in \mathbb{Z}$ applying cohomology

$$\mathrm{R}^n F(X) = \mathrm{H}^n(\mathrm{R}F(X)) \cong \mathrm{H}^n(Q(F(Y^\bullet))) = \mathrm{H}^n(F(Y^\bullet))$$

□

Remark 9.2.2. In particular Proposition 9.2.2 applies if C has enough injectives.

9.2.1 Derived functor of the composition

In this subsection we consider two left-exact functors $F : \mathsf{C} \to \mathsf{C}'$ and $G : \mathsf{C}' \to \mathsf{C}''$ between abelian categories.

Theorem 9.2.2. Suppose there exist full additive subcategories $\mathsf{D} \subseteq \mathsf{C}$ and $\mathsf{D}' \subseteq \mathsf{C}'$ such that D is F-injective, D' is G-injective and $F(\mathsf{D}) \subseteq \mathsf{D}'$.
Then D is $(G \circ F)$-injective and there exists an isomorphism

$$\mathrm{R}(G \circ F) \xrightarrow{\sim} \mathrm{R}G \circ \mathrm{R}F$$

Proof. It is clear that D is $(G \circ F)$-injective.

To see the isomorphism, consider $X \in \mathrm{K}^+(\mathsf{C})$ and $Y \in \mathrm{K}^+(\mathsf{D})$ together with a quasi-isomorphism $X \to Y$. Then $\mathrm{R}F(X)$ is represented by the complex $F(Y)$, which belongs to $\mathrm{K}^+(\mathsf{D}')$.

Therefore $\mathrm{R}G(\mathrm{R}F(X))$ is represented by $(G \circ F)(Y)$, which also represents $\mathrm{R}(G \circ F)(Y)$ since $Y \in \mathsf{Comp}^+(\mathsf{D})$ and D is $(G \circ F)$-injective. □

Corollary 9.2.1. Let D be a F-injective subcategory of C. If G is exact then D is $(G \circ F)$-injective and there exists an isomorphism

$$\mathrm{R}(G \circ F) \xrightarrow{\sim} \mathrm{R}G \circ \mathrm{R}F$$

Proof. Since G is exact we have that C' is G-injective. Therefore we can apply Theorem 9.2.2 with $\mathsf{D}' = \mathsf{C}'$. □

Corollary 9.2.2. Suppose there exist full additive subcategories $\mathsf{D} \subseteq \mathsf{C}$ and $\mathsf{D}' \subseteq \mathsf{C}'$ such that D is F-injective, D' is G-injective and $F(\mathsf{D}) \subseteq \mathsf{D}'$.
If $X \in \mathsf{C}$ is F-acyclic then for all $n \in \mathbb{Z}$

$$\mathrm{R}^n(G \circ F)(X) \cong (\mathrm{R}^n G)(F(X))$$

Proof. Since X is F-acyclic we have $\mathrm{R}F(X) \cong F(X)$ in $\mathrm{D}^+(\mathsf{C}')$. Therefore by Theorem 9.2.2 we have

$$\mathrm{R}(G \circ F)(X) \cong (\mathrm{R}G \circ \mathrm{R}F)(X) \cong \mathrm{R}G(F(X))$$

□

9.3 Bifunctors

We now consider three abelian categories together with an additive bifunctor

$$F : \mathsf{C} \times \mathsf{C}' \to \mathsf{C}''$$

We will also assume that F is left-exact with respect to each of its arguments.

Let $X \in \mathsf{Comp}^+(\mathsf{C})$ and $X' \in \mathsf{Comp}^+(\mathsf{C})$. Then the double complex $F(X, X') \in \mathsf{Comp}^2(\mathsf{C}'')$ satisfies the finiteness condition and thus $\mathrm{tot}(F(X, X')) \in \mathsf{Comp}^+(\mathsf{C}'')$ is a well-defined and defines a functor

$$\begin{aligned} \mathrm{K}^+F : \mathrm{K}^+(\mathsf{C}) \times \mathrm{K}^+(\mathsf{C}') &\to \mathrm{K}^+(\mathsf{C}'') \\ (X, X') &\mapsto \mathrm{tot}(F(X, X')) \end{aligned}$$

When the context is clear we will write F instead of K^+F.

Definition 9.3.1. If the functor

$$\mathrm{K}^+F : \mathrm{K}^+(\mathsf{C}) \times \mathrm{K}^+(\mathsf{C}') \to \mathrm{D}^+(\mathsf{C}'')$$

admits a right localization with respect to quasi-isomorphisms in $\mathrm{K}^+(\mathsf{C})$ and $\mathrm{K}^+(\mathsf{C}')$ we say that F admits a **right derived functor** and denote this functor by

$$\mathrm{R}F : \mathrm{D}^+(\mathsf{C}) \times \mathrm{D}^+(\mathsf{C}') \to \mathrm{D}^+(\mathsf{C}'')$$

There is a dual definition of **left derived functor** for a right exact bifunctor.

Definition 9.3.2. Let D, D' be additive subcategories of C and C' respectively. We say that $(\mathsf{D}, \mathsf{D}')$ is **F-injective** if

1. for all $X' \in \mathsf{D}'$, D is $F(-, X')$-injective.
2. for all $X \in \mathsf{D}$, D' is $F(X, -)$-injective.

There is a dual definition of being **G-projective** for a right exact bifunctor G.

Theorem 9.3.1. Let $F : \mathsf{C} \times \mathsf{C}' \to \mathsf{C}''$ be a left exact bifunctor between abelian categories. If there exists $(\mathsf{D}, \mathsf{D}')$ F-injective subcategories then F admits a right derived functor

$$\mathrm{R}F : \mathrm{D}^+(\mathsf{C}) \times \mathrm{D}^+(\mathsf{C}') \to \mathrm{D}^+(\mathsf{C}'')$$

Proof. Let $X \in \mathrm{D}^+(\mathsf{C})$ and $X' \in \mathrm{D}^+(\mathsf{C}')$. If X or X' is quasi-isomorphic to 0 then all rows or all columns or $F(X, X')$ are exact.

Therefore by Corollary 7.4.2 $\mathrm{tot}(F(X, X'))$ is quasi-isomorphic to zero. Finally applying Theorem 8.2.3 with $\mathsf{D} = \mathrm{K}^+(\mathsf{D})$ and $\mathsf{D}' = \mathrm{K}^+(\mathsf{D}')$ we get the result. □

Theorem 9.3.2. Let $F : \mathsf{C} \times \mathsf{C}' \to \mathsf{C}''$ be a left exact bifunctor between abelian categories. Let D, D' be additive subcategories of C and C' respectively. If there

exists $(\mathsf{D}, \mathsf{D}')$ F-injective subcategories then for all $X \in \mathsf{C}, Y \in \mathsf{C}'$ we have

$$\mathrm{R}F(X,Y) \cong \mathrm{R}_{II}F(X,Y) \cong \mathrm{R}_I F(X,Y)$$

where $\mathrm{R}_I F, \mathrm{R}_{II} F$ are the derived functors of F_I and F_{II} respectively.

Proof. Let $X \in \mathsf{C}$ and $Y \in \mathsf{C}'$. Consider resolutions $0 \to X \to I_X^\bullet$, $0 \to Y \to I_Y^\bullet$ with $I_X^\bullet \in \mathsf{Comp}(\mathsf{D})$ and $I_Y^\bullet \in \mathsf{Comp}(\mathsf{D}')$.

The object $\mathrm{R}F(X,Y)$ is the image by the localization functor Q of the object $\mathrm{tot}(F(I_X^\bullet, I_Y^\bullet)) \in \mathrm{K}^+(\mathsf{C}'')$. Consider the double complex

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & 0 & \longrightarrow & F(I_X^0, Y) & \longrightarrow & F(I_X^1, Y) & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & F(X, I_Y^0) & \longrightarrow & F(I_X^0, I_Y^0) & \longrightarrow & F(I_X^1, I_Y^0) & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & F(X, I_Y^1) & \longrightarrow & F(I_X^0, I_Y^1) & \longrightarrow & F(I_X^1, I_Y^1) & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & \vdots & & \vdots & & \vdots
\end{array}$$

By hypothesis all rows and columns except the first (the ones starting with $0 \to 0$) are exact. Therefore by Corollary 7.4.3 $\mathrm{tot}(F(I_X^\bullet, I_Y^\bullet))$ is quasi-isomorphic to the cohomology of the 0-column, which computes $\mathrm{R}_{II} F(X,-)(Y)$, and to the cohomology of the 0-row, which computes $\mathrm{R}_I F(X,Y)$. □

Lemma 9.3.1. Let $F : \mathsf{C} \times \mathsf{C}' \to \mathsf{C}''$ be a left exact bifunctor between abelian categories. Let D' be a cogenerating subcategory of C' and let $I \in \mathsf{C}$.
If $F(I,-)$ and $F(-,Y)$ for all $Y \in \mathsf{D}'$ are exact then for any $Y' \in \mathsf{C}'$ and any short exact sequence

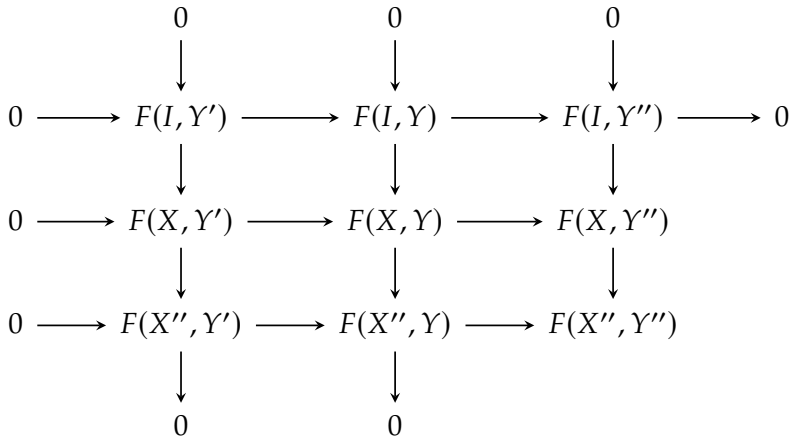
$$0 \longrightarrow I \longrightarrow X \longrightarrow X'' \longrightarrow 0$$

in C the sequence

$$0 \longrightarrow F(I,Y') \longrightarrow F(X,Y') \longrightarrow F(X'',Y') \longrightarrow 0$$

is exact.

Proof. Choose a short exact sequence $0 \to Y' \to Y \to Y'' \to 0$ with $Y' \in \mathsf{D}'$. Consider the commutative diagram



The first row is exact since $F(I,-)$ is exact. The second and third rows are exact since F is left-exact. The second column is exact since $Y \in \mathsf{D}'$.

Therefore by the Snake lemma the first column is exact. □

Theorem 9.3.3. Let $F : \mathsf{C} \times \mathsf{C}' \to \mathsf{C}''$ be a left exact bifunctor between abelian categories.
Define the additive subcategories D_F and D'_F of C and C' respectively by

$$\mathsf{D}_F \coloneqq \{X \in \mathsf{C} \mid F(X,-) \text{ exact } \}$$
$$\mathsf{D}'_F \coloneqq \{X' \in \mathsf{C}' \mid F(-,X') \text{ exact } \}$$

If D_F and D'_F are cogenerating then for any $Y \in \mathsf{C}'$ D_F is $F(-,Y)$-injective and for any $X \in \mathsf{C}$ D'_F is $F(X,-)$-injective

Proof. We will see that for any $Y \in \mathsf{C}$ D_F is $F(-,Y)$-injective. The other case follows analogously.

1. D_F is cogenerating by hypothesis.
2. For any exact sequence
$$0 \longrightarrow I \longrightarrow X \longrightarrow X'' \longrightarrow 0$$
in C with $I \in \mathsf{D}_F$ the sequence
$$0 \longrightarrow F(I,Y') \longrightarrow F(X,Y') \longrightarrow F(X'',Y') \longrightarrow 0$$
is exact by Lemma 9.3.1.
3. Consider an exact sequence as above with $I, X \in \mathsf{D}_F$. Consider the double complex

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F(I,Y') & \longrightarrow & F(I,Y) & \longrightarrow & F(I,Y'') & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F(X,Y') & \longrightarrow & F(X,Y) & \longrightarrow & F(X,Y'') & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F(X'',Y') & \longrightarrow & F(X'',Y) & \longrightarrow & F(X'',Y'') & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

By hypothesis the first and second rows are exact. Therefore since by Lemma 9.3.1 all columns are exact, the whole diagram is exact.

□

Proposition 9.3.1. Let $F : \mathsf{C} \times \mathsf{C}' \to \mathsf{C}''$ be a left exact bifunctor between abelian categories and let $G : \mathsf{C}'' \to \mathsf{C}'''$ be a left-exact functor.
If $(\mathsf{D}, \mathsf{D}')$ is F-injective, D'' is G-injective and $F(\mathsf{D}, \mathsf{D}') \subseteq D''$ then the right derived functor $\mathrm{R}(G \circ F)$ exists and moreover $\mathrm{R}(G \circ F) \cong \mathrm{R}\,G \circ \mathrm{R}\,F$.

Proof. Analogous to Theorem 9.2.2. □

Example 9.3.1. Suppose that C has enough injectives. Then

$$\mathrm{R}\,\mathrm{Hom}_{\mathsf{C}} : \mathrm{D}^{-}(\mathsf{C})^{\mathrm{op}} \times \mathrm{D}^{+}(\mathsf{C}) \to \mathrm{D}^{+}(\mathsf{Ab})$$

exists and can be computed as follows.
Let $X^\bullet \in \mathsf{Comp}^{-}(\mathsf{C})$ and $Y^\bullet \in \mathsf{Comp}^{+}(\mathsf{C})$. There exists a quasi-isomorphism $Y^\bullet \to I^\bullet$ in $\mathrm{K}^{+}(\mathsf{C})$ with I^j injective. Then

$$\mathrm{R}\,\mathrm{Hom}_{\mathsf{C}}(X^\bullet, Y^\bullet) \cong \mathrm{Hom}^\bullet_{\mathsf{C}}(X^\bullet, I^\bullet)$$

If C has enough projectives and $X^\bullet \to P^\bullet$ is a quasi-isomorphism in $\mathrm{K}^{-}(\mathsf{C})$ with P^j projective then

$$\mathrm{R}\,\mathrm{Hom}_{\mathsf{C}}(X^\bullet, Y^\bullet) \cong \mathrm{Hom}^\bullet_{\mathsf{C}}(P^\bullet, Y^\bullet)$$

Example 9.3.2. Let A be a R-algebra. By choosing the category of projective modules as D and D' in Theorem 9.3.1 we get that the bifunctor

$$- \overset{L}{\otimes}_A - : \mathrm{D}^{-}(\mathsf{Mod}A^{\mathrm{op}}) \times \mathrm{D}^{-}(\mathsf{Mod}A) \to \mathrm{D}^{-}(\mathsf{Mod}(R))$$

is well defined.
Furthermore if $M^\bullet \to P^\bullet$ and $N^\bullet \to Q^\bullet$ are quasi-isomorphisms with P^j, Q^j projective then

$$N^\bullet \overset{L}{\otimes}_A M^\bullet \cong \mathrm{tot}(N^\bullet \otimes_A P^\bullet) \cong \mathrm{tot}(Q^\bullet \otimes_A M^\bullet)$$

PART III

Sheaves

Chapter 10

Sheaves

10.1 Presheaves

Definition 10.1.1 (Presheaf). Let X be a topological space and let $\mathsf{Op}(X)$ be its posetal category of open sets. Let C be a category. The category of C-valued **presheaves** over X is the functor category

$$\mathsf{PSh}_X(\mathsf{C}) := \mathsf{Fun}(\mathsf{Op}(X)^{\mathrm{op}}, \mathsf{C})$$

The elements of this category are functors $\mathcal{F} : \mathsf{Op}(X)^{\mathrm{op}} \to \mathsf{C}$ and morphisms are natural transformations $\phi : \mathcal{F} \to \mathcal{G}$.

Less abstractly, a C-valued presheaf $\mathcal{F}$ is the collection of data:

- For each open set $U \subseteq X$ an object $\mathcal{F}(U) \in \mathsf{C}$ called *sections of $\mathcal{F}$ over U*.
- For every inclusion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{G}(U)$ in C called *restriction* such that
  - For every open set $U \subseteq X$, $\mathrm{res}_{U,U} = \mathbb{1}_U$.
  - For every inclusion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms

$$\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Op}(X)}$$

in C making the following diagram commute

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ {\scriptstyle \mathrm{res}_{U,V}}\downarrow & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow[\phi(V)]{} & \mathcal{G}(V) \end{array}$$

Notation. When the context is clear, we will drop the subscript of $\mathsf{PSh}_X(\mathsf{C})$ and denote the category of C-valued presheaves over X as $\mathsf{PSh}(\mathsf{C})$.

Since $\mathsf{PSh}_X(\mathsf{C})$ is a functor category, if C has limits or colimits of a certain shape then $\mathsf{PSh}_X(\mathsf{C})$ also has them and they are computed pointwise. This means that limits and colimits can be computed *open set by open set*.

Proposition 10.1.1. Let X be a topological space. Let $\mathcal{I} \to \mathsf{PSh}_X(\mathsf{C}), i \mapsto \mathcal{F}_i$ be a diagram. If C has limits (resp. colimits) of shape $\mathcal{I}$ then $\lim_i \mathcal{F}_i$ (resp. $\operatorname{colim}_i \mathcal{F}_i$) exists in $\mathsf{PSh}_X(\mathsf{C})$. Moreover for every open set $U \subseteq X$

$$\left(\lim_i \mathcal{F}_i\right)(U) = \lim_i \left(\mathcal{F}_i(U)\right) \qquad \left(\operatorname*{colim}_i \mathcal{F}_i\right)(U) = \operatorname*{colim}_i \left(\mathcal{F}_i(U)\right)$$

Proof. Particular case of Proposition 4.3.1. □

In particular if C is an abelian category then kernels and cokernels exist in $\mathsf{PSh}_X(\mathsf{C})$ and are computed *open set by open set*, making $\mathsf{PSh}_X(\mathsf{C})$ an abelian category.

Example 10.1.1. Consider $X = \{*\}$ the one-point topological space. The category $\mathsf{Op}(X)$ has

$$\mathrm{Ob}(\mathsf{Op}(X)) = \{\varnothing, X\} \qquad \mathrm{Hom}(\mathsf{Op}(X)) = \{\varnothing \hookrightarrow X\}$$

Therefore every $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ consists of two objects $\mathcal{F}(\varnothing), \mathcal{F}(X) \in \mathsf{C}$ and a restriction morphism $\rho : \mathcal{F}(X) \to \mathcal{F}(\varnothing)$.
This makes $\mathsf{PSh}_{\{*\}}(\mathsf{C}) \cong \mathsf{Arr}(\mathsf{C})$.

10.1.1 Presheaves and continuous maps

Definition 10.1.2 (Direct image presheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let $\mathcal{F}$ be a C-valued presheaf over X.
The **direct image presheaf** of $\mathcal{F}$ via f, $f_*\mathcal{F} \in \mathsf{PSh}_Y(\mathsf{C})$, is defined as

$$\forall V \in \mathsf{Op}(Y) \quad f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}V).$$

Given open subsets $W \subseteq V \subseteq Y$, the restriction maps $\mathrm{res}_{V,W}$ comes directly from the diagram

$$\begin{array}{ccc} f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \\ \vdots & & \downarrow{\scriptstyle \mathrm{res}_{f^{-1}V, f^{-1}W}} \\ f_*\mathcal{F}(W) & = & \mathcal{F}(f^{-1}W) \end{array}$$

Notice that this construction defines a functor $f_* : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$.

Example 10.1.2 (Global section)**.** Let X be a topological space. Consider the one-point topological space $\{*\}$ and the unique continuous map $f : X \to \{*\}$. Let $\mathcal{F}$ be a C-valued presheaf over X.
Consider the direct image presheaf of $\mathcal{F}$ via f. Since this is a C-valued presheaf over $\{*\}$ by Example 10.1.1 we have that

$$f_*\mathcal{F} \cong (\mathcal{F}(\{*\}) \to \mathcal{F}(\varnothing))$$

Using the definition of the direct image presheaf we have

$$f_*\mathcal{F}(\varnothing) = \mathcal{F}(f^{-1}(\varnothing)) = \mathcal{F}(\varnothing)$$
$$f_*\mathcal{F}(\{*\}) = \mathcal{F}(f^{-1}(\{*\})) = \mathcal{F}(X)$$

Therefore we define the **global sections** of $\mathcal{F}$ as

$$\Gamma(X, \mathcal{F}) := f_*\mathcal{F} \cong \mathcal{F}(X) \to \mathcal{F}(\varnothing)$$

This defines a functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Arr}(\mathsf{C})$.
In particular, if $\mathcal{F}(\varnothing)$ is a terminal object in C we can identify $\mathcal{F}(X) \to \mathcal{F}(\varnothing)$ with $\mathcal{F}(X)$, getting a functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$.

Example 10.1.3 (Skyscraper presheaf)**.** Let X be a topological space and $p \in X$. Consider the one-point topological space $\{*\}$ and the continuous map $\iota_p : \{*\} \to X, * \mapsto p$. Consider a morphism $\rho : S \to T$ in C, which by Example 10.1.1 corresponds to a presheaf $\mathcal{R} \in \mathsf{PSh}_{\{*\}}(\mathsf{C})$.
Consider the direct image presheaf of $\mathcal{R}$ via ι_p:

$$\forall U \in \mathsf{Op}(X) \quad \iota_{p,*}\mathcal{R}(U) = \mathcal{R}(\iota_p^{-1}(U)) = \begin{cases} \mathcal{R}(\{*\}) & p \in U \\ \mathcal{R}(\varnothing) & p \notin U \end{cases} = \begin{cases} S & p \in U \\ T & p \notin U \end{cases}$$

Note that if $V \subseteq U$ the restriction map is given by

$$\mathrm{res}_{U,V} = \begin{cases} \mathbb{1}_S & p \in V \\ \mathbb{1}_T & p \notin U \\ \rho & p \in U \wedge p \notin V \end{cases}$$

We define the **skyscraper presheaf** supported at ρ, $\iota_{p,*}\rho \in \mathsf{PSh}_X(\mathsf{C})$, as

$$\iota_{p,*}\underline{\rho}_{\mathrm{pre}} \coloneqq \iota_{p,*}\mathcal{R} \tag{10.1.1}$$

This defines a functor $\iota_{p,*} : \mathsf{Arr}(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$.
In particular if $T = \mathcal{R}(\varnothing)$ is a terminal object in C we can identify $\rho : S \to T$ with S, getting a functor $\iota_{p,*} : \mathsf{C} \to \mathsf{PSh}_X(\mathsf{C})$. In this situation we write $\iota_{p,*}\underline{S}_{\mathrm{pre}}$.

For sufficiently nice categories (those with direct limits), we have a *dual* construction.

Definition 10.1.3 (Inverse image presheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let C be a category with direct limits. Let $\mathcal{G}$ be a C-valued presheaf over Y.
The **inverse image presheaf** of $\mathcal{G}$, $f_{\mathrm{pre}}^{-1}\mathcal{G} \in \mathsf{PSh}_X(\mathsf{C})$, of $\mathcal{G}$ via f is the presheaf

$$\forall U \in \mathsf{Op}(X) \quad \left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)(U) \coloneqq \varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(Y)}} \mathcal{G}(V)$$

Notice that this defines a functor $f_{\mathrm{pre}}^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$.

Remark 10.1.1. For $\mathsf{C} = \mathsf{Set}$ we can describe $f^{-1}\mathcal{G}(U)$ as

$$\left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)(U) = \{(s, V) \mid f(U) \subseteq V \in \mathsf{Op}(Y) \wedge s \in \mathcal{G}(V)\}$$

modulo the relation

$$(s, V) \sim (s', V') \iff \exists W \in \mathsf{Op}(Y) \mid W \subseteq V \cap V' \wedge f(U) \subseteq W \wedge s|_W = s'|_W$$

In the following examples, we assume that the category C has direct limits.

Example 10.1.4 (Constant presheaf). Let X be a topological space. Consider the one-point topological space $\{*\}$. Consider a morphism $\rho : S \to T$ in C. Note that by Example 10.1.1 this corresponds to a presheaf $\mathcal{R} \in \mathsf{PSh}_{\{*\}}(\mathsf{C})$.
Consider the inverse image presheaf of $\mathcal{R}$ via the map $f : X \to \{*\}$.

- If $U \in \mathsf{Op}(X)$ is not the empty set we have that

$$(f_{\mathrm{pre}}^{-1}\mathcal{R})(U) = \varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(\{*\})}} \mathcal{R}(V) = \varinjlim_{\substack{V \ni * \\ V \in \mathsf{Op}(\{*\})}} \mathcal{R}(V) = \mathcal{R}(\{*\}) = S$$

- If $U = \varnothing$ then we take the limit over the diagram $S \xrightarrow{\rho} T$, getting

$$(f_{\mathrm{pre}}^{-1}\mathcal{R})(\varnothing) = T$$

Therefore

$$\forall U \in \mathsf{Op}(X) \quad \left(f_{\mathrm{pre}}^{-1}\mathcal{R}\right)(U) = \begin{cases} S & U \neq \varnothing \\ T & U = \varnothing \end{cases}$$

Note that for $V \subseteq U$ open subsets of X the restriction maps are given by

$$\mathrm{res}_{U,V} = \begin{cases} \mathbb{1}_S & V \neq \varnothing \\ \mathbb{1}_T & U = \varnothing \\ \rho & V = \varnothing \wedge U \neq \varnothing \end{cases}$$

Therefore we can define the **constant presheaf** supported as ρ as

$$\underline{\rho}_{\mathrm{pre}}(U) \coloneqq f_{\mathrm{pre}}^{-1}\mathcal{R}$$

This defines a functor $\underline{\cdot}_{\mathrm{pre}} : \mathsf{Arr}(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$
Note that if $T = \mathcal{R}(\varnothing)$ is a terminal object in C we can identify $\rho : S \to T$ with S, getting a functor $\underline{\cdot}_{\mathrm{pre}} : \mathsf{C} \to \mathsf{PSh}_X(\mathsf{C})$. In this situation we write $\underline{S}_{\mathrm{pre}}$.

Example 10.1.5 (Stalk). Let X be a topological space and $p \in X$. Consider the one-point topological space $\{*\}$ and the inclusion $f : \{*\} \hookrightarrow X, * \mapsto p$. Let $\mathcal{F}$ be a C-valued presheaf over X.
Consider the inverse image presheaf of $\mathcal{F}$ via f. Since this is a C-valued presheaf over $\{*\}$ by Example 10.1.1

$$f_{\mathrm{pre}}^{-1}\mathcal{F} \cong \left(f_{\mathrm{pre}}^{-1}\mathcal{F}(\{*\}) \to f_{\mathrm{pre}}^{-1}\mathcal{F}(\varnothing)\right)$$

Using the definition of the inverse image presheaf we have that

$$f_{\mathrm{pre}}^{-1}\mathcal{F}(\{*\}) = \varinjlim_{\substack{U \supseteq f(\{*\}) \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) = \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U)$$

$$f_{\mathrm{pre}}^{-1}\mathcal{F}(\varnothing) = \varinjlim_{\substack{U \supseteq f(\varnothing) \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) = \varinjlim_{U \in \mathsf{Op}(X)} \mathcal{F}(U) = \mathcal{F}(\varnothing)$$

Therefore

$$f_{\text{pre}}^{-1}\mathcal{F} \cong \left(\varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) \to \mathcal{F}(\varnothing) \right)$$

We define the **stalk** of $\mathcal{F}$ at p as

$$\mathcal{F}_p \coloneqq f_{\text{pre}}^{-1}\mathcal{F}$$

This defines a functor $-_p : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Arr}(\mathsf{C})$.
In particular if $\mathcal{F}(\varnothing)$ is a terminal object in C we can identify

$$\left(\varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) \to \mathcal{F}(\varnothing) \right) \cong \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U)$$

getting a functor $-_p : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$.

Lemma 10.1.1. Let $f : X \to Y$ be a continuous map between two topological space X, Y and $p \in X$. Let C be a category with direct limits. Let $\mathcal{G}$ be a C-valued presheaf over Y. Then $(f_{\text{pre}}^{-1}\mathcal{G})_p = \mathcal{G}_{f(p)}$.

Proof.

$$\begin{aligned}
(f_{\text{pre}}^{-1}\mathcal{G})_p &= \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} f_{\text{pre}}^{-1}\mathcal{G}(U) \to \varinjlim_{U \in \mathsf{Op}(X)} f_{\text{pre}}^{-1}\mathcal{G}(U) \\
&= \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \left(\varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(Y)}} \mathcal{G}(V) \right) \to \varinjlim_{U \in \mathsf{Op}(X)} \left(\varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(Y)}} \mathcal{G}(V) \right) \\
&= \varinjlim_{\substack{V \ni f(p) \\ V \in \mathsf{Op}(Y)}} \mathcal{G}(V) \to \varinjlim_{V \in \mathsf{Op}(Y)} \mathcal{G}(V) \\
&= \mathcal{G}_{f(p)}
\end{aligned}$$

□

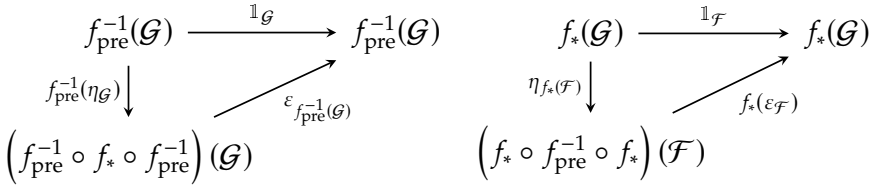
Theorem 10.1.1. Let $f : X \to Y$ be a continuous function between two topological spaces X, Y. Let C be a category with direct limits. Then there exists an adjunction

$$f_{\text{pre}}^{-1} \dashv f_* : \mathsf{PSh}_X(\mathsf{C}) \underset{f_*}{\overset{f_{\text{pre}}^{-1}}{\leftrightarrows}} \mathsf{PSh}_Y(\mathsf{C})$$

Proof. We want to construct natural transformations

$$\begin{aligned}
\eta &: \mathbb{1}_{\mathsf{PSh}_Y(\mathsf{C})} \implies f_* \circ f_{\text{pre}}^{-1} \\
\varepsilon &: f_{\text{pre}}^{-1} \circ f_* \implies \mathbb{1}_{\mathsf{PSh}_X(\mathsf{C})}
\end{aligned}$$

making for all $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and $\mathcal{G} \in \mathsf{PSh}_Y(\mathsf{C})$ the following diagrams commute



Note that, by definition of the direct image presheaf and inverse image presheaf we have, for $U \in \mathsf{Op}(X)$ and $V \in \mathsf{Op}(Y)$

$$\left(f_{\mathrm{pre}}^{-1} \circ f_*\right)(\mathcal{F})(U) = \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Op}(Y)}} (f_*\mathcal{F})(W) = \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Op}(Y)}} \mathcal{F}(f^{-1}(W))$$

$$\left(f_* \circ f_{\mathrm{pre}}^{-1}\right)(\mathcal{G})(V) = (f_{\mathrm{pre}}^{-1}\mathcal{G})(f^{-1}(V)) = \varinjlim_{\substack{W \supseteq (f \circ f^{-1})(V) \\ W \in \mathsf{Op}(Y)}} \mathcal{G}(W)$$

Since we have $(f \circ f^{-1})(V) \subseteq V$, we can define $\eta_{\mathcal{G}}(V)$ as the canonical injection into the direct limit

$$\eta_{\mathcal{G}}(V) : \mathcal{G}(V) \to \varinjlim_{\substack{W \supseteq (f \circ f^{-1})(V) \\ W \in \mathsf{Op}(Y)}} \mathcal{G}(W)$$

On the other hand, since for every $W \subseteq f(U)$ open we have $f^{-1}(W) \subseteq (f^{-1} \circ f)(U) \subseteq U$ there exists a restriction morphism

$$\mathrm{res}_{f^{-1}(W),U} : \mathcal{F}(f^{-1}(W)) \to \mathcal{F}(U)$$

Using this maps we obtain a morphism

$$\varepsilon_{\mathcal{F}}(U) : \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Op}(Y)}} \mathcal{F}(f^{-1}(W)) \to \mathcal{F}(U)$$

□

In particular, we get the following particular cases of Theorem 10.1.1 for a category C with direct limits.

Corollary 10.1.1. Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued presheaf over X and let $\rho : S \to T$ be a morphism in C. Then

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{\rho}_{\mathrm{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{Arr}(\mathsf{C})}(\rho, \Gamma(X, \mathcal{F}))$$

In particular, if $\mathcal{F}(\varnothing)$ and T are terminal objects in C we get

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{S}_{\mathrm{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{C}}(S, \Gamma(X, \mathcal{F}))$$

Corollary 10.1.2. Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued presheaf over X and let $\rho : S \to T$ be a morphism in C. Then

$$\mathrm{Hom}_{\mathsf{Arr}(\mathsf{C})}(\mathcal{F}_p, \rho) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{\rho}_{\mathrm{pre}})$$

In particular, if $\mathcal{F}(\varnothing)$ and T are terminal objects in C we get

$$\mathrm{Hom}_{\mathsf{C}}(\mathcal{F}_p, S) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{S}_{\mathrm{pre}})$$

10.2 Sheaves

Write motivation on why sheaves capture more *geometric* information

We now turn our attention to a particular class of presheaves. Let's first tackle the case where the evaluation category is Set.

Definition 10.2.1. Let X be a topological space. The category of **sheaves of sets** over X is the full subcategory $\mathsf{Sh}_X(\mathsf{Set})$ of $\mathsf{PSh}_X(\mathsf{Set})$ whose objects verify, for every open set $U \subseteq X$ and every open cover $\{U_i\}_{i\in\mathcal{I}}$ of U

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections $(\forall i, j \in \mathcal{I} \quad s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j})$ there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = s_i$.

Presheaves that only satisfy axiom SH.1 are called **separated presheaves**.

Notation. From now on, to distinguish a presheaf from a sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

Let's try to make Definition 10.2.1 work on a more general evaluation category. To do this, the first thing we need to do is to write axioms SH.1 and SH.2 in terms of universal properties.

SH.1) Consider the morphism

$$\begin{aligned} e : \mathscr{F}(U) &\to \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \\ s &\mapsto \{s|_{U_i}\}_{i\in\mathcal{I}} \end{aligned}$$

given by the restriction maps res_{U,U_i}.
The condition SH.1 is equivalent to e being injective, which (in Set) means that e is a monomorphism.

SH.2) Consider the morphisms

$$p, q : \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j)$$

where p takes the section on U_i and sends it to the intersection and q takes the section on U_j and sends it to the intersection.
The condition SH.2 is then equivalent to saying that $\mathscr{F}$ is the equalizer of the diagram consisting of p and q.

Therefore a presheaf of sets $\mathscr{F}$ is a sheaf of sets if for every open set U and open cover $\{U_i\}_{i\in\mathcal{I}}$ of U the following diagram is an equalizer sequence

$$\mathscr{F}(U) \longrightarrow \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j)$$

Let's try to go now to a more general category C. First of all, note that the equalizer sequence makes sense if products indexed by $\mathcal{I}$ and $\mathcal{I}\times\mathcal{I}$ exist in C. Furthermore applying the functor $\mathrm{Hom}_{\mathsf{C}}(-,-):\mathsf{C}\times\mathsf{C}\to\mathsf{Set}$ by the Yoneda lemma we have that the diagram is an equalizer if and only if for every object $C\in\mathsf{C}$ the diagram

$$\mathrm{Hom}_{\mathsf{C}}(C,\mathscr{F}(U)) \longrightarrow \prod_{i\in\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C,\mathscr{F}(U_i)) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C,\mathscr{F}(U_i \cap U_j))$$

is an equalizer diagram in Set.

Definition 10.2.2 (Sheaf). Let X be a topological space. Let C be a category with products indexed by $\mathcal{I}$ and $\mathcal{I}\times\mathcal{I}$.
The category of C-valued **sheaves** over X is the full subcategory $\mathsf{Sh}_X(\mathsf{C})$ of the category of C-valued presheaves over X whose objects verify that, for every open subset $U\subseteq X$ and every open cover $\{U_i\}_{i\in\mathcal{I}}$ the diagram

$$\mathrm{Hom}_{\mathsf{C}}(C,\mathscr{F}(U)) \longrightarrow \prod_{i\in\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C,\mathscr{F}(U_i)) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C,\mathscr{F}(U_i \cap U_j))$$

is an equalizer sequence in Set.

Remark 10.2.1. From now on, every time we work with C-valued sheaves we will tacitly assume that C has sufficient products so that Definition 10.2.2 makes sense.

Remark 10.2.2. Note that the empty set $\varnothing\in\mathsf{Op}(X)$ is covered via the *empty cover* $\{U_i\}_{i\in\varnothing}$. Therefore for any sheaf $\mathscr{F}$ we have that $\mathscr{F}(\varnothing)$ is an element of the empty product, i.e: $\mathscr{F}(\varnothing)$ is a terminal object in C.
Therefore following Example 10.1.1 we have that, if $\{*\}$ is the one-point topological space,

$$\mathsf{Sh}_*(\mathsf{C})\cong\mathsf{C}$$

Remark 10.2.3. If C has small limits then the equalizer sequence in Definition 10.2.2 can be written directly in C.
In this case, a C-valued presheaf is a sheaf if for every open set $U\subseteq X$ and open cover $\{U_i\}_{i\in\mathcal{I}}$ the following diagram

$$\mathscr{F}(U) \longrightarrow \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j)$$

is an equalizer sequence in C.

10.2.1 Sheaves of Set-like categories

Let's consider the case where our category C comes with a functor $F:\mathsf{C}\to\mathsf{Set}$.

Here, for every C-valued presheaf $\mathcal{F}$ we have an associated *underlying presheaf of sets* $F\mathcal{F}$.

We want to know in which categories we can check the sheaf conditions in the underlying presheaf of sets.

We have the following diagram

$$\begin{array}{ccccc}
\mathscr{F}(U) & \longrightarrow & \prod_{i\in I}\mathscr{F}(U_i) & \rightrightarrows & \prod_{(i,j)\in I\times I}\mathscr{F}(U_i\cap U_j) \\
\downarrow & & \downarrow & & \downarrow \\
(F\mathscr{F})(U) & \longrightarrow & \prod_{i\in I}(F\mathscr{F})(U_i) & \rightrightarrows & \prod_{(i,j)\in I\times I}(F\mathscr{F})(U_i\cap U_j)
\end{array}$$

If we suppose that the top row is an equalizer sequence in C ($\mathscr{F}$ is a C-valued sheaf) then F needs to preserve limits to make the bottom row an equalizer sequence in Set ($F\mathscr{F}$ is a sheaf of sets).

Similarly, if we suppose that the bottom row is an equalizer sequence in Set ($F\mathscr{F}$ is a sheaf of sets) then F needs to reflect limits to make the top row an equalizer sequence in C ($\mathscr{F}$ is a C-valued sheaf).

Note that this does not happen in general for all concrete categories. For example the functor $F : \mathsf{Top} \to \mathsf{Set}$ sending every topological space to its underlying set does not reflect limits.

Indeed, consider $X = (\mathbb{R}\times\mathbb{R}, \mathcal{T})$ the real plane with the usual product topology and $Y = (\mathbb{R}\times\mathbb{R}, \mathcal{T}_d)$ the real plane with the discrete topology. Let $Z = (\mathbb{R}, \mathcal{T})$ be the real line with the usual topology.

Consider the diagram in Top

$$\begin{array}{ccc}
 & Y & \\
{\scriptstyle p_1}\swarrow & & \searrow{\scriptstyle p_2} \\
Z & & Z
\end{array}$$

Note that this is not a limit in Top. Indeed the diagram

$$\begin{array}{ccccc}
 & & X & & \\
 & {\scriptstyle \pi_1}\swarrow & & \searrow{\scriptstyle \pi_2} & \\
 & & Y & & \\
 & {\scriptstyle p_1}\swarrow & & \searrow{\scriptstyle p_2} & \\
Z & & & & Z
\end{array}$$

cannot be completed since there does not exist a continuous map $X \to Y$.

However, applying the forgetful functor to the diagram we get

$$\begin{array}{ccc}
 & \mathbb{R}^2 & \\
\swarrow & & \searrow \\
\mathbb{R} & & \mathbb{R}
\end{array}$$

which is a limit in Set (it is the diagram of a product).

For other categories such as Ban (the category of Banach spaces) the forgetful functor $F : \mathsf{Ban} \to \mathsf{Set}$ does not preserve limits.

Luckily *algebraic categories* such as $\mathsf{Grp}, \mathsf{Ab}, \mathsf{Ring}, \mathsf{Vect}_{\mathbb{K}}, \dots$ do preserve and reflect limits. Therefore in this categories we can *forget* the structure when checking for sheaf conditions.

10.3 Sheafification

We have established the definition of presheaves and sheaves. It is clear from the definitions that every sheaf is a presheaf. This defines a "forgetful" functor:

$$\iota : \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$$

Since sheaves are the objects with "local-to-global" gluing properties, we often wish to associate to an arbitrary presheaf $\mathcal{F}$ the "best possible" sheaf that approximates it. In the language of category theory, this problem is framed as finding a left adjoint to the inclusion functor.

Definition 10.3.1 (Sheafification). Let X be a topological space and C a category. The **sheafification** (or associated sheaf) functor, denoted by $-^{\#} : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$, is the left adjoint to the inclusion functor $\iota : \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$.
That is, for every presheaf $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and every sheaf $\mathcal{G} \in \mathsf{Sh}_X(\mathsf{C})$, there is a natural bijection:

$$\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathcal{F}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota\mathcal{G})$$

The unit of this adjunction gives a canonical morphism $\theta : \mathcal{F} \mapsto \iota(\mathcal{F}^{\#})$ in $\mathsf{PSh}_X(\mathsf{C})$.

The existence of the sheafification functor ... Therefore, we will first assume its existence and later show that it indeed exists for certain categories.

First of all, we can translate the concepts of direct image presheaf and inverse image presheaf to sheaves.

Definition 10.3.2. Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let $\mathscr{F}$ be a C-valued sheaf over X.
The **direct image sheaf** of $\mathscr{F}$ via f, $f_*\mathscr{F} \in \mathsf{Sh}_Y(\mathsf{C})$ is defined as

$$f_*\mathscr{F} = (f_*^{\mathrm{pre}} \iota\mathscr{F})^{\#}$$

where f_*^{pre} is the direct image presheaf of $\iota\mathscr{F} \in \mathsf{PSh}_X(\mathsf{C})$.
Therefore we have

$$\forall Y \in \mathsf{Op}(Y) \quad (f_*\mathscr{F})(V) = \left(f_*^{\mathrm{pre}} \iota\mathscr{F}\right)^{\#}(V) = (\iota\mathscr{F})^{\#}(f^{-1}V) = \mathscr{F}(f^{-1}V)$$

i.e: the direct image sheaf coincides with the direct image presheaf.
Note that this defines a functor $f_* : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Y(\mathsf{C})$.

We can also translate Example 10.1.2 and Example 10.1.3 to sheaves. Furthermore, since by Remark 10.2.2 $\mathscr{F}(\varnothing)$ is a terminal object in C the definitions simplify.

Definition 10.3.3 (Global section). Let X be a topological space. Let $\mathscr{F}$ be a C-valued sheaf over X.
Using Example 10.1.2 we can define the **global section** of $\mathscr{F}$ as

$$\Gamma(X, \mathscr{F}) := \mathscr{F}(X)$$

This induces a functor $\Gamma(X, -) : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$.

Definition 10.3.4 (Skyscraper sheaf). Let X be a topological space and $p \in X$. Let C be a category with a terminal object $* \in \mathsf{C}$.
Using Example 10.1.3 we can define the **skyscraper sheaf** supported at $S \in \mathsf{C}$ as

$$\forall U \in \mathsf{Op}(X) \quad \iota_{p,*}\underline{S} = \begin{cases} S & p \in U \\ * & p \notin U \end{cases}$$

This induces a functor $\iota_{p,*} : \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$.

Definition 10.3.5 (Inverse image sheaf). Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let C be a category with direct limits. Let $\mathscr{G}$ be a C-valued sheaf over Y.
The **inverse image sheaf** of $\mathscr{G}$, $f^{-1}\mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$, of $\mathscr{G}$ via f is the sheaf

$$f^{-1}\mathscr{G} := \left(f^{-1}_{\mathrm{pre}}\iota\mathscr{G}\right)^{\#}$$

Therefore we have

$$\forall U \in \mathsf{Op}(X) \quad \left(f^{-1}\mathscr{G}\right)(U) = \left(f^{-1}_{\mathrm{pre}}\iota\mathscr{G}\right)^{\#}(U) = \left(\varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(Y)}} \iota\mathscr{G}(V)\right)^{\#}$$

In this occassion, unlike Definition 10.3.2, we are stuck with taking sheafification.
Notice that this defines a functor $f^{-1} : \mathsf{Sh}_Y(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$.

Again, we can translate Example 10.1.4 and Example 10.1.5. We assume C is a category with direct limits as in the presheaf case.

Definition 10.3.6 (Constant sheaf). Let X be a topological space. The **constant sheaf** supported at $S \in \mathsf{C}$ is

$$\underline{S} := \left(\underline{S}_{\mathrm{pre}}\right)^{\#}$$

Note that this induces a functor $\underline{\cdot} : \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$.

Definition 10.3.7 (Stalk). Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued sheaf over X.
We define the **stalk** of $\mathscr{F}$ at p as

$$\mathscr{F}_p := \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathscr{F}(U)$$

Note that this induces a funtor $-_p : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$.

The adjunction of Theorem 10.1.1 also translates directly

Theorem 10.3.1. Let $f : X \to Y$ be a continuous function between two topological spaces X, Y. Let C be a category with direct limits. Then there exists an adjunction

$$f^{-1} \dashv f_* : \mathsf{PSh}_X(\mathsf{C}) \underset{f_*}{\overset{f^{-1}}{\rightleftarrows}} \mathsf{PSh}_Y(\mathsf{C})$$

Proof. Using the adjunctions $-^{\#} \dashv \iota$ and $f^{-1}_{\text{pre}} \dashv f_*$ we have

$$\begin{aligned}\text{Hom}_{\text{Sh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) &= \text{Hom}_{\text{PSh}_X(\mathsf{C})}(f^{-1}\iota\mathscr{G}, \iota\mathscr{F}) \\ &= \text{Hom}_{\text{PSh}_X(\mathsf{C})}(\iota\mathscr{G}, f_*\iota\mathscr{F}) \\ &= \text{Hom}_{\text{Sh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{F})\end{aligned}$$

□

Therefore we have the analogous adjunctions of Corollary 10.1.2 and Corollary 10.1.2 for sheaves.

Proposition 10.3.1. Let X be a topological space and $p \in X$. For any C-valued presheaf $\mathcal{F} \in \text{PSh}_X(\mathsf{C})$ we have that $\mathcal{F}_p = (\mathcal{F}^{\#})_p$.

Proof. Note that the stalk functor $-_p : \text{Sh}_X(\mathsf{C}) \to \mathsf{C}$ can be written as the $-^{\text{pre}}_p \circ \iota$ where $\iota : \text{Sh}_X(\mathsf{C}) \to \text{PSh}_X(\mathsf{C})$ is the inclussion functor and $-^{\text{pre}}_p : \text{PSh}_X(\mathsf{C}) \to \text{Arr}(\mathsf{C})$ is the presheaf stalk functor.

Note that $\iota^{\text{pre}}_{p,*} : \text{Arr}(\mathsf{C}) \to \text{PSh}_X(\mathsf{C})$ is, by Corollary 10.1.2, right adjoint to $-^{\text{pre}}_p$. Furthermore, by Definition 10.3.4 we have that $\iota_{p,*} : \mathsf{C} \to \text{Sh}_X(\mathsf{C})$ verifies $\iota_{p,*} = -^{\text{pre}}_p \circ \iota$.

Therefore if $F = (-^{\text{pre}}_p \circ \iota) \circ -^{\#}$ we have that its right adjoint is

$$R_F = \iota \circ \iota^{\text{pre}}_{p,*} = \iota_{p,*}$$

Therefore by *uniqueness* of the adjoint we have that $\mathcal{F}_p \cong \mathcal{F}^{\#}_p$. □

Proposition 10.3.2. Sheaf morphisms are determined by their induced maps on stalks

Proof. Let $\varphi, \psi : \mathscr{F} \to \mathscr{G}$ be two sheaf morphisms such that, for every point $p \in X$ $\varphi_x = \psi_x$. We want to show that, for every open subset $U \subseteq X$ the component maps $\varphi_U, \psi_U : \mathscr{F}(U) \to \mathscr{G}(U)$ are equal.

Let $U \subseteq X$ be an open subset and $s \in \mathscr{F}(U)$ be an arbitrary section. Consider $t_1 \coloneqq \varphi_U(s), t_2 \coloneqq \psi_U(s)$.

By the definition of stalk we have a commutative diagram

$$\begin{array}{ccc} \mathscr{F}(U) & \xrightarrow{\varphi_U} & \mathscr{G}(U) \\ \downarrow & & \downarrow \\ \mathscr{F}_p & \xrightarrow{\varphi_p} & \mathscr{G}_p \end{array}$$

Then for any $p \in U$ we have

$$\begin{aligned}(t_1)_p &= (\varphi_U(s))_p = \varphi_p(s_p) \\ (t_2)_p &= (\psi_U(s))_p = \psi_p(s_p)\end{aligned}$$

and therefore by hypothesis $(t_1)_p = (t_2)_p$ □

Existence of sheafification

We will study first the construction of the sheafification functor for the case where $\mathsf{C} = \mathsf{Set}$, and then prove that this construction can be carried in other categories.

Construction (Godement construction of sheafification)**.** Let $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{Set})$.

For every open set $U \subseteq X$, let

$$\Lambda(\mathcal{F})(U) := \prod_{p \in U} \mathcal{F}_p$$

Considering for $V \subseteq U$ the morphism

$$\begin{aligned} \mathrm{res}_{U,V} : \Lambda(\mathcal{F})(U) &\to \Lambda(\mathcal{F})(V) \\ \prod_{p \in U} \mathcal{F}_p &\mapsto \prod_{p \in V} \mathcal{F}_p \end{aligned}$$

as the restriction maps, we see that $\Lambda(\mathcal{F})$ is a Set-valued presheaf.

Let's check that $\Lambda(\mathcal{F})$ verifies the sheaf axioms. To do this, consider an open set $U \subseteq X$.

SH.1) Let $s, t \in \mathcal{F}(U)$ be two sections such that there exists an open covering $\{U_i\}_{i \in \mathcal{I}}$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = t|_{U_i}$.
Note that for any $x \in U$ since there exists $j \in \mathcal{I}$ such that $x \in U_j$ we have that $s_x = t_x$. Therefore, since this holds for all $x \in U$, we have that $s = t$.

SH.2) Let $\{U_i\}_{i \in \mathcal{I}}$ be an open covering of U and let $\{s_i \mid s_i \in \Lambda(\mathcal{F})(U_i)\}_{i \in \mathcal{I}}$ be a family of sections such that for any $i, j \in \mathcal{I}$ we have $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$.
Define a section $s \in \Lambda(\mathcal{F})(U)$ by choosing, for each $x \in U$, an index $k \in \mathcal{I}$ such that $x \in U_k$ and setting $s_x = (s_k)_x$. Note that this is well-defined by the compatibility condition of the family. Furthermore, for all $i \in \mathcal{I}$ by construction we have that $s|_{U_i} = s_i$.

Note that there is a presheaf morphism $\theta : \mathcal{F} \to \Lambda(\mathcal{F})$ defined for any $s \in \mathcal{F}(U)$ by sending s to its germ at every point:

$$\theta(U)(s) = (s_p)_{p \in U} \in \prod_{p \in U} \mathcal{F}_p$$

where s_p denotes the image of s in the stalk $\mathcal{F}_p$.

We define the sheafification $\mathcal{F}^\#$ of $\mathcal{F}$ as the sub-presheaf of $\Lambda(\mathcal{F})$ consisting of sections that are *locally compatible* with $\mathcal{F}$. Explicitly, for every open set $U \subseteq X$ an element $\sigma = (\sigma_p)_{p \in U} \in \Lambda(\mathcal{F})(U)$ belongs to $\mathcal{F}^\#(U)$ if and only if:

> For every $p \in U$, there exists an open neighborhood $V \subseteq U$ of p and a section $t \in \mathcal{F}(V)$ such that for all $q \in V$, $\sigma_q = t_q$ (as elements of $\mathcal{F}_q$).

We need to check that our definition of $\mathcal{F}^\#$ verifies the universal propery of sheafification. To do this, consider a sheaf $\mathcal{G}$ and a morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$. We wish to show that there exists a unique sheaf morphism $\psi : \mathcal{F}^\# \to \mathcal{G}$ such that $\phi \circ \theta = \psi$.

Let's check existence first by defining ψ explicitly. Let $U \subseteq X$ be an open set and $s \in \mathcal{F}^{\#}(U)$. By definition there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U and sections $t_i \in \mathcal{F}(V_i)$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = \theta(t_i)$.

We want ψ to verify $\psi(s)|_{U_i} = \psi(s|_{U_i}) = \psi(\theta_{t_i})$. This forces us to define $\psi(\theta(t_i)) = \phi(t_i) \in \mathcal{G}(U_i)$.

Let's see that $\{\phi(t_i)\}_{i\in\mathcal{I}}$ can be glued in $\mathcal{G}$. Note that for all $y \in U_i \cap U_j$ we have $(t_i)_y = (t_j)_y$. Therefore using the induced stalk morphism $\phi_y : \mathcal{F}_y \to \mathcal{G}_y$ we have $\phi_y((t_i)_y) = \phi_y((t_j)_y)$. Therefore $(\phi(t_i))_y = (\phi(t_j))_y$ for all $y \in U_i \cap U_j$, meaning that $\phi(t_i)|_{U_i\cap U_j} = \phi(t_j)|_{U_i\cap U_j}$.

Applying SH.2 to this family we get that there is a unique section $t \in \mathcal{G}(U)$ such that $\forall i \in \mathcal{I}\, t|_{U_i} = \phi(t_i)$. Therefore we can define $\psi_U(s) = t$, which verifies the universal property by construction.

To check uniqueness, since sheaf morphisms are determined by their induced maps on stalks and for all points $x \in X$ we have $\theta_x = \mathbb{1}$, ψ verifies $\forall x \in X \quad \psi_x = \phi_x$ making it unique.

Now that we have seen how to construct the sheafification of a sheaf of sets, we will see that it also exists for *algebraic structures*.

Definition 10.3.8. A **type of algebraic structure** is a category C together with a functor $F : \mathsf{C} \to \mathsf{Set}$ verifying:

1. F is faithful.
2. C has limits and F commutes with them.
3. C has direct limits and F commutes with them.
4. F reflects isomorphisms.

Example 10.3.1. The following categories, together with their respective forgetful functors are types of algebraic structures:

- The category of pointed sets.
- The category of (abelian) groups.
- The category of monoids.
- The category of rings.
- The category of R-modules for a fixed ring R.
- The category of Lie algebras over a fixed field.

Remark 10.3.1. If (C, F) is a type of algebraic structure then C-valued sheaves are well defined.

We state some properties of this types of categories without proof.

Lemma 10.3.1 (Properties of types of algebraic structures). Let (C, F) be a type of algebraic structure. Then:

1. C has a terminal object 0 and $F(0) = \{*\}$.
2. C has products and $F\left(\prod A_i\right) = \prod F\left(A_i\right)$.
3. C has fibre products and $F(A \times_B C) = F(A) \times_{F(B)} F(C)$.
4. C has equalizers and if $E \to A$ is the equalizer of $a, b : A \to B$ then $F(E) \to F(A)$ is the equalizer of $F(a), F(b) : F(A) \to F(B)$.
5. $A \to B$ is a monomorphism if and only if $F(A) \to F(B)$ is injective.

6. If $F(a) : F(A) \to F(B)$ is surjective then a is an epimorphism.
7. Given $A_1 \to A_2 \to \cdots$, $\operatorname{colim} A_i$ exists and $F(\operatorname{colim} A_i) = \operatorname{colim} F(A_i)$. And more generally for any direct limit.

Remark 10.3.2. It follows from the definition of a type of algebraic structure and Lemma 10.3.1 that F both preserves and reflects limits.
Therefore by subsection 10.2.1 we have that $\mathscr{F}$ is a C-valued sheaf if and only if its underlying presheaf of sets $F(\mathscr{F})$ is a sheaf of sets.

Lemma 10.3.2. Let (C, F) be a type of algebraic structure. Let $f : A \to B$ and $g : C \to B$ be two morphisms in C. Then if $F(g)$ is injective and $\operatorname{im} F(f) \subseteq \operatorname{im} F(g)$ we have that f factors as $f = g \circ t$ for some morphism $t : A \to C$ in C.

Proposition 10.3.3. prop:sheafification-of-algebraic-structure] Let X be a topological space. Let (C, F) be a type of algebraic structure. Let $\mathcal{F}$ be a C-valued presheaf over X. Then there exists a C-valued sheaf $\mathcal{F}^\#$ and a morphism $\mathcal{F} \to \mathcal{F}^\#$ such that

1. The map $\mathcal{F} \to \mathcal{F}^\#$ identifies the underlying sheaf of sets of $\mathcal{F}^\#$ with the underlying sheafification of the underlying presheaf of sets of $\mathcal{F}$.
2. For any C-valued sheaf $\mathcal{G}$ and any morphism $\mathcal{F} \to \mathcal{G}$ there exists a unique factorization $\mathcal{F} \to \mathcal{F}^\# \to \mathcal{G}$.

Proof. Consider the *product sheaf* defined as, for any open subset $U \subseteq X$

$$\prod(\mathcal{F})(U) = \prod_{x \in U} \mathcal{F}_x$$

This is well-defined because C has products. Notice that we have a map $\mathcal{F} \to \prod(\mathcal{F})$ defined by $\mathcal{F}(U) \mapsto \prod(\mathcal{F})(U)$.

For any open set $U \subseteq X$ we define $\mathcal{F}^\#(U)$ as the fiber product in C of the diagram

$$\begin{array}{ccc} & & \prod(\mathcal{F})(U) \\ & & \downarrow \\ \prod_{x \in U} \mathcal{F}_x & \longrightarrow & \prod_{x \in U} \prod(\mathcal{F})_x \end{array}$$

We have that

$$F(\mathcal{F}^\#(U)) = F\left(\prod(\mathcal{F})(U)\right) \times_{F(\prod(\mathcal{F})_x)} \prod F(\mathcal{F}_x)$$

Therefore by **[stacks-project]** we have property 1.

For property 2. consider a sheaf $\mathcal{G}$ and a morphism $\mathcal{F} \to \mathcal{G}$. By the universal property of sheafification there exists a map of sheaves of sets $\psi_{\mathsf{Set}} : F(\mathcal{F}^\#) \to F(\mathcal{G})$. Therefore using Lemma 10.3.2 we have

$$\begin{array}{ccc} \mathcal{F}^\#(U) & \dashrightarrow & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \prod_{x \in U} \mathcal{F}_x & \longrightarrow & \prod_{x \in U} \mathcal{G}_x \end{array}$$

□

Chapter 11

Ringed spaces

11.1 Sheaves of modules

Definition 11.1.1. Let X be a topological space. Let $\mathcal{O}_X$ be a presheaf of rings over X. A **presheaf of $\mathcal{O}_X$-modules** is a presheaf of abelian groups $\mathcal{F}$ together with a map of presheaves of sets $\mathcal{O} \times \mathcal{F} \to \mathcal{F}$ such that for every open subset $U \subseteq X$ the map $\mathcal{O}(U) \times \mathcal{F}(U) \to \mathcal{F}(U)$ defines the structure of an $\mathcal{O}(U)$-module on $\mathcal{F}(U)$.
A **morphism** of presheaves of $\mathcal{O}_X$-modules $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves of abelian groups $\varphi : \mathcal{F} \to \mathcal{G}$ such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{O}_X \times \mathcal{F} & \longrightarrow & \mathcal{F} \\ {\scriptstyle \mathbb{1}\times\varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ \mathcal{O}_X \times \mathcal{G} & \longrightarrow & \mathcal{G} \end{array}$$

We denote the category of presheaves of $\mathcal{O}_X$-modules over X as $\mathsf{PMod}(\mathcal{O}_X)$.

Notation. When the context is clear, we will drop the X subindex and say presheaves of $\mathcal{O}$-modules and $\mathsf{PMod}(\mathcal{O})$.

Definition 11.1.2. Let X be a topological space. Let $\mathcal{O}_1 \to \mathcal{O}_2$ be a morphism of presheaves of rings over a topological space X. We define the **restriction** functor

$$-_{\mathcal{O}_1} : \mathsf{PMod}(\mathcal{O}_2) \to \mathsf{PMod}(\mathcal{O}_1)$$

by using for any presheaf of $\mathcal{O}_2$-modules $\mathcal{F}$ the composition map $\mathcal{O}_1 \times \mathcal{F} \to \mathcal{O}_2 \times \mathcal{F} \to \mathcal{F}$.

Definition 11.1.3. Let X be a topological space. For any presheaves of rings $\mathcal{O}_1, \mathcal{O}_2$ define the **change of rings** functor

$$\mathcal{O}_2 \otimes_{\mathrm{pre},\mathcal{O}_1} - : \mathsf{PMod}(\mathcal{O}_1) \to \mathsf{PMod}(\mathcal{O}_2)$$

as, for any presheaf of $\mathcal{O}_1$-modules $\mathcal{G}$ and open subset $U \subseteq X$

$$(\mathcal{O}_2 \otimes_{\mathrm{pre},\mathcal{O}_1} \mathcal{G})(U) := \mathcal{O}_2(U) \otimes_{\mathcal{O}_1(U)} \mathcal{G}(U)$$

Lemma 11.1.1. For any $\mathcal{O}_1 \to \mathcal{O}_2$ morphism of presheaves of rings the restriction

functor and the change of rings functor are adjoint. In formula

$$\mathrm{Hom}_{\mathcal{O}_1}\left(\mathcal{G}, \mathcal{F}_{\mathcal{O}_1}\right) \cong \mathrm{Hom}_{\mathcal{O}_2}(\mathcal{O}_2 \oplus_{\mathrm{pre},\mathcal{O}_1} \mathcal{G}, \mathcal{F})$$

Proof. This follows directly from the adjuntion of the corresponding functors in Ring, see for example [**<empty citation>**]. □

Definition 11.1.4. Let X be a topological space. Let $\mathscr{O}_X$ be a sheaf of rings. The category of **sheaves of $\mathscr{O}_X$-modules** $\mathsf{Mod}(\mathscr{O}_X)$ is the full subcategory of $\mathsf{PMod}(\iota\mathscr{O}_X)$ consisting of presheaves of $\iota\mathscr{O}_X$-modules where the the underlying presheaves of abelian modules are sheaves.

Since, at the end of the day, presheaves of $\mathcal{O}$-modules and sheaves of $\mathscr{O}$-modules are presheaves and sheaves of abelian groups respectively, all the constructions we have seen in the past chapters apply.

For example, in the case of presheaves of $\mathcal{O}$-modules we have.

Lemma 11.1.2. Let X be a topological space and $x \in X$. Let $\mathcal{O}$ be a presheaf of rings over X. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$-modules.
The canonical map $\mathcal{O}_x \times \mathcal{F}_x \to \mathcal{F}_x$ coming from the map $\mathcal{O} \times \mathcal{F} \to \mathcal{F}$ defines an $\mathcal{O}_x$-module structure on the abelian group $\mathcal{F}_x$.

Lemma 11.1.3. Let X be a topological space and $x \in X$. Let $\mathcal{O} \to \mathcal{O}'$ be a morphism of presheaves of rings over X. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$-modules over X.
A $\mathcal{O}'_x$-modules we have that

$$\mathcal{F}_x \otimes_{\mathcal{O}_x} \mathcal{O}'_x \cong (\mathcal{F} \oplus_{\mathrm{pre},\mathcal{O}} \mathcal{O}')_x$$

The case of sheafification is a little bit more tricky. We have an analogous of **??**.

Proposition 11.1.1. Let X be a topological space. Let $\mathcal{O}$ be a presheaf of rings over X and $\mathcal{O}^\#$ be its sheafification. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$-modules. Let $\mathcal{F}^\#$ be the sheafification of $\mathcal{F}$ as a presheaf of abelian groups.
There exists a map of sheaves of sets $\mathcal{O}^\# \times \mathcal{F}^\# \to \mathcal{F}^\#$ making the diagram

$$\begin{array}{ccc} \mathcal{O} \times \mathcal{F} & \longrightarrow & \mathcal{F} \\ \downarrow & & \downarrow \\ \mathcal{O}^\# \times \mathcal{F}^\# & \longrightarrow & \mathcal{F}^\# \end{array}$$

commute and which makes $\mathcal{F}^\#$ into a sheaf of $\mathcal{O}^\#$-modules. Furthermore, for any sheaf of $\mathcal{O}^\#$-modules $\mathscr{G}$, all morphisms of presheaves of $\mathcal{O}$-modules $\mathcal{F} \to \mathscr{G}$ factor uniquely as $\mathcal{F} \to \mathcal{F}^\# \to \mathscr{G}$ where $\mathcal{F}^\# \to \mathscr{G}$ is a morphism of sheaves of $\mathcal{O}^\#$-modules.
This defines a sheafification functor

$$-^\# : \mathsf{PMod}(\mathcal{O}) \to \mathsf{Mod}(\mathcal{O}^\#)$$

Lemma 11.1.4. Let X be a topological space. Let $\mathcal{O}$ be a presheaf of rings over X.
The functors $-^\# : \mathsf{PMod}(\mathcal{O}) \to \mathsf{Mod}(\mathcal{O}^\#)$ and $\iota : \mathsf{Mod}(\mathscr{O}) \to \mathsf{PMod}(\iota\mathscr{O})$ are adjoint.
In formula

$$\mathrm{Hom}_{\mathsf{PMod}(\mathcal{O})}(\mathcal{F}, \iota\mathscr{G}) \cong \mathrm{Hom}_{\mathsf{Mod}(\mathcal{O}^\#)}(\mathcal{F}^\#, \mathscr{G})$$

Using this adjuntion we can *translate* Definition 11.1.2 and Definition 11.1.3.

Definition 11.1.5. Let X be a topological space. Let $\mathscr{O}_1 \to \mathscr{O}_2$ be a morphism of sheaves of rings. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_2$-modules.
We define the **restriction** of $\mathscr{F}$ to $\mathscr{O}_1$ as

$$\mathscr{F}_{\mathscr{O}_1} := \left(\iota\left(\mathscr{F}_{\mathscr{O}_1}\right)\right)^{\#} = \mathscr{F}_{\mathscr{O}_1}$$

where $\iota(\mathscr{F}_{\mathscr{O}_1})$ is the restriction as a presheaf of $\iota\mathscr{O}_2$-modules.
I.e: the restriction of sheaves of modules coincides with the restriction of presheaves of modules. This induces a functor

$$-_{\mathscr{O}_1} : \mathsf{Mod}(\mathscr{O}_2) \to \mathsf{Mod}(\mathscr{O}_1)$$

Definition 11.1.6. Let X be a topological space. Let $\mathscr{O}_1, \mathscr{O}_2$ be sheaves of rings over X. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_2$-modules.
We define the **change of basis** of $\mathscr{F}$ into $\mathscr{O}_2$ as

$$\mathscr{O}_2 \otimes_{\mathscr{O}_1} \mathscr{F} := \left(\iota\mathscr{O}_2 \otimes_{\mathrm{pre},\iota\mathscr{O}_1} \iota\mathscr{F}\right)^{\#}$$

This induces a functor

$$\mathscr{O}_2 \otimes_{\mathscr{O}_1} - : \mathsf{Mod}(\mathscr{O}_1) \to \mathsf{Mod}(\mathscr{O}_2)$$

We also have an analogous of Lemma 11.1.1

Lemma 11.1.5. Let X be a topological space. Let $\mathscr{O}_1 \to \mathscr{O}_2$ be a morphism of sheaves of rings over X.
The functors $-_{\mathscr{O}_1}$ and $\mathscr{O}_2 \otimes_{\mathscr{O}_1} -$ are adjoint. In formula, if $\mathscr{F}$ is a sheaf of $\mathscr{O}_2$-modules and $\mathscr{G}$ is a sheaf of $\mathscr{O}_1$-modules we have

$$\mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_1)}\left(\mathscr{G}, \mathscr{F}_{\mathscr{O}_1}\right) \cong \mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_2)}\left(\mathscr{O}_2 \otimes_{\mathscr{O}_1} \mathscr{G}, \mathscr{F}\right)$$

Lemma 11.1.6. Let X be a topological space and $x \in X$. Let $\mathscr{O} \to \mathscr{O}'$ be a morphism of sheaves of rings over X. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}$-modules.
As $\mathscr{O}'_x$-modules we have

$$\mathscr{F}_x \otimes_{\mathscr{O}_x} \mathscr{O}'_x \cong (\mathscr{F} \otimes_{\mathscr{O}} \mathscr{O}')_x$$

Theorem 11.1.1. $\mathsf{Mod}(\mathscr{O})_X$ is an abelian category

Proof. In Proposition 10.1.1 we have seen that limits and colimits of presheaves can be computed *open set by open set*. Therefore since the category of modules over a ring is abelian, the category $\mathsf{PMod}(\mathscr{O}_X)$ inherits this property.

Let's see how we can see that $\mathsf{Mod}(\mathscr{O}_X)$ is also abelian. It is clear that $\mathsf{Mod}(\mathscr{O}_X)$ is an additive category. Therefore, we only need to check that kernels and cokernels exist in $\mathsf{Mod}(\mathscr{O}_X)$.

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves of $\mathscr{O}$-modules. We have an associated morphism of presheaves of $\iota\mathscr{O}$-modules $\iota\varphi : \iota\mathscr{F} \to \iota\mathscr{G}$. Using that $\mathsf{PMod}(\iota\mathscr{O}_X)$ is an abelian category the morphism $\iota\varphi$ admits a kernel $\ker_{\mathrm{pre}} \iota\varphi$ and cokernel $\mathrm{coker}_{\mathrm{pre}} \iota\varphi$ in $\mathsf{PMod}(\iota\mathscr{O}_X)$. Therefore we can define the kernel and cokernel of φ using the

sheafification functor

$$\ker \varphi := \left(\ker_{\text{pre}} \iota\varphi\right)^{\#} = \ker_{\text{pre}} \left(\iota\varphi\right)^{\#} = \ker_{\text{pre}} \varphi$$
$$\operatorname{coker} \varphi := \left(\operatorname{coker}_{\text{pre}} \iota\varphi\right)^{\#}$$

where the equalities in $\ker \varphi$ come from the fact that right adjoints preserve limits. □

11.2 Ringed spaces

Definition 11.2.1 (Ringed space). A **ringed space** is a pair $(X, \mathscr{O}_X)$ consisting of a topological space X and a sheaf of rings $\mathscr{O}_X$ over X.
A **morphism** of ringed spaces $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a pair $(f, f^{\#})$ where $f : X \to Y$ is a continuous map and $f^{\#} : \mathscr{O}_y \to f_*\mathscr{O}_X$ is a morphism of sheaves.

11.2.1 Ringed spaces and continuous functions

We wish to translate what we did for sheaves in subsection 10.1.1 to sheaves of $\mathscr{O}_X$-modules.

The direct image is direct.

Definition 11.2.2. Let $(f, f^{\#}) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of ringed spaces.
Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules.
We define the **direct image** of $\mathscr{F}$ via f as the sheaf of $\mathscr{O}_Y$-modules which has the direct image sheaf $f_*\mathscr{F}$ as sheaf of abelian groups and with module structure given by the map $f^{\#} : \mathscr{O}_Y \to f_*\mathscr{O}_X$.
This defines a functor

$$f_* : \mathsf{Mod}(\mathscr{O}_X) \to \mathsf{Mod}(\mathscr{O}_Y)$$

However, as in the case for the inverse image for sheaves, the inverse image of a sheaf of $\mathscr{O}_X$-modules needs more work.

Explain why ⊗

Definition 11.2.3. Let $(f, f^{\#}) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of ringed spaces.
Let $\mathscr{G}$ be an $\Gamma(X, \mathscr{O}_X)$-module, where
We define the **inverse image** of $\mathscr{G}$ via f to be the sheaf of $\mathscr{O}_X$-modules

$$f^*\mathscr{G} := \mathscr{O}_X \otimes_{f^{-1}\mathscr{O}_Y} f^{-1}\mathscr{G}$$

where f^{-1} is the inverse image sheaf.
This defines a functor

$$f^* : \mathsf{Mod}(\mathscr{O}_Y) \to \mathsf{Mod}(\mathscr{O}_X)$$

Theorem 11.2.1. Let $(f, f^{\#}) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of ringed spaces.
The functors f_* and f^* are adjoint.
In formula, if $\mathscr{F}$ is a sheaf of $\mathscr{O}_X$-modules and $\mathscr{G}$ is a sheaf of $\mathscr{O}_Y$-modules we have

$$\operatorname{Hom}_{\mathsf{Mod}(\mathscr{O}_X)}\left(f^*\mathscr{G}, \mathscr{F}\right) \cong \operatorname{Hom}_{\mathsf{Mod}(\mathscr{O}_Y)}\left(\mathscr{G}, f_*\mathscr{F}\right)$$

Proof.

$$\begin{aligned}\mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_X)}\left(f^*\mathscr{G},\mathscr{F}\right) &\cong \mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_X)}\left(\mathscr{O}_X \otimes_{f^{-1}\mathscr{O}_Y} f^{-1}\mathscr{G},\mathscr{F}\right)\\ &\cong \mathrm{Hom}_{\mathsf{Mod}(f^{-1}\mathscr{O}_Y)}\left(f^{-1}\mathscr{G},\mathscr{F}_{f^{-1}\mathscr{O}_Y}\right)\\ &\cong \mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_Y)}\left(\mathscr{G},f_*\mathscr{F}\right)\end{aligned}$$

□

With the adjunction between f_* and f^* out of the way, let's now ...

Let $(X,\mathscr{O}_X)$ be a ringed space and $p \in X$.

Consider the one-point topological space $\{*\}$. Recall that by Remark 10.2.2 we have that $\mathsf{Sh}_{\{*\}}(\mathsf{Ring}) \cong \mathsf{Ring}$. Therefore to make $\{*\}$ into a ringed space we only need to *accompany* it by a ring.

We are interested in the direct and inverse images of the morphisms of ringed spaces induced by the maps

$$\pi : X \to \{*\} \qquad \iota_p : \{*\} \to X$$

Note that

$$\begin{aligned}\pi^\# &: \mathscr{O}_\pi \to \pi_*\mathscr{O}_X = \Gamma(X,\mathscr{O}_X)\\ \iota_p^\# &: \mathscr{O}_X \to \iota_{p,*}\mathscr{O}_{\iota_p} = \mathscr{O}_{\iota_p,p}\end{aligned}$$

Therefore we choose $\mathscr{O}_\pi = \Gamma(X,\mathscr{O}_X)$ and $\mathscr{O}_{\iota_p} = \mathscr{O}_{X,p}$.

Definition 11.2.4 (Global sections). Let $(X,\mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. Consider the morphism of ringed spaces

$$(\pi,\pi^*) : (X,\mathscr{O}_X) \to (\{*\},\Gamma(X,\mathscr{O}_X))$$

We define the **global sections** of $\mathscr{F}$ as the direct image of $\mathscr{F}$ via π:

$$\Gamma(X,\mathscr{F}) \coloneqq \pi_*\mathscr{F}$$

Note that this defines a functor

$$\Gamma(X,-) : \mathsf{Mod}(\mathscr{O}_X) \to \mathsf{Mod}(\Gamma(X,\mathscr{O}_X))$$

Definition 11.2.5 ("Constant" sheaf). Let $(X,\mathscr{O}_X)$ be a ringed space. Let M be a $\Gamma(X,\mathscr{O}_X)$ module. Consider the morphism of ringed spaces

$$(\pi,\pi^\#) : (X,\mathscr{O}_X) \to (\{*\},\Gamma(X,\mathscr{O}_X))$$

We define the **"constant" sheaf** supported at M as inverse image of M via π:

$$\mathscr{O}_X \otimes_{\Gamma(X,\mathscr{O}_X)} M \coloneqq \pi^*M = \mathscr{O}_X \otimes_{\pi^{-1}\Gamma(X,\mathscr{O}_X)} \pi^{-1}M = \mathscr{O}_X \otimes_{\underline{\Gamma(X,\mathscr{O}_X)}} \underline{M}$$

Note that this defines a functor

$$\mathscr{O}_X \otimes_{\Gamma(X,\mathscr{O}_X)} - : \mathsf{Mod}(\Gamma(X,\mathscr{O}_X)) \to \mathsf{Mod}(\mathscr{O}_X)$$

Corollary 11.2.1. Let $(X, \mathscr{O}_X)$ be a ringed space. The global sections functor and constant sheaf functor of sheaves of $\mathscr{O}_X$-modules are adjoint.
In formula, if $\mathscr{F} \in \mathsf{Mod}(\mathscr{O}_X)$ and $M \in \mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$

$$\mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_X)}(\mathscr{O}_X \otimes_{\Gamma(X,\mathscr{O}_X)} M, \mathscr{F}) \cong \mathrm{Hom}_{\mathsf{Mod}(\Gamma(X,\mathscr{O}_X))}(M, \Gamma(X, \mathscr{F}))$$

Definition 11.2.6 (Skyscraper sheaf). Let $(X, \mathscr{O}_X)$ be a ringed space with $p \in X$. Let M be a $\mathscr{O}_{X,p}$-module. Consider the morphism of ringed spaces

$$(\iota_p, \iota_p^\#) : (\{*\}, \mathscr{O}_{X,p}) \to (X, \mathscr{O}_X))$$

We define the **skyscraper sheaf** supported at M as the direct image of M via ι_p:

$$\iota_{p,*}\underline{M} \coloneqq \iota_{p,*}M$$

Note that this defines a functor

$$\iota_{p,*} : \mathsf{Mod}(\mathscr{O}_{X,p}) \to \mathsf{Mod}(\mathscr{O}_X)$$

Definition 11.2.7 (Stalk). Let $(X, \mathscr{O}_X)$ be a ringed space with $p \in X$. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. Consider the morphism of ringed spaces

$$(\iota_p, \iota_p^\#) : (\{*\}, \mathscr{O}_{X,p}) \to (X, \mathscr{O}_X))$$

We define the **stalk** of $\mathscr{F}$ at p as inverse image of $\mathscr{F}$ via ι_p:

$$\mathscr{F}_p \coloneqq \iota_p^*\mathscr{F} = \mathscr{O}_{X,p} \otimes_{\iota_p^{-1}\mathscr{O}_X} \iota_p^{-1}\mathscr{F} = \mathscr{O}_{X,p} \otimes_{\mathscr{O}_{X,p}} \mathscr{F}_p = \mathscr{F}_p$$

Note that this defines a functor

$$-_p : \mathsf{Mod}(\mathscr{O}_X) \to \mathsf{Mod}(\mathscr{O}_{X,p})$$

Corollary 11.2.2. Let $(X, \mathscr{O}_X)$ be a ringed space with $p \in X$. The skyscraper functor and stalk functor of sheaves of $\mathscr{O}_X$-modules are adjoint.
In formula, if $\mathscr{F} \in \mathsf{Mod}(\mathscr{O}_X)$ and $M \in \mathsf{Mod}(\mathscr{O}_{X,p})$

$$\mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_{X,p})}(\mathscr{F}_p, M) \cong \mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_X)}(\mathscr{F}, \iota_{p,*}\underline{M})$$

Lemma 11.2.1. Let $(f, f^\#) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of ringed spaces and $p \in X$. Let $\mathscr{G}$ be a sheaf of $\mathscr{O}_Y$-modules.
As $\mathscr{O}_{X,p}$ modules we have

$$(f^*\mathscr{G})_p = \mathscr{G}_{f(p)} \otimes_{\mathscr{O}_{Y,f(p)}} \mathscr{O}_{X,p}$$

11.3 Vector bundles

Definition 11.3.1 (Topological vector bundle). Let X be a topological space. Let $k \in \{\mathbb{R}, \mathbb{C}\}$ be either the field of real or complex numbers.
A **topological k-vector bundle** over X (denoted by $E \xrightarrow{\pi} X$) consists of

- A topological space E.
- A continuous map $\pi : E \to X$.
- For every point $x \in X$ the structure of a finite-dimensional k-vector space

over $E_x := \pi^{-1}(\{x\})$.

That admits a *local trivialization*, i.e: there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of X and for each $i \in \mathcal{I}$ an $n_i \in \mathbb{N}$ and a homeomorphism

$$\phi_i : U_i \times k^{n_i} \xrightarrow{\sim} \pi^{-1}(U_i)$$

satisfying

1. $\pi \circ \phi_i = \mathrm{pr}_1$, meaning that $\phi_i(\{x\} \times k^{n_i}) \subseteq \pi^{-1}(\{x\})$.
2. For all $x \in X$ the map $\phi_i(x) : k^{n_i} \to E_x$ is linear.

If all the numbers n_i are equal to some $n \in \mathbb{N}$ we say that the vector bundle has rank n.
A **morphism** of topological k-vector bundles from $E_1 \xrightarrow{\pi_1} X$ to $E_2 \xrightarrow{\pi_2} X$ is a continuous map $f : E_1 \to E_2$ such that $\pi_2 \circ f = \pi_1$ and for every $x \in X$ the map $f|_x : (E_1)_x \to (E_2)_x$ is linear.
We denote by $\mathsf{Vect}_k(X)$ the category of topological k-vector bundles over X.

Notation. A topological k-vector bundle of rank n is:

- A **real topological vector bundle** if $k = \mathbb{R}$.
- A **complex topological vector bundle** if $k = \mathbb{C}$.
- A **line bundle** if $n = 1$.

Example 11.3.1 (Trivial topological vector bundle)**.** Let X be a topological space and $n \in \mathbb{N}$. The product topological space $X \times k^n \xrightarrow{\mathrm{pr}_1} X$ is a topological k-vector bundle of rank n called the **trivial** topological vector bundle over X.

Remark 11.3.1. Let $E \xrightarrow{\pi} X$ be a topological k-vector bundle of rank n with local trivialization

$$\{\phi_i : U_i \times k^n \xrightarrow{\sim} \pi^{-1}(U_i)\}_{i\in\mathcal{I}}$$

For each $i, j \in \mathcal{I}$ the composite function

$$\phi_i^{-1} \circ \phi_j : (U_i \cap U_j) \times k^n \to (U_i \cap U_j) \times k^n$$

is well-defined and satisfies

$$\left(\phi_i^{-1} \circ \phi_j\right)(x, y) = (x, g_{i,j}(x)y)$$

for some $\mathrm{GL}_n(k)$-valued function

$$g_{i,j} : U_i \cap U_j \to \mathrm{GL}_n(k)$$

We call these functions $g_{i,j}$ the **transition functions** of the bundle.
Note that this functions satisfy $g_{i,i}(x) = \mathrm{Id}_n$ for all $x \in U_i$ and, for all open subsets U_i, U_j, U_k such that the bundle trivializes and $U_i \cap U_j \cap U_k \neq \varnothing$

$$\forall x \in U_i \cap U_j \cap U_k \quad g_{i,j}(x)g_{j,k}(x)g_{k,i}(x) = \mathrm{Id}_n$$

This is called a *Čech cocycle*.

Lemma 11.3.1. Let (E, X, π, k^n) be a fiber bundle with a $\mathrm{GL}_n(k)$ cocycle acting in the standard way on the fiber k^n. Then there exists a topological k-vector bundle with transition functions determined by the cocycle.

Example 11.3.2.

Examples of vector bundles: Cylinder, Möbius, ...

Definition 11.3.2 (Section of a topological vector bundle)**.** Let $E \xrightarrow{\pi} X$ be a topological k-vector bundle.
For any open set $U \subseteq X$ a **section** of $E \xrightarrow{\pi} X$ over U is a continuous map $\sigma : U \to E$ such that $\pi \circ \sigma = \mathbb{1}_U$.
The set of all sections of $E \xrightarrow{\pi} X$ over U is denoted by $\Gamma(U, E \xrightarrow{\pi} X)$.

Remark 11.3.2. The set $\Gamma(U, E \xrightarrow{\pi} X)$ admits the structure of a k-vector space.

Definition 11.3.3. Let $E \xrightarrow{\pi} X$ be a topological k-vector bundle. The **sheaf of sections** of $E \xrightarrow{\pi} X$ is

$$\begin{aligned} \mathcal{E} : \mathsf{Op}^{\mathrm{op}}(X) &\to \mathsf{Vect}_k \\ U &\mapsto \Gamma(U, E \xrightarrow{\pi} X) \end{aligned}$$

Proposition 11.3.1. Let $E \xrightarrow{\pi} X$ be a topological k-vector bundle. Let $C_X(k)$ be the sheaf of continuous k-valued functions over X.
Then the sheaf of sections $\mathcal{E}$ is a sheaf of $C_X(k)$-modules.

Proof. Let $U \subseteq X$ be an open subset. Let $f \in \mathscr{O}_X(U)$ and $\sigma \in \mathcal{E}(U)$.

We define the action $f \cdot \sigma$ pointwise: for every $x \in U$

$$(f \cdot \sigma)(x) := f(x) \cdot \sigma(x)$$

where the operation in the right is scalar multiplication in E_x.

Let's verify that $f \cdot \sigma$ is a valid section of $E \xrightarrow{\pi} X$.

1. **Linearity:** Since $(f \cdot \sigma)(x)$ is a scalar multiple of a vector in E_x, it remains in the fiber $\pi^{-1}(\{x\})$. Thus $\pi \circ (f \cdot \sigma) = \mathbb{1}_U$.
2. **Continuity:** Since the bundle admits local trivializations, scalar multiplication $k \times E \to E$ is a continuous map. Because f and σ are both continuous, their pointwise product $f \cdot \sigma$ is continuous.

The module axioms follow directly from the vector space structure on E_p. □

Definition 11.3.4 (Locally free sheaf)**.** A sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ is **locally free** of rank n if there exist an open cover $\{U_i\}_{i \in \mathcal{I}}$ of X such that for every $i \in \mathcal{I}$ there is an isomorphism of $\mathscr{O}_X|_{U_i}$-modules

$$\mathscr{F}|_{U_i} \cong \bigoplus_{j=1}^{n} \mathscr{O}_X|_{U_i} =: (\mathscr{O}_X|_{U_i})^{\oplus n}$$

We denote by $\mathsf{LocFree}_n(\mathscr{O}_X)$ the full subcategory of $\mathsf{Mod}(\mathscr{O}_X)$ consisting of locally free sheaves of $\mathscr{O}_X$-modules.

Theorem 11.3.1. Let $k \in \{\mathbb{R}, \mathbb{C}\}$. There is an equivalence of categories between the category of topological k-vector bundles of rank n over X and the category of locally free sheaves of $C_X(k)$-modules of rank n.
Under this correspondence

- The trivial topological k-vector bundle $X \times k^n$ corresponds to $(C_X(k))^{\oplus n}$.
- A section of $E \xrightarrow{\pi} X$ corresponds to a global section of $C_X(k)$.

Proof. In Proposition 11.3.1 we have seen that $\mathcal{E}$ defines a functor

$$\mathsf{Vect}_n(X) \longrightarrow \mathsf{LocFree}_n(C_X(k))$$

Therefore we only need to construct the inverse correspondence.

Let $\mathcal{E}$ be a locally free sheaf of $C_X(k)$-modules. Then there exists an open covering $\mathcal{U} = \{U_i\}_{i \in \mathcal{I}}$ of X such that for every $U_i \in \mathcal{U}$ there is an isomorphism of sheaves of $C_X(k)$-modules

$$\tau|_i : \mathcal{E}|_{U_i} \xrightarrow{\sim} \left(C_X(k)|_{U_i}\right)^{\oplus n}$$

Therefore on $U_i \cap U_j$ we have an isomorphism of locally free sheaves of $C_X(k)$-modules

$$\tau_j \circ \tau_i^{-1} : \left(C_X(k)|_{U_i \cap U_j}\right)^{\oplus n} \xrightarrow{\sim} \left(C_X(k)|_{U_i \cap U_j}\right)^{\oplus n}$$

Note that this isomorphism is given by an $n \times n$ matrix $M_{i,j}$ consisting of elements of $C_X(k)|_{U_i \cap U_j}$ that is invertible at every point.

This matrices are not arbitrary. Indeed they satisfy the *cocycle condition*:

$$M_{i,j} \cdot M_{j,k} \cdot M_{k,i} = \mathrm{Id}_n$$

Therefore we can assign to $\mathcal{E}$ the vector bundle of rank n, trivial on the open sets U_i of the covering, and whose transitions are the matrics $M_{i,j}$. Less abstractly, E is obtained by gluing $U_i \times \mathbb{R}^n$ via the identification

$$U_i \cap U_j \times \mathbb{R}^n \subset U_i \times \mathbb{R}^n \xrightarrow{\sim} U_i \times U_j \times \mathbb{R}^n \subset U_j \times \mathbb{R}^n$$

given by $\mathbb{1} \times M_{i,j}$.

This defines a functor

$$\mathsf{LocFree}_n(C_X(k)) \longrightarrow \mathsf{Vect}_n(X)$$

that by construction is *inverse* to the previous functor. □

Definition 11.3.5 (Smooth vector bundle). Let M be a smooth manifold. Let $k \in \{\mathbb{R}, \mathbb{C}\}$ be either the field of real or complex numbers.

A **smooth k-vector bundle** is a topological k-vector bundle $E \xrightarrow{\pi} M$ such that

1. E is a smooth manifold.
2. The projection map $\pi : E \to M$ is smooth.
3. The local trivialization maps $\phi_i : U_i \times k^{n_i} \xrightarrow{\sim} \pi^{-1}(U_i)$ are diffeomorphism.

A **morphism** of smooth k-vector bundles from $E_1 \xrightarrow{\pi_1} M$ to $E_2 \xrightarrow{\pi_2} M$ is morphism of topological vector bundles $f : E_1 \to E_2$ such that f is smooth.
We denote by $\mathsf{Vect}_k^\infty(X)$ the category of smooth k-vector bundles over X.

Definition 11.3.6 (Holomorphic vector bundle)**.** Let M be a complex manifold.

A **holomorphic vector bundle** is a topological $\mathbb{C}$-vector bundle $E \xrightarrow{\pi} M$ such that

1. E is a complex manifold.
2. The projection map $\pi : E \to M$ is an holomorphism.
3. The local trivialization maps $\phi_i : U_i \times \mathbb{C}^{n_i} \xrightarrow{\sim} \pi^{-1}(U_i)$ are biholomorphisms[a].

A **morphism** of holomorphic vector bundles from $E_1 \xrightarrow{\pi_1} M$ to $E_2 \xrightarrow{\pi_2} M$ is morphism of topological vector bundles $f : E_1 \to E_2$ such that f is holomorphic. We denote by $\mathsf{Vect}^{\omega}_{\mathbb{C}}(X)$ the category of holomorphic vector bundles over X.

[a] A map $f : \mathbb{C} \to \mathbb{C}$ is biholomorphic is it is holomorphic, bijective and its inverse is holomorphic.

Remark 11.3.3. Theorem 11.3.1 can be proved analogously to the case of smooth and holomorphic vector bundles.

PART IV

Sheaf cohomology

Chapter 12

Cohomology of sheaves

Let $(X, \mathscr{O}_X)$ be a ringed space.

We want to define the cohomology of X with coefficients in a sheaf of $\mathscr{O}_X$-modules as the right derived functor of the global sections functor

$$\Gamma(X, -) : \mathsf{Mod}(\mathscr{O}_X) \to \mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$$

To do this, we need to check the following:

1. $\mathsf{Mod}(\mathscr{O}_X)$ and $\mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$ are abelian categories.
2. The functor $\Gamma(X, -)$ is left exact.
3. $\mathsf{Mod}(\mathscr{O})_X$ has enough injectives.

In Theorem 11.1.1 we have proved that the category of sheaves of $\mathscr{O}_X$-modules is abelian. Furthermore, since $\Gamma(X, \mathscr{O}_X)$ is a ring, $\mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$ is also abelian.

Recall that by Corollary 11.2.1 $\Gamma(X, -)$ is right adjoint. Therefore by **??** $\Gamma(X, -)$ is left exact.

It only remains to show that $\mathsf{Mod}(\mathscr{O}_X)$ has enough injectives.

Lemma 12.0.1. Let $(X, \mathscr{O}_X)$ be a ringed space. The category $\mathsf{Mod}(\mathscr{O}_X)$ of sheaves of $\mathscr{O}_X$-modules over X has enough injectives.

Proof. We will use the following claim

> The category $\mathsf{Mod}(R)$ of modules over a ring R has enough injectives.

For a proof, see [Alu09, Corollary VIII.6.4].

Let $\mathscr{F}$ be an $\mathscr{O}_X$-module. Let $\mathscr{F}_p \in \mathsf{Mod}(\mathscr{O}_{X,p})$ be that stalk of $\mathscr{F}$ at a point $p \in X$.

Since $\mathsf{Mod}(\mathscr{O}_{X,p})$ has enough injectives we can choose an object I_p and a monomorphism

$$i_p : \mathscr{F}_p \rightarrowtail I_p$$

Consider the sheaf

$$\mathscr{I} := \bigoplus_{p \in X} \iota_{p,*} \underline{I_p}$$

where $\iota_{p,*}I_p$ is the skyscraper sheaf. More explicitly, for every open set $U \subseteq X$ the sheaf $\mathscr{I}$ is defined as

$$\mathscr{I}(U) = \bigoplus_{p \in U} I_p$$

Recall that by Corollary 11.2.2 the skyscraper functor is left-adjoint to the stalk functor, which is exact. Therefore by **??** $\mathscr{I}$ is an injective object.

Furthermore, there exists a monomorphism $\phi : \mathscr{F} \to \mathscr{I}$ defined as, for any open set $U \subseteq X$

$$\begin{aligned} \phi(U) : \mathscr{F}(U) &\to \mathscr{I}(U) \\ s &\mapsto \bigoplus_{p \in U} j_p(s_p) \end{aligned}$$

□

Having seen this, we now can finally define sheaf cohomology.

Definition 12.0.1 (Sheaf Cohomology). Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules over a topological space X. The i-th **cohomology** of X with coefficients in $\mathscr{F}$ as the i-th right derived functor of the global sections functor:

$$\mathrm{H}^i(X, \mathscr{F}) \coloneqq R^i\Gamma(X, \mathscr{F})$$

Therefore, given a sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ to compute its cohomology we *only* need to construct an injective (or $\Gamma(X, -)$ acyclic by **??**) resolution of $\mathscr{F}$

$$0 \longrightarrow \mathscr{F}^{(0)} \longrightarrow \mathscr{F}^{(1)} \longrightarrow \cdots$$

and compute the cohomology of the $\Gamma(X, \mathscr{O}_X)$-modules

$$0 \longrightarrow \Gamma(X, \mathscr{F}^{(0)}) \longrightarrow \Gamma(X, \mathscr{F}^{(1)}) \longrightarrow \cdots$$

Since injective resolutions are usually hard to find (note how Lemma 12.0.1 doesn't give an *easy* construction) we will mostly work with $\Gamma(X, -)$ acyclic resolutions.

12.1 The Godement resolution

Definition 12.1.1 (Flasque sheaf). A sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ is **flasque** (or **flabby**) if for any inclussion of open subsets $V \subseteq U$ the restriction map $\mathrm{res}_{U,V} : \mathscr{F}(U) \to \mathscr{F}(V)$ is surjective.

Proposition 12.1.1. Flasque sheaves are $\Gamma(X, -)$ acyclic.

Proof. Let $\mathscr{F}$ be a flasque sheaf. Since $\mathsf{Mod}(\mathscr{O}_X)$ has enough injectives we have a short exact sequence

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{I} \longrightarrow \mathscr{G} \longrightarrow 0$$

where $\mathscr{I}$ is an injective object.

We will first show that, for any open subset $U \subseteq X$, the map $\mathscr{I}(U) \twoheadrightarrow \mathscr{G}(U)$ is surjective. Let $s \in \mathscr{G}(U)$. Let $V, W \subseteq U$ be two open subsets and $\tau_V \in \mathscr{I}(V), \tau_W \in \mathscr{I}(W)$ be two sections lifting s to $\mathscr{I}$. Consider

$$\tau_V|_{V\cap U} - \tau_W|_{V\cap W} \in \mathscr{F}(V \cap W)$$

Since $\mathscr{F}$ is flasque there exists a section $\chi_V \in \mathscr{F}(V)$ such that $\chi_V|_{V\cap W} = \tau_V - \tau_W$. Therefore if $\tau'_V = \tau_V - \chi_V$ we have $\tau'_V|_{V\cap W} = \tau_W|_{V\cap W}$, i.e: there exists a section $\tau \in \mathscr{I}(V \cup W)$ such that $\tau|_V = \tau'|_V$ and $\tau'|_W = \tau|_W$. Clearly τ is sent to $\sigma|_{V\cap W}$.

Consider now a pair (W, τ) which is maximal (for the obvious order relation), where W is an open subset of U and τ is a section of $\mathscr{I}$ lifting σ to W. Since σ lifts locally on $\mathscr{I}$ and using the previous result we see that $W = U$.

But then the long exact sequence in cohomology of our sequence and the fact that for $i > 0 \quad \mathrm{H}^i(X, \mathscr{I}) = 0$ imply that $\mathrm{H}^1(X, \mathscr{F}) = 0$ and

$$\forall k - 1 \geq 1 \quad \mathrm{H}^k(X, \mathscr{F}) = \mathrm{H}^{k-1}(X, \mathscr{G})$$

Furthermore surjectivity of the maps $\mathscr{I}(U) \twoheadrightarrow \mathscr{G}(U)$ implies that $\mathscr{G}$ is flasque.

It follows by induction that

$$\forall k > 0 \quad \mathrm{H}^k(X, \mathscr{F}) = 0$$

□

Using flasque sheaves we can construct a canonical (and functorial) resolution of a sheaf of $\mathscr{O}_X$-modules to compute its cohomology.

Construction (Godement resolution)**.** Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. The **Godement resolution** of $\mathscr{F}$ is constructed by considering the inclussion of $\mathscr{F}$ into the *Godement sheaf* $\mathscr{F}_{\mathrm{God}}$

$$U \mapsto \bigoplus_{p\in U} \mathscr{F}_p =: \mathscr{F}_{\mathrm{God}}(U)$$

Note that $\mathscr{F}_{\mathrm{God}}$ is flasque.

We then inject the quotient $\mathscr{F}_{\mathrm{God}} / \mathscr{F}$ into $\left(\mathscr{F}_{\mathrm{God}} / \mathscr{F}\right)_{\mathrm{God}}$ and so on.

Example 12.1.1. Let's try to compute sheaf cohomology using the Godement resolution.
Let $X = \{p, q\}$ with the *Sierpiński* topology $\mathcal{T} = \{\varnothing, \{p\}, X\}$. Consider the constant sheaf supported at the integers $\mathscr{F} = \underline{\mathbb{Z}}$.
To compute the cohomology of $\mathscr{F}$, we are going to construct the complex of global sections:

$$0 \longrightarrow \Gamma(X, \mathscr{F}_{\mathrm{God}}) \xrightarrow{d^0} \Gamma\left(X, \left(\mathscr{F}_{\mathrm{God}} / \mathscr{F}\right)_{\mathrm{God}}\right) \xrightarrow{d^1} \cdots$$

Note that the stalks of $\mathscr{F}$ are:

$$\mathscr{F}_p = \mathscr{F}(\{p\}) = \mathbb{Z}, \qquad \mathscr{F}_q = \mathscr{F}(X) = \mathbb{Z}.$$

Consider the Godement sheaf associated to $\mathscr{F}$:

$$\mathscr{F}_{\text{God}}(U) = \bigoplus_{x \in U} \mathscr{F}_x.$$

We have that:

$$\Gamma(X, \mathscr{F}_{\text{God}}) = \mathscr{F}_{\text{God}}(X) = \mathscr{F}_p \oplus \mathscr{F}_q = \mathbb{Z} \oplus \mathbb{Z}.$$

This means that we have a map d^{-1} as follows:

$$d^{-1} : \underbrace{\Gamma(X, \mathscr{F})}_{\mathbb{Z}} \to \underbrace{\Gamma(X, \mathscr{F}_{\text{God}})}_{\mathbb{Z}\oplus\mathbb{Z}}, \quad d^{-1}(k) = (k, k).$$

Let $\mathscr{Q}^1 = {}^{\mathscr{F}_{\text{God}}}\big/_{\mathscr{F}}$ be the cokernel sheaf. If we compute its stalks, we have that:

$$(\mathscr{Q}^1)_p = {}^{\mathscr{F}_{\text{God},p}}\big/_{\mathscr{F}_p} = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0,$$
$$(\mathscr{Q}^1)_q = {}^{\mathscr{F}_{\text{God},q}}\big/_{\mathscr{F}_q} = {}^{\mathbb{Z}\oplus\mathbb{Z}}\big/_{\langle (k,k) \rangle} \cong \mathbb{Z}.$$

Therefore, the global sections of the Godement sheaf of $\mathscr{Q}^1$ are:

$$\Gamma\left(X, (\mathscr{Q}^1)_{\text{God}}\right) = (\mathscr{Q}^1)_p \oplus (\mathscr{Q}^1)_q = 0 \oplus \mathbb{Z} = \mathbb{Z}.$$

This produces a differential map:

$$d^0 : \underbrace{\Gamma(X, \mathscr{F}_{\text{God}})}_{\mathbb{Z}\oplus\mathbb{Z}} \to \underbrace{\Gamma\left(X, (\mathscr{Q}^1)_{\text{God}}\right)}_{\mathbb{Z}}, \quad d^0(m, n) = m - n.$$

Let $\mathscr{Q}^2$ be the cokernel of the map $\mathscr{F}_{\text{God}} \to (\mathscr{Q}^1)_{\text{God}}$. If we compute its stalks, we have:

$$(\mathscr{Q}^2)_p = {}^{0}\big/_{0} = 0,$$
$$(\mathscr{Q}^2)_q = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0.$$

Therefore, we have that:

$$\Gamma\left(X, (\mathscr{Q}^2)_{\text{God}}\right) = 0 \oplus 0 = 0.$$

This means that our resolution of global sections is:

$$0 \longrightarrow \mathbb{Z} \oplus \mathbb{Z} \xrightarrow[(m,n)\mapsto m-n]{d^0} \mathbb{Z} \longrightarrow 0$$

Therefore, the cohomology groups of X with coefficients in $\underline{\mathbb{Z}}$ are:

$$\mathrm{H}^0(X, \underline{\mathbb{Z}}) = \ker(d^0) = \{(m, n) \in \mathbb{Z} \oplus \mathbb{Z} \mid m - n = 0\} \cong \mathbb{Z},$$
$$\mathrm{H}^1(X, \underline{\mathbb{Z}}) = \operatorname{coker}(d^0) = {}^{\mathbb{Z}}\big/_{\operatorname{im}(d^0)} = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0,$$
$$\mathrm{H}^i(X, \underline{\mathbb{Z}}) = 0 \quad \forall i > 1.$$

Notice that if the Godement resolution makes computing sheaf cohomology difficult on a *simple* space, it renders the task nearly impossible on more *complicated* spaces.

To address this, the upcoming chapters will explore how we can select different,

more efficient resolutions to compute cohomology on sufficiently nice topological spaces.

Chapter 13

Cohomology and homotopy

Our goal in this chapter will be to prove that sheaf cohomology of locally constant sheaves are a homotopy invariant.

We will denote $I := [0,1]$ the unit interval.

13.1 Sheaves on the interval $[0,1]$

Lemma 13.1.1. Let $\mathscr{F} \in \mathsf{Mod}(\underline{R}_X)$. Then

1. For $j > 1$ we have $\mathrm{H}^j(I;\mathscr{F}) = 0$.
2. If $\mathscr{F}(I) \to \mathscr{F}_t$ is an epimorphism for all $t \in I$ then $\mathrm{H}^1(I;\mathscr{F}) = 0$.

Proof. Let $j \geq 1$ and $s \in \mathrm{H}^j(I;\mathscr{F})$.

For $0 \leq t_1 < t_2 \leq 1$ consider the morphism

$$f_{t_1,t_2} : \mathrm{H}^j(I;\mathscr{F}) \to \mathrm{H}^j([t_1,t_2],\mathscr{F})$$

and let

$$J := \{t \in [0,1] \mid f_{0,t}(s) = 0\}$$

Since $\mathrm{H}^j(\{0\},\mathscr{F}) = 0$ we have that $0 \in J$. Furthermore since $f_{0,t}(s) = 0$ implies $f_{0,t'}(s) = 0$ for $t' \in [0,t]$, J is an interval and thus connected.

We want to see that J is both open and closed.

Open By ... we have that

$$\mathrm{H}^j([0,t_0];\mathscr{F}) \cong \varinjlim_{t>t_0} \mathrm{H}^j([0,t];\mathscr{F})$$

Therefore J is open.

Closed For $t \in [0,t_0]$ consider the Mayer-Vietoris sequence

$$\cdots \longrightarrow \mathrm{H}^j([0,t_0];\mathscr{F}) \longrightarrow \mathrm{H}^j([0,t];\mathscr{F}) \oplus \mathrm{H}^j([t,t_0];\mathscr{F}) \longrightarrow \mathrm{H}^j(\{t\},\mathscr{F}) \longrightarrow \cdots$$

For $j > 1$ or for $j = 1$ and assuming that $\mathrm{H}^0(I;\mathscr{F}) \twoheadrightarrow \mathrm{H}^0(\{t\},\mathscr{F})$ is surjective we get

$$\mathrm{H}^j([0,t_0],\mathscr{F}) \cong \mathrm{H}^j([0,t];\mathscr{F}) \oplus \mathrm{H}^j([t,t_0];\mathscr{F})$$

Let $t_0 = \sup\{t \mid t \in J\}$. Then $f_{0,t}(s) = 0$ for all $t < t_0$. On the other hand by ...

$$\varinjlim_{t<t_0} \mathrm{H}^j([t, t_0], \mathscr{F}) \cong \mathrm{H}^j(\{t_0\}; \mathscr{F}) = 0$$

Therefore there exists $t < t_0$ with $f_{t,t_0}(s) = 0$ which implies $f_{0,t_0}(s) = 0$ meaning that $t_0 \in J$.

□

13.2 Invariance by homotopy

Definition 13.2.1. Let X, Y be two topological spaces

- Two continuous maps $f_0, f_1 : X \rightrightarrows Y$ are **homotopic** if there exists a continuous map $h : I \times X \to Y$ such that, for all $x \in X$

$$f_0(x) = h(0, x) \quad f_1(x) = h(1, x)$$

  We denote this by $f_0 \simeq f_1$.
- A continuous map $f : X \to Y$ is a **homotopy equivalence** if there exists a continuous map $g : Y \to X$ such that

$$g \circ f \simeq \mathrm{id}_X \quad f \circ g = \mathrm{id}_Y$$

  In this case we say that X and Y are **homotopic** and denote it by $X \simeq Y$.
- A topological space Z is **contractible** if $Z \simeq \{*\}$.

Theorem 13.2.1 (Invariance by homotopy)**.** Let $f_0, f_1 : X \rightrightarrows Y$ be two homotopic maps and let $\mathscr{G}$ be a locally constant sheaf

Chapter 14

de Rham Cohomology

Definition 14.0.1 (Partition of unity). Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. A **partition of unity** of $\mathscr{F}$ subordinate to a locally finite open covering $\mathcal{U} = \{U_i\}_{i\in\mathcal{I}}$ is a collection of endomorphisms of $\mathscr{O}_X$-modules

$$\{f_i : \mathscr{F} \to \mathscr{F}\}_{i\in\mathcal{I}}$$

satisfying

1. For all $i \in \mathcal{I}$
$$\operatorname{supp} f_i := \overline{\{x \in X \mid (f_i)_x \neq 0\}} \subset U_i$$
2. $$\sum_{i\in\mathcal{I}} f_i = \mathbb{1}_{\mathscr{F}}$$
Note that this sum is well defined since for any point $x \in X$ there is an open neighbourhood where all but finitely many of the maps f_i are the zero maps, i.e: the sum is locally finite.

Definition 14.0.2 (Fine sheaf). A sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ is **fine** if for every open cover $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ of X there exists a partition of unity $\{f_i\}_{i\in\mathcal{I}}$ subordinate to this covering.

Proposition 14.0.1. Fine sheaves are $\Gamma(X, -)$ acyclic.

Proof. Let $\mathscr{F}$ be a fine sheaf. Using the Godement resolution we have that $\mathscr{F}$ admits a flasque resolution $\mathscr{F}[0] \to \mathscr{I}^\bullet$.

Using Proposition 12.1.1 by **??** we have that

$$\mathrm{H}^k(X, \mathscr{F}) = \ker\left(\Gamma(X, \mathscr{I}^k) \to \Gamma(X, \mathscr{I}^{k+1})\right) \Big/ \operatorname{im}\left(\Gamma(X, \mathscr{I}^{k-1}) \to \Gamma(X, \mathscr{I}^k)\right)$$

Let $\alpha \in \ker\left(\Gamma(X, \mathscr{I}^k) \to \Gamma(X, \mathscr{I}^{k+1})\right)$. Since $\mathscr{I}^\bullet$ is locally exact in degree greater than 0 we have that α comes from $\mathscr{I}^{k-1}$, i.e: there exists an open cover $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ of X such that $\alpha|_{U_i} = d\beta_i$ with $\beta_i \in \mathscr{I}^{k-1}(U_i)$.

Consider a partition of unity $\{f_i\}_{i\in\mathcal{I}}$ subordinate to $\mathscr{U}$ and let

$$\beta := \sum_{i\in\mathcal{I}} f_i\beta_i$$

where the sum is locally finite. Note that $f_i\beta_i$ is a section of $\mathscr{I}^{k-1}(X)$ with values in $\mathscr{I}_p^{k-1}(X)$:

$$(f_i\beta_i)_p = \begin{cases} 0 & p \notin U_i \\ f_i\beta_i & p \in U_i \end{cases}$$

Since f_i has support in U_i this defines a section in $\mathscr{I}^{k-1}(U_i)$.

We have that $d\beta = \alpha$ since $\alpha = \sum_{i\in\mathcal{I}} f_i\alpha|_{U_i}$. Therefore $\ker\left(\Gamma(X,\mathscr{I}^k) \to \Gamma(X,\mathscr{I}^{k+1})\right) = \operatorname{im}\left(\Gamma(X,\mathscr{I}^{k-1}) \to \Gamma(X,\mathscr{I}^k)\right)$, which means

$$\forall k > 0 \quad \mathrm{H}^k(X,\mathscr{F}) = 0$$

□

Definition 14.0.3. Let M be a smooth manifold of dimension m.
The **bundle of k-forms** $\Lambda^k(\mathrm{T}^*M)$ is the smooth vector bundle defined with:

- *Total space*: $\Lambda^k(\mathrm{T}^*M) = \coprod_{p\in M} \Lambda^k(\mathrm{T}_p^*M)$.
- *Projection map*: $\pi : \Lambda^k(\mathrm{T}^*M) \to M,\ \pi(p,\omega) = p$.
- *Fiber space*: For every $p \in M$ the fiber $E_p = \pi^{-1}(\{p\}) = \Lambda^k(\mathrm{T}_p^*M)$ is a real vector space of dimension $\binom{m}{k}$.

To define the local trivialization, consider a local chart $(U, \{x^i\}_{i=1}^m)$ of M. Then the differentials $\{\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}\}_{1\le i_1<\cdots<i_k\le m}$ form a basis for each fiber with base point in U and the map $\phi^{-1} : \pi^{-1}(U) \to U \times \mathbb{R}^{\binom{m}{k}}$ is given by sending a form ω to its coefficients in this base.

Notation. Let $I = (i_1, \ldots, i_k)$ be a multi-index. We will write

$$\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \mathrm{d}x^{i_k}$$

In this way the basis of the local trivialization is $\{\mathrm{d}x^I\}_I$.

Definition 14.0.4 (Smooth k-form)**.** Let M be a smooth manifold of dimension m.
The **sheaf of k-forms** $\Omega^k(M)$ over M is the sheaf of sections of the bundle of k-forms $\Lambda^k(\mathrm{T}^*M)$.
More concretely, a k-form over an open set U is a map $\omega : U \to \Lambda^k(\mathrm{T}^*M)$ such that $\pi \circ \omega = \mathbb{1}_U$.

Example 14.0.1. Let M be a smooth manifold of dimension m.
Note that $\Lambda^0(\mathrm{T}^*M) = M \times \mathbb{R}$ is the trivial vector bundle. Therefore a 0-form is a map $\omega : M \to M \times \mathbb{R}$ such that $\pi \circ \omega = \mathbb{1}_M$, i.e: ω is a smooth function.
This means that $\Omega^0(M) \cong C^\infty(M)$.

Example 14.0.2. Let M be a smooth manifold of dimension m.
Suppose that $k > m$. Since the fibers of the bundle of k-forms $\Lambda^k(\mathrm{T}^*M)$ have dimension $\binom{m}{k}$ this means that at least one index i in the multi-index I will be repeated. Therefore since $\mathrm{d}x^i \wedge \mathrm{d}x^i = 0$ we have that $\mathrm{d}x^I = 0$, i.e: $\Lambda^k(\mathrm{T}^*M) = 0$.
In particular $\Omega^k(M) = 0$ for all $k > m$.

Definition 14.0.5 (Exterior derivative)**.** Let ω be a k-form over a smooth manifold M.

If $(U, \{x^i\}_{i=1}^m)$ is a local chart then there exist functions $\alpha_I \in C^\infty(M)$ such that

$$\omega = \sum_I \alpha_I \mathrm{d}x^I$$

where $I = (i_1, \dots, i_k)$ is a multi-index and $\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \mathrm{d}x^{i_k}$.
We can define the **exterior derivative** of ω as the $k+1$-form

$$\mathrm{d}\omega = \sum_I \mathrm{d}\alpha_I \wedge \mathrm{d}x^I = \sum_I \left(\sum_{j=1}^m \frac{\partial \alpha_I}{\partial x^j} \mathrm{d}x^j \right) \wedge \mathrm{d}x^I$$

Lemma 14.0.1. Let M be a smooth manifold of dimension m.
Then for any k-form the exterior derivative verifies $\mathrm{d}^2\omega = 0$, i.e: the sequence

$$\Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \Omega^2(M) \xrightarrow{\mathrm{d}} \cdots$$

is a complex of sheaves of $C^\infty(M)$-modules.

Proof. It suffices to show that for any smooth function $f \in C^\infty(M) \cong \Omega^0(M)$ we have $\mathrm{d}^2 f = 0$. Indeed, if $\mathrm{d}^2 f = 0$ then for any *basic* form $f\mathrm{d}x^I$ we have

$$\mathrm{d}(f\mathrm{d}x^I) = \mathrm{d}(\mathrm{d}f \wedge \mathrm{d}x^I) = \mathrm{d}^2 f \wedge \mathrm{d}x^I - \mathrm{d}f \wedge \mathrm{d}^2 x^I = 0$$

Let $f \in C^\infty(M)$ be a smooth function. Its exterior derivative is

$$\mathrm{d}f = \sum_{i=1}^m \frac{\partial f}{\partial x^i} \mathrm{d}x^i$$

Differentiating again we get

$$\mathrm{d}^2 f = \sum_{j=1}^m \frac{\partial}{\partial x^j} \left(\sum_{i=1}^m \frac{\partial f}{\partial x^i} \right) \wedge \mathrm{d}x^j = \sum_{i,j=1}^m \frac{\partial^2 f}{\partial x^i \partial x^j} \mathrm{d}x^i \wedge \mathrm{d}x^j$$

By Clairaut's theorem we have that $\frac{\partial^2 f}{\partial x^i \partial x^j} = \frac{\partial^2 f}{\partial x^j \partial x^i}$ and by the antisymmetry of the wedge product $\mathrm{d}x^i \wedge \mathrm{d}x^j = -\mathrm{d}x^j \wedge \mathrm{d}x^j$. Therefore all the terms cancel leaving

$$\mathrm{d}^2 f = 0$$

□

Definition 14.0.6 (de Rham complex)**.** Let M be a smooth manifold of dimension m.
The **de Rham complex** of M is the complex of sheaves of $C^\infty(M)$-modules

$$\Omega^\bullet(M) := \Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \cdots \xrightarrow{\mathrm{d}} \Omega^m(M) \xrightarrow{\mathrm{d}} 0$$

where $\Omega^k(M)$ is the sheaf of k-forms over M and d is the exterior derivative.

Lemma 14.0.2 (Poincaré)**.** Let $U \subseteq \mathbb{R}^n$ be a contractible open set and $k \geq 1$. If $\omega \in \Omega^k(U)$ is a k-form such that $\mathrm{d}\omega = 0$ then there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(U)$ such that $\mathrm{d}\eta = \omega$.

Lemma 14.0.3. The de Rham complex is exact.

Proof. For $k \geq 1$ consider the relevant part of the complex

$$\Omega^{k-1} \xrightarrow{\mathrm{d}^{k-1}} \Omega^k \xrightarrow{\mathrm{d}^k} \Omega^{k+1}$$

We need to show that for any $p \in M$ the complex

$$\left(\Omega^{k-1}\right)_p \xrightarrow{\mathrm{d}_p^{k-1}} \left(\Omega^k\right)_p \xrightarrow{\mathrm{d}_p^k} \left(\Omega^{k+1}\right)_p$$

is exact, i.e: $\operatorname{im} \mathrm{d}_p^{k-1} = \ker \mathrm{d}_p^k$.

For the $[\subseteq]$ inclusion, let $\mathrm{d}\eta \in \operatorname{im} \mathrm{d}^{k-1} \subseteq \Omega^k$. By Lemma 14.0.1 we have that $\mathrm{d}(\mathrm{d}\eta) = 0$, i.e: $\mathrm{d}\eta \in \ker \mathrm{d}^k$.

For the $[\supseteq]$ inclusion, let $[\omega]_p \in \ker \mathrm{d}_p^k$, i.e: $[\omega]_p \in \left(\Omega^k\right)_p$ verifies $\mathrm{d}_p^k[\omega]_p = 0$. By definition there exists an open neighbourhood U of p and a k-form ω such that $\mathrm{d}^k\omega|_U = 0$.

Since M is a smooth manifold of dimension m we can choose a local chart (V, ψ) around p such that $V \subseteq U$ and $\psi(V)$ is an open ball in $\mathbb{R}^m$. Since open balls are contractible, by Poincaré's lemma there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(V)$ such that $\mathrm{d}^{k-1}\eta = \omega|_V$.

Therefore there exists $[\eta]_p \in \left(\Omega^{k-1}\right)_p$ such that $\mathrm{d}_p^{k-1}[\eta]_p = [\omega]_p$, i.e: $[\omega]_p \in \operatorname{im} d_p^{k-1}$. □

Lemma 14.0.4. Let M be a smooth paracompact manifold. Then for all k the sheaf of k-forms $\Omega^k(M)$ is fine.

Proof. Let $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ be an arbitrary open cover of M.

Since M is paracompact there exists a partition of unity $\{f_i\}_{i\in\mathcal{I}} \subset C^\infty(M)$ subordinate to $\mathscr{U}$. This means that

$$\begin{aligned} \forall i \in \mathcal{I} \quad & \operatorname{supp} f_i \subseteq U_i \\ \forall p \in M \quad & \sum_{i\in\mathcal{I}} f_i(p) = 1 \end{aligned}$$

and this last sum is finite.

For all indices $i \in \mathcal{I}$ consider the map

$$\begin{aligned} \eta_i : \Omega^k(M) &\to \Omega^k(M) \\ \omega &\mapsto f_i \cdot \omega \end{aligned}$$

Note that since $\operatorname{supp} f_i \subseteq U_i$ we have $\operatorname{supp} \eta_i \subseteq U_i$. Furthermore for any k-form ω

$$\sum_{i\in\mathcal{I}} \eta_i(\omega) = \sum_{i\in\mathcal{I}} f_i \cdot \omega = \left(\sum_{i\in\mathcal{I}} f_i\right) \cdot \omega = 1 \cdot \omega = \omega$$

Therefore $\sum_{i\in\mathcal{I}} \eta_i = \mathbb{1}$, making $\{\eta_i\}_{i\in\mathcal{I}}$ a partition of unity subordinate to $\mathscr{U}$. □

Theorem 14.0.1. Let M be a smooth paracompact manifold. Then

$$\mathrm{H}^\bullet(M, \underline{\mathbb{R}}) \cong \mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R})$$

Proof. Consider the resolution

$$0 \longrightarrow \underline{\mathbb{R}} \longrightarrow \Omega^\bullet(M)$$

where $\Omega^\bullet$ is the de Rham complex of M.

By Lemma 14.0.3 we know that $\Omega^\bullet(M)$ is exact, therefore we only need to show that this resolution is exact at $\Omega^0(M)$, i.e: that for all $p \in M$

$$0 \longrightarrow \underline{\mathbb{R}}_p \xrightarrow{\iota_p} \Omega^0_p \xrightarrow{\mathrm{d}_p} \Omega^1_p \longrightarrow 0$$

is a short exact sequence, i.e: $\operatorname{im} \iota_p = \ker \mathrm{d}_p$.

- $\subseteq$ Let $[c]_p \in \underline{\mathbb{R}}_p$. By definition this is a locally constant function $c : U \to \mathbb{R}$ on some open neighbourhood U of p.
  Since c is locally constant we have that $\mathrm{d}c|_U = 0$, i.e $\mathrm{d}_p[c]_p = 0$.
- $\supseteq$ Let $[f]_p \in \Omega^0_p$ such that $\mathrm{d}_p[f]_p = 0$. This means that there exists an open neighbourhood U of p and a function $f \in \Omega^0(V) \cong C^\infty(M)$ such that $\mathrm{d}f|_U = 0$. Therefore f is locally constant in each of the connected components of U, i.e: $[f]_p \in \underline{\mathbb{R}}_p$.

Therefore since by Lemma 14.0.4 the resolution is made of fine sheaves and fine sheaves are $\Gamma(X, -)$ acyclic by definition of sheaf cohomology

$$\mathrm{H}^\bullet(M, \underline{\mathbb{R}}) = \mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R})$$

□

Corollary 14.0.1. Let M be a smooth paracompact manifold of dimension m. Then for all $k > m$

$$\mathrm{H}^k(M, \underline{\mathbb{R}}) = 0$$

Example 14.0.3 (The cohomology of S^1)**.** Consider the circle S^1 as a smooth manifold of dimension 1.
We are going to compute its sheaf cohomology with coefficients in $\underline{\mathbb{R}}$, $\mathrm{H}^\bullet(S^1, \underline{\mathbb{R}})$, using the de Rham complex

$$\Omega^0(S^1) \xrightarrow{\mathrm{d}^0} \Omega^1(S^1) \xrightarrow{\mathrm{d}^1} 0$$

On the one hand

$$\mathrm{H}^0(S^1, \underline{\mathbb{R}}) = \ker d^0 = \{f \in \Omega^0(S^1) \cong C^\infty(S^1) \mid \mathrm{d}f = 0\}$$

Since S^1 is connected, the only functions with 0 derivative are the constant functions. Therefore

$$\mathrm{H}^0(S^1, \underline{\mathbb{R}}) = \mathbb{R}$$

On the other

$$\mathrm{H}^1(S^1, \underline{\mathbb{R}}) = \ker d^1 \,/\, \operatorname{im} d^0 = \{\omega \in \Omega^1(S^1) \mid \mathrm{d}\omega = 0\} \,/\, \{\mathrm{d}f \mid f \in \Omega^0(S^1) \cong C^\infty(S^1)\}$$

Note that since $\Omega^2(M) = 0$ all 1-forms $\omega \in \Omega^1(S^1)$ verify $\mathrm{d}\omega = 0$. Therefore

$$\{\omega \in \Omega^1(S^1) \mid \mathrm{d}\omega = 0\} = \Omega^1(S^1)$$

Consider now the *integration* map

$$I : \Omega^1(S^1) \to \mathbb{R}$$
$$\omega \mapsto \int_{S^1} \omega$$

Let $\mathrm{d}\theta$ be the differential of the angle. Since $I(\mathrm{d}\theta) = 2\pi$ we have that I is surjective. Furthermore by Stoke's theorem we have that $I(\omega) = 0$ if and only if $w = \mathrm{d}f$ for some $f \in C^\infty(S^1)$.
Therefore by the first isomorphism theorem we have that

$$\mathrm{H}^1(S^1, \underline{\mathbb{R}}) = {}^{\Omega^1(S^1)}\big/_{\ker I} \cong \operatorname{im} I = \mathbb{R}$$

In summary, we have that

$$\mathrm{H}^k(S^1, \underline{\mathbb{R}}) = \begin{cases} \mathbb{R} & k \in \{0, 1\} \\ 0 & k > 1 \end{cases}$$

Appendix A

Algebra

Definition A.0.1 (Algebra)**.**

Definition A.0.2 (Local ring)**.**

Definition A.0.3 (Module)**.**

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